Decision Overmatch: Accelerating Military Advantage with AI-Augmented Decentralized Autonomous Organizations

Aaron Luhning

2026-03-23

Decision Overmatch

Accelerating Military Advantage with AI-Augmented Decentralized Autonomous Organizations

Aaron Luhning

[Institution Name]

v0.2

March 2026

DRAFT - For Advisor Review

Abstract

[See 00-abstract.md for current abstract.]

Document History

Version	Date	Changes
0.1	2026-01-24	Initial draft - all sections complete
0.2	2026-03-23	Major update: robot bridge/vision/swarm, knowledge graph brain visualization, Decide tab, resource DIDs, training assessment, 8 contributions, 50+ phases documented

---

Strategic Research Requirement Master's Program

Abstract

Coalition military operations face an accelerating coordination crisis. Siloed decision-making, fragmented decision cycles, and gaps between strategic intent and tactical execution hinder effective coalition operations. Traditional hierarchical command and control structures strain under information overload, coalition complexity, and the imperative for rapid decision-making in contested environments.

This research explores how AI-augmented Decentralized Autonomous Organizations (DAOs) can provide a governance framework for military command and control. BASTION (Blockchain Autonomous Strategy & Tactical Intelligence Operational Network) integrates blockchain-based DAOs with AI agents to enable secure, transparent, and resilient coordination across strategic, operational, and tactical levels. Smart contracts encode policy constraints that execute automatically, ensuring compliance with national caveats and classification restrictions.

A five-tier authority model governs 65 Military Decision Making Process (MDMP) activities, with three decision categories permanently locked to human-only authority. Governance gates enforce doctrinal phase progression, assumption lifecycle management triggers automatic replanning, and adversary modeling provides analytical depth within governance-bounded wargaming. The platform implements doctrine-first design through six workflow tabs aligned to JP 5-0 (Understand, Design, Plan, Decide, COP, Assess).

BASTION fields 131 AI agents—31 specialized agents plus 102 Joint Planning Process staff role agents. A resource registry with blockchain-anchored Decentralized Identifiers manages military assets with real-time readiness tracking. A global training/operational mode toggle enables “train as you fight” doctrine with full data isolation. An adaptive knowledge graph brain visualizes semantic intelligence networks, surfacing entity relationships and confidence-scored intelligence chains across hierarchical problem sets.

A physical demonstration validates the approach using NVIDIA Jetson Orin Nano edge computing and Sphero RVR+ robotic platforms. Robot vision detects threats autonomously, doctrinal swarm formations coordinate using UDP peer mesh, and a DAO assigns mission objectives to robot platforms. Three graduated human authority positions demonstrate that BASTION maintains a 100% approval threshold invariant for strike authorization while enabling autonomous execution at lower authority levels.

This research makes eight novel contributions: DAO governance for military C2; AI agent augmentation of decentralized decisions; multi-echelon coordination from strategic to tactical; edge AI and robotic proof-of-concept with vision and swarm capabilities; semantic knowledge graph with adaptive brain visualization; blockchain-anchored resource identity with plugin architecture; training/operational governance parity; and hierarchical problem set inheritance for theater-to-tactical context propagation.

Keywords: Decentralized Autonomous Organizations, Artificial Intelligence, Military Command and Control, Blockchain, Coalition Operations, Human-Machine Teaming, MDMP, Common Operating Picture, Edge Computing, Knowledge Graphs

1. Introduction

Modern military operations face an accelerating crisis of coordination. As adversaries develop capabilities to contest every domain simultaneously, the traditional model of hierarchical command and control (C2) strains under the weight of information overload, coalition complexity, and the imperative for rapid decision-making. This paper introduces BASTION (Blockchain Autonomous Strategy & Tactical Intelligence Operational Network), a novel framework that integrates AI-augmented Decentralized Autonomous Organizations (DAOs) with military command structures to address fundamental coordination challenges in coalition operations.

1.1 The Problem: Siloed Decision-Making in Coalition Operations

Effective military and coalition operations are hindered by systemic challenges, foremost among them siloed decision-making.[1] Fragmented and stove-piped decision cycles, where branches, nations, or tactical units operate independently without cohesive governance and dynamic coordination, create costly delays. These delays are particularly acute at the operational level, where the gap between strategic intent and tactical execution determines mission success or failure.

Intelligence sharing exemplifies this dysfunction. Despite technological capabilities that could enable real-time information exchange, efforts to share intelligence remain constrained by national policies and caveats.[2] A sensor operator in one nation may observe critical tactical information that could save lives in a partner nation’s area of operations, yet classification restrictions, incompatible systems, and bureaucratic release authorities prevent that information from reaching those who need it. The result is critical gaps in situational awareness, eroded trust between coalition partners, and decision-making that proceeds at the speed of the slowest approval chain rather than the speed of relevance.

The lack of transparency and accountability compounds these issues. When resources flow through multiple national channels and decisions traverse complex approval hierarchies, tracking outcomes becomes nearly impossible. Commanders struggle to answer basic questions: Which assets are available? Who authorized this action? What happened to the supplies we requested three weeks ago? This opacity results in wasted effort, duplicated requests, and friction that degrades the trust essential for coalition operations.[3]

1.2 Why This Problem Matters

The consequences of coordination failure extend beyond operational inefficiency to strategic defeat. In contested environments where adversaries operate inside friendly decision cycles, the inability to coordinate rapidly across coalition boundaries directly threatens the ability to achieve objectives. Modern peer adversaries have studied Western coalition operations and explicitly designed their strategies to exploit coordination gaps.[4]

The Russia-Ukraine conflict beginning in 2022 provides a contemporary case study. International efforts to support Ukraine's defense required coordination across dozens of nations, each with distinct approval processes, legal frameworks, and political constraints. Materiel donations that were approved in principle took weeks or months to reach the front lines. Training programs suffered from lack of visibility into what other nations were providing. Maintenance and sustainment chains fragmented across national boundaries. The coalition's aggregate capability far exceeded what could be effectively employed due to coordination friction.[5]

These challenges will only intensify. Future conflicts will likely involve larger coalitions operating across multiple domains simultaneously. The speed of modern warfare, driven by autonomous systems, cyber operations, and hypersonic weapons, compresses decision timelines from hours to minutes. Coalition structures designed for deliberate planning processes cannot adapt to this tempo. Without new approaches to coordination, coalitions will find themselves outmaneuvered by more agile adversaries willing to accept greater centralization and risk.[6]

1.3 Why This Problem Is Hard

Three fundamental sources of friction make military coordination resistant to traditional solutions: security requirements, trust deficits, and speed imperatives. These constraints often conflict, creating impossible tradeoffs that current systems cannot resolve.

Security requirements impose classification barriers that fragment information by design. Information classified at higher levels cannot flow to those without appropriate clearances. National caveats restrict sharing even among cleared personnel from partner nations. While these restrictions serve legitimate purposes, they create information silos that degrade collective situational awareness. Systems designed to enforce these restrictions often do so clumsily, blocking legitimate information sharing while providing insufficient protection against sophisticated adversaries.[7]

Trust between coalition partners cannot be assumed and must be continuously verified. Partners need assurance that their contributions will be used as intended, that commitments will be honored, and that sensitive information will be protected. Traditional approaches to establishing trust rely on personal relationships, institutional reputation, and legal agreements. These mechanisms work poorly in ad hoc coalitions formed rapidly in response to emerging crises, where there may be no history of cooperation and insufficient time to build relationships through gradual exposure.[8]

Speed requirements conflict directly with the deliberate processes that security and trust traditionally require. Human approval chains, while providing accountability, introduce latency that can exceed the relevant decision window. The time required for a multi-national approval process may exceed the time available to engage a fleeting target or respond to a developing threat. Systems that require human approval for every action cannot match the tempo of autonomous adversary systems operating without such constraints.[9]

Centralized command and control structures, while familiar and proven in many contexts, are ill-suited to address these challenges simultaneously. Centralized C2 creates single points of failure vulnerable to disruption. It concentrates decision authority at echelons removed from tactical reality. It cannot adapt dynamically to disconnected, intermittent, limited bandwidth (DDIL) environments where communication with higher headquarters may be impossible for extended periods. Compounding this is inefficient resource management driven by poor visibility into available assets and real-time requirements across the coalition.[10]

1.4 Why Previous Approaches Have Not Succeeded

Several significant initiatives have attempted to address military coordination challenges, yet each falls short of providing a complete solution.

Joint All-Domain Command and Control (JADC2) represents the U.S. Department of Defense's vision for connecting sensors and shooters across all domains.[11] JADC2 focuses on technical interoperability, creating the "pipes" through which information can flow. However, JADC2 does not address governance: it can connect systems, but it does not determine who has authority to act, how decisions should be made, or how to verify that actions comply with policy. JADC2 enables coordination but does not provide the governance framework to manage it.

NATO Federated Mission Networking (FMN) provides a framework for coalition information sharing.[12] FMN addresses interoperability through standardized interfaces and certification processes. However, FMN relies on a fundamentally centralized trust model where participating nations must trust the federation infrastructure and governance bodies. This works for established alliances with mature trust relationships but scales poorly to ad hoc coalitions or scenarios where some partners have limited trust in others. FMN also lacks mechanisms for autonomous policy enforcement; compliance depends on human implementers following documented procedures.

Military AI systems increasingly provide decision support for commanders, analyzing large datasets, identifying patterns, and recommending courses of action.[13] These systems augment human decision-making but operate within existing command structures rather than transforming them. They do not address the fundamental governance challenges of coalition operations: who decides, based on what authority, with what verification. AI systems that recommend actions still require human commanders to navigate the same approval chains and coordination processes that create current delays.

Commercial DAOs have demonstrated the viability of decentralized governance for coordinating resources and decisions among parties who do not fully trust each other.[14] Smart contracts encode rules that execute automatically, providing transparency and accountability without centralized control. However, commercial DAOs are designed for business contexts where the primary currencies are financial tokens and the tempo allows for extended deliberation. Military operations require different governance models that account for human authority over lethal decisions, policy compliance, classification constraints, and operational tempo measured in seconds rather than days.

Each of these approaches addresses part of the problem. None provides an integrated solution that combines connectivity (JADC2), coalition interoperability (FMN), intelligent analysis (military AI), and decentralized governance (DAOs) into a coherent framework suitable for military operations.

1.5 The BASTION Approach

This research proposes BASTION (Blockchain Autonomous Strategy & Tactical Intelligence Operational Network), a framework that integrates AI-augmented Decentralized Autonomous Organizations with military command and control structures. BASTION addresses coordination challenges through four key components working in concert.

AI-augmented DAOs provide governance structures that coordinate resources and decisions across organizational boundaries. Unlike commercial DAOs designed for deliberate token-based voting, BASTION's DAOs are architected for military tempo. AI agents augment human decision-makers by analyzing proposals, assessing risks, monitoring compliance, and executing routine decisions within delegated authority. Human commanders retain authority over significant decisions while AI agents handle the coordination overhead that currently consumes staff attention.

Blockchain infrastructure provides the transparency, immutability, and trust verification that coalition operations require. Every decision, resource commitment, and action is recorded in an append-only ledger that all authorized participants can verify independently. Partners can confirm that agreements are being honored without trusting any single authority to maintain records honestly. The blockchain serves not as a cryptocurrency platform but as a shared source of truth for coalition coordination.

Smart contracts encode policy constraints that execute automatically, ensuring compliance without requiring human verification of every action. National caveats can be expressed as code that governs how a nation's contributed resources may be employed. Classification rules can be enforced automatically, preventing unauthorized disclosure while enabling maximum legitimate sharing. Rules of engagement can be verified computationally before actions are taken, providing confidence that operations remain within authorized boundaries.

Three levels of human authority ensure appropriate human control over AI agent actions. Decisions are categorized by type and assigned to one of three authority positions:

- **Human-in-the-loop:** Humans must approve each action before execution. Required for lethal decisions, strategic commitments, and actions with significant consequence potential.
- **Human-on-the-loop:** AI agents may execute actions autonomously, but humans monitor and may intervene. Appropriate for routine coordination where speed benefits outweigh individual decision review.
- **Human-out-of-the-loop:** AI agents execute autonomously within policy constraints, with human review conducted periodically or on exception. Suitable only for well-bounded decisions with clear policy guidance and low consequence potential.

A physical demonstration validates these concepts using tangible hardware. A Jetson Orin Nano edge computing platform and Sphero RVR+ robotic vehicle operate within a physical area of operations model, demonstrating DAO-coordinated autonomous target identification, resource selection, and effects delivery. This demonstration grounds theoretical contributions in observable system behavior, proving that BASTION's architecture can function in scenarios approaching real-world conditions.

1.6 Research Question

This research addresses the following question:

> **How can interconnected, AI-augmented Decentralized Autonomous Organizations (DAOs) provide a secure, transparent, and resilient governance framework that enables effective C2, accelerates decision-making, optimizes resource management, and supports autonomous, policy-compliant coordination across diverse national and organizational boundaries?**

Answering this question requires demonstrating that DAO-based governance can satisfy military requirements for security, accountability, and human control while providing benefits in speed, transparency, and resilience that traditional centralized approaches cannot match.

1.7 Contributions

This research makes the following contributions:

- **Novel integration of DAO governance with military C2:** This work is among the first to systematically apply decentralized autonomous organization concepts to military command and control, adapting commercial DAO patterns for the unique requirements of defense operations including classification constraints, national caveats, and human authority over lethal decisions.

- **AI agent augmentation of decentralized decision processes:** The research demonstrates how AI agents can enhance DAO governance by providing analysis, risk assessment, and autonomous execution within policy constraints, while maintaining appropriate human oversight through tiered authority levels.

- **Multi-level coordination framework spanning strategic, operational, and tactical levels:** BASTION provides a unified governance approach that functions consistently across military echelons, enabling seamless coordination from strategic resource allocation through tactical execution.

- **Physical proof-of-concept demonstration with edge AI and robotics:** Beyond theoretical analysis, this research includes a working demonstration using NVIDIA Jetson Orin Nano edge computing and Sphero RVR+ robotic platforms. The demonstration validates robot vision-enabled threat detection, doctrinal swarm formations (wedge, line, column, echelon, vee, staggered column), and DAO-coordinated mission assignment — proving the architecture's viability across software and hardware domains.

- **Semantic knowledge graph with adaptive brain visualization:** BASTION's JSON-LD knowledge graph captures entities, relationships, and confidence-scored intelligence chains across hierarchical problem sets. An adaptive neural canvas renders this graph as an interactive brain visualization, enabling analysts to explore semantic intelligence networks and temporal knowledge evolution.

- **Blockchain-anchored resource identity and training/operational governance parity:** A resource registry with Decentralized Identifiers (did:near:resource-{id}) treats military assets as first-class blockchain entities. A global training/operational mode toggle provides identical DAO governance in both exercise and operational contexts, enabling authentic "train as you fight" doctrine.

1.8 Development Scope and Version Note

This paper reflects BASTION capabilities as of Phase 53 (March 2026), encompassing over 50 completed development phases since the initial architecture described in v0.1 (January 2026). Significant capabilities added between v0.1 and v0.2 include: the Decide tab with governance-gated decision dashboard and RACI matrix, the operational design workspace with Center of Gravity analysis, robot bridge integration with vision and swarm capabilities, adaptive knowledge graph brain visualization, resource DID registry with plugin architecture, training/operational mode toggle, and hierarchical problem set inheritance with automatic context propagation. These capabilities are documented in the methodology and results sections; this version note provides temporal context for readers comparing this version to future releases.

1.9 Paper Organization

The remainder of this paper is organized as follows:

Section 2 (Background) provides foundation in three domains: Web3 technologies and DAOs, military coordination frameworks and challenges, and AI applications in defense. This section establishes the technical and operational context necessary to understand BASTION's design.

Section 3 (Methodology) describes BASTION's architecture, including the DAO governance structures, AI agent framework, blockchain infrastructure, and security mechanisms. This section explains the design decisions that enable BASTION to address the challenges identified in this introduction.

Section 4 (Results) presents the end-to-end coordination flow and demonstration outcomes. This section provides evidence that BASTION achieves its intended functionality and demonstrates the three human authority positions in operation.

Section 5 (Discussion) analyzes BASTION's implications, limitations, and potential for future development. This section addresses risks, ethical considerations, and research directions that extend beyond the current work.

Section 6 (Conclusion) summarizes findings and their significance for military coordination in coalition operations.

Appendices provide implementation status (Appendix A) and a detailed demonstration script (Appendix B) for readers interested in technical depth or replication.

---

2. Background: Decentralized Autonomous Organizations and Web3 Technologies

This section introduces the foundational technologies that underpin BASTION's approach to military coordination. For readers unfamiliar with blockchain or decentralized systems, the following subsections provide the necessary context to understand how these technologies enable novel governance frameworks for command and control.

2.1 Blockchain Fundamentals

What is a Blockchain?

A blockchain is a distributed ledger technology that maintains an identical copy of data across a network of computers rather than storing information in a single central location. When new information is added to a blockchain, every computer in the network receives and verifies this update, creating multiple redundant copies of the same data. This distributed architecture provides several properties critical to secure, multi-party coordination.

The term "blockchain" derives from its underlying data structure: information is grouped into discrete units called blocks, and each block is cryptographically linked to the previous block, forming an unbroken chain. This linking occurs through cryptographic hashing, a mathematical process that converts data of any size into a fixed-length string of characters. Even the smallest change to the original data produces a completely different hash value. When each block contains the hash of the previous block, any attempt to alter historical data would break the chain of hashes, immediately revealing the tampering.

This architectural design produces three key properties. First, immutability: once data is recorded on a blockchain, it cannot be altered or deleted without detection. The chain of cryptographic hashes ensures that any modification to historical records would be evident to all network participants. Second, transparency: all transactions recorded on the blockchain are visible to participants in the network, creating a shared source of truth. Third, distributed trust: the network operates without requiring participants to trust any single central authority, as consensus mechanisms ensure agreement on the current state of the ledger.

Consensus mechanisms are protocols that enable distributed network participants to agree on which transactions are valid and in what order they occurred. Proof-of-stake consensus, used by many modern blockchains including NEAR Protocol, requires network validators to commit economic stake as collateral [CITATION NEEDED: NEAR Protocol technical documentation]. Validators who attempt to approve fraudulent transactions risk losing their stake, aligning economic incentives with honest behavior. This approach provides security without the energy-intensive computation required by earlier proof-of-work mechanisms [CITATION NEEDED: Energy comparison studies between PoW and PoS].

Why Blockchain Matters for Coordination

Traditional multi-party coordination requires either mutual trust between all parties or reliance on a trusted intermediary. Coalition military operations often lack both: national interests may conflict, and no single nation may be willing to cede control to another. Blockchain technology offers an alternative model where coordination rules are encoded transparently and enforced automatically, without requiring parties to trust each other or any central authority.

The immutable audit trail provided by blockchain creates accountability for all decisions and actions. Every vote, approval, or resource allocation is permanently recorded with a timestamp and cryptographic proof of its origin. This comprehensive record enables after-action review, supports attribution in disputes, and deters bad actors who know their actions will be permanently visible.

Blockchain-based coordination also provides resilience against disruption. Because the ledger is replicated across many nodes, the system continues to function even if individual nodes are compromised or communication links are severed. This property is particularly valuable in contested environments where adversaries may target communication infrastructure or attempt to corrupt centralized command systems.

2.2 Decentralized Autonomous Organizations (DAOs)

What is a DAO?

A Decentralized Autonomous Organization (DAO) is an organization whose governance rules are encoded in software and executed automatically by a blockchain network. Unlike traditional organizations where decisions flow through hierarchical management structures, DAOs coordinate collective action through transparent proposals, voting mechanisms, and automated execution.

In a DAO, members submit proposals for consideration by the organization. These proposals might request funding for a project, propose changes to organizational rules, or authorize specific actions. Other members review the proposal and cast votes according to the organization's voting rules. When a proposal receives sufficient support, the associated action is executed automatically by the underlying blockchain without requiring any intermediary to implement the decision.

This structure eliminates the need for trusted executives or administrators who might delay, modify, or fail to implement decisions. The rules governing the organization are visible to all members, votes are recorded immutably, and outcomes follow deterministically from the voting results. Members can verify that the organization operates according to its stated rules rather than relying on promises from leadership.

DAOs exist on a spectrum of autonomy. Some DAOs require human approval for every action. Others operate semi-autonomously, executing certain routine decisions automatically while requiring human intervention for significant matters. The most autonomous DAOs implement complex rule sets that handle a wide range of scenarios without human involvement. The appropriate level of autonomy depends on the organization's purpose and the consequences of potential errors [CITATION NEEDED: DAO Research Trends systematic review, MDPI 2025].

Smart Contracts

Smart contracts are the software programs that encode a DAO's rules and execute its decisions. The term "smart contract" refers not to legal contracts but to programs that automatically execute predefined actions when specific conditions are met. These programs run on blockchain infrastructure, inheriting the blockchain's properties of immutability and transparency.

Once deployed to a blockchain, a smart contract cannot be altered. This immutability ensures that the rules governing an organization remain exactly as agreed upon by its members. No administrator can secretly modify voting thresholds, no insider can create exceptions for favored parties, and no external actor can corrupt the governance process. The rules are literally set in code, visible to all and enforced by mathematics rather than by fallible human institutions.

Smart contracts can encode complex conditional logic: if a proposal receives a two-thirds majority vote and at least five council members participate, then transfer these resources to that recipient. They can implement time delays, requiring that approved actions wait for a review period before execution. They can require multiple signatures from different parties before authorizing sensitive operations. All of these constraints operate automatically and cannot be bypassed without deploying an entirely new contract with different rules.

DAO Governance Mechanisms

DAOs implement various mechanisms to ensure decisions reflect the collective will of their members while maintaining operational efficiency. Proposal submission typically requires meeting certain criteria: the proposer might need to hold a minimum stake in the organization or receive endorsements from existing members. These requirements prevent spam while ensuring that serious proposals receive consideration.

Voting on proposals can follow different rules depending on the decision’s importance. Routine operational decisions might require simple majority approval. Constitutional changes that alter the organization’s fundamental rules might require supermajority thresholds of two-thirds or even three-fourths agreement. The most consequential decisions might require unanimous consent from designated council members.

Quorum requirements ensure that decisions reflect broad participation rather than just the preferences of a small active minority. A proposal might require not only majority support among those who vote but also that a minimum percentage of all eligible members participate in the vote. This prevents decisions from being made when most members are unaware or uninvolved.

Time-locked execution provides a safeguard against hasty decisions and allows for error correction. When a proposal is approved, its execution might be delayed for a period during which members can review the decision. Some DAOs implement veto mechanisms allowing designated parties to block execution during this window if they identify problems.

Multi-signature requirements ensure that critical decisions cannot be made unilaterally. Certain actions might require approval from multiple independent parties, each holding a separate cryptographic key. This requirement ensures that sensitive operations proceed only when multiple designated authorities agree.

The balance between autonomous execution and human oversight represents a fundamental design choice. Fully autonomous execution maximizes speed and consistency but risks implementing flawed decisions. Required human approval introduces delays but provides opportunities for judgment and course correction. Effective DAO design matches the level of autonomy to the consequences of potential errors.

2.3 Web3 Principles

User-Owned Data

Web3 represents a paradigm shift in how users interact with digital systems. In the current web model, users create accounts on platforms controlled by corporations, which then own and control the data those users generate. Web3 inverts this model: users own their data and grant selective access to applications rather than surrendering ownership to platforms.

This ownership model relies on cryptographic key pairs. Users hold private keys that only they control, while public keys serve as their identity across multiple systems. When users interact with a Web3 application, they sign their actions with their private key, proving their identity without revealing sensitive credentials to the application. This architecture enables users to maintain a consistent identity across different systems while retaining control over their data and its use.

The interoperability enabled by this model allows applications to share information and coordinate actions without requiring users to create separate accounts or transfer data manually between services. A credential issued by one organization can be verified by another without either party needing to trust or communicate with the other, provided both recognize the cryptographic standards in use.

Decentralization Benefits

Decentralized systems distribute control across multiple independent parties rather than concentrating authority in a single entity. This distribution provides resilience: the system continues to operate even when individual components fail. No single point of failure exists that an adversary could target to disable the entire system.

Decentralization also provides censorship resistance. When no single party controls the system, no single party can arbitrarily exclude participants or suppress information. This property matters when coordination must occur across organizational or national boundaries where any central authority might face pressure to favor certain parties.

Perhaps most importantly for coalition operations, decentralization reduces trust requirements. Parties that might not fully trust each other can still coordinate effectively when the rules of their interaction are encoded transparently and enforced automatically. Each party can verify for themselves that the system operates as promised rather than relying on assurances from others.

2.4 Defense Applications of Blockchain

Current and Emerging Uses

Military organizations worldwide have begun exploring blockchain applications, though deployment remains limited compared to commercial adoption [CITATION NEEDED: Blockchain Applications in the Military Domain systematic review, MDPI 2025]. The most mature applications focus on supply chain tracking, where blockchain's immutable record-keeping can verify the provenance and handling of components throughout complex logistics networks. The U.S. Department of Defense has piloted blockchain systems to track parts through the defense supply chain, ensuring authenticity and flagging potential counterfeit components [CITATION NEEDED: DoD blockchain pilot documentation].

Cybersecurity represents another active area of defense blockchain research. Blockchain-based identity management can reduce reliance on centralized authentication servers that present attractive targets for adversaries. Distributed logging and monitoring systems can resist tampering that might allow attackers to cover their tracks. Several nations are developing or have deployed blockchain-based systems for securing military communications and identity verification [CITATION NEEDED: Leveraging Blockchain for Military Automation and Cybersecurity, SSRN 2025].

Coalition information sharing presents particular opportunities for blockchain adoption. When multiple nations must share intelligence or coordinate operations, blockchain can provide a shared record that no single nation controls while maintaining the security classifications and access controls each nation requires. NATO has explored federated blockchain approaches for this purpose [CITATION NEEDED: NATO C2COE or NCIA documentation on blockchain initiatives].

Research institutions and defense agencies have published substantial work examining blockchain's potential military applications. A comprehensive systematic review identified 43 peer-reviewed articles on blockchain applications in military domains, with applications spanning command and control, logistics, identity management, and cybersecurity [CITATION NEEDED: Blockchain Applications in the Military Domain, MDPI Technologies 13(1):23, 2025]. The majority of these applications focus on blockchain's data integrity properties rather than its governance capabilities.

Gap: DAO Governance for Command and Control

Despite growing interest in blockchain for defense applications, a notable gap exists in the literature and in practice: no existing systems combine DAO governance mechanisms with military command and control. Current blockchain defense applications use the technology primarily as a secure database, leveraging immutability and transparency for record-keeping but not exploiting the autonomous governance capabilities that distinguish DAOs from simple distributed ledgers.

This gap is significant because the coordination challenges facing coalition military operations extend beyond data integrity to governance itself. The challenge is not merely ensuring that records are accurate but ensuring that decisions are made according to agreed-upon rules, that resources are allocated fairly and efficiently, and that actions are authorized appropriately across national boundaries.

Similarly, while artificial intelligence is increasingly applied to military decision support, existing systems lack decentralized governance frameworks. AI tools can accelerate analysis and recommend courses of action, but the decisions ultimately flow through traditional hierarchical command structures. No existing systems combine AI augmentation with DAO governance to create a framework that is both intelligent and decentralized.

BASTION addresses this gap by integrating DAO governance with AI agent augmentation specifically for command and control applications. See Section 3 (Methodology) for a detailed description of how BASTION implements these principles through its layered architecture of strategic, operational, and tactical

DAOs. See Section 4 (Results) for a demonstration of DAO governance in action through the physical proof-of-concept system.

2.5 Summary

This section has introduced the foundational technologies that enable BASTION's approach to military coordination:

- **Blockchain** provides immutable, transparent record-keeping distributed across multiple nodes, enabling coordination without central authority.
- **Decentralized Autonomous Organizations (DAOs)** build on blockchain to enable collective governance through encoded rules, transparent proposals, and automated execution.
- **Smart contracts** encode governance rules that execute automatically and cannot be altered after deployment.
- **Web3 principles** enable user-owned data and identity that can operate across organizational boundaries.

Defense applications of blockchain currently focus on data integrity for supply chains, cybersecurity, and information sharing. However, no existing systems apply DAO governance mechanisms to military command and control. This gap represents the opportunity that BASTION addresses: combining decentralized governance with AI augmentation to enable effective coordination across diverse national and organizational boundaries.

The following section (Section 2.2) examines the military coordination context, including traditional command and control structures and the specific challenges that modern coalition operations present. Together, these background sections provide the foundation for understanding BASTION's methodology and contributions in Section 3.

2.5 Military Coordination Frameworks

Military operations, particularly those involving coalition partners, require sophisticated coordination mechanisms that span multiple organizational levels and national boundaries. Understanding these frameworks is essential to appreciating the coordination challenges that modern command and control (C2) systems must address.

Levels of Warfare

Military doctrine recognizes three interconnected levels of warfare, each with distinct decision-making requirements and time horizons [CITATION NEEDED].

Strategic Level. The strategic level of warfare addresses national and multinational objectives, translating political aims into military goals. At this level, national leadership and senior military commanders determine which military objectives will achieve political ends, allocate resources among theaters of operation, and establish overall priorities. Strategic decisions typically involve long time horizons measured in months or years and require coordination across government agencies, allied nations, and international organizations [CITATION NEEDED]. Examples include decisions about force structure, alliance commitments, and campaign objectives.

Operational Level. The operational level serves as the crucial bridge between strategic objectives and tactical actions. Operational commanders design and conduct campaigns and major operations to achieve strategic objectives within a theater of operations. This level is where the gap between strategic intent and tactical execution is most acute, as operational planners must translate broad strategic guidance into specific, achievable military tasks [CITATION NEEDED]. The operational level typically operates on time horizons of weeks to months and requires coordination across multiple tactical units, service branches, and often coalition partners. Joint Force Commanders and their staffs operate primarily at this level.

Tactical Level. The tactical level involves the actual conduct of battles and engagements. Tactical commanders employ forces to accomplish military objectives through direct engagement with enemy forces. Decisions at this level often occur in real-time or near-real-time, with time horizons measured in hours or days. Tactical units must execute orders received from operational commanders while adapting to rapidly changing battlefield conditions [CITATION NEEDED].

Information Flow. Effective military operations require bidirectional information flow across these levels. Strategic guidance flows downward as orders and directives, becoming increasingly specific as it moves from strategic to tactical levels. Simultaneously, tactical information flows upward as reports and assessments, enabling operational and strategic commanders to understand the current situation and adjust plans accordingly. This constant flow of information is essential for maintaining unity of effort, but it also creates significant coordination challenges when systems are disconnected or degraded [CITATION NEEDED].

Command and Control (C2)

Command and Control, commonly abbreviated as C2, refers to the exercise of authority and direction by a designated commander over assigned and attached forces in the accomplishment of a mission [CITATION NEEDED]. C2 encompasses the processes and systems commanders use to plan, direct, coordinate, and control forces and operations.

Traditional Centralized C2. Historically, military C2 has relied on hierarchical, centralized structures. A single commander receives authority from higher headquarters and exercises that authority over subordinate units through a defined chain of command. This model provides clear lines of authority, accountability, and unity of command. Centralized C2 has proven effective in relatively stable environments where communications are reliable and commanders have time to process information and issue detailed orders [CITATION NEEDED].

Vulnerabilities in Contested Environments. Modern adversaries increasingly target C2 systems as a means of disrupting military effectiveness. Electronic warfare, cyber attacks, and kinetic strikes against communication nodes can sever the links between commanders and subordinate units [CITATION NEEDED]. When centralized C2 systems are disrupted, forces may be unable to receive orders, report status, or coordinate with adjacent units. This vulnerability is particularly acute against peer or near-peer adversaries with sophisticated capabilities to deny, degrade, or disrupt communications.

DDIL Challenges. Disconnected, Degraded, Intermittent, and Limited-bandwidth (DDIL) environments present fundamental challenges to traditional C2 approaches. In such environments, forces may be unable to maintain continuous communication with higher headquarters. Bandwidth constraints limit the amount of data that can be transmitted, forcing commanders to prioritize information. Intermittent connectivity means that forces may alternate between periods of communication and periods of isolation [CITATION NEEDED]. These conditions demand C2 approaches that can function even when units cannot maintain constant contact with higher headquarters.

Joint All-Domain Command and Control (JADC2)

The United States Department of Defense has recognized the need to modernize C2 capabilities to address these challenges. Joint All-Domain Command and Control (JADC2) represents the

DOD’s concept for connecting sensors and shooters across all domains: land, sea, air, space, and cyberspace [CITATION NEEDED].

Core Vision. JADC2 aims to enable any sensor to share data with any shooter, allowing commanders to make faster, better-informed decisions. By creating a seamless network of connected systems, JADC2 seeks to accelerate decision cycles and enable forces to respond more rapidly to threats and opportunities. The concept envisions using artificial intelligence and machine learning to process vast amounts of sensor data and present commanders with actionable information [CITATION NEEDED].

Implementation Challenges. Despite significant investment, JADC2 remains largely aspirational. A 2025 Government Accountability Office report found that the Department of Defense has not yet established a clear vision, strategy, or goals for JADC2, hindering progress toward achieving the concept’s objectives [CITATION NEEDED: GAO-25-106454]. Technical challenges include integrating legacy systems with modern networks, ensuring interoperability across services and domains, and protecting the network from cyber threats. Organizational challenges include aligning service-specific programs with joint requirements and defining clear lines of authority and responsibility.

What JADC2 Provides and Lacks. JADC2 focuses primarily on connecting sensors and enabling information sharing across domains. While this connectivity is essential, JADC2 does not inherently address the governance and decision-making frameworks needed to coordinate actions across organizational boundaries. The concept emphasizes technical connectivity but does not provide a decentralized governance mechanism for multi-stakeholder decision-making [CITATION NEEDED].

2.6 Coalition Coordination Challenges

Coalition operations introduce additional coordination challenges beyond those inherent in single-nation military operations. When multiple nations operate together, differences in doctrine, policy, and authority create friction that can impede effective coordination.

Multi-National Operations

Modern military operations increasingly involve coalition partners. Whether through formal alliances like the North Atlantic Treaty Organization (NATO) or ad hoc coalitions formed for specific operations, nations routinely operate alongside partners with different capabilities, procedures, and national interests [CITATION NEEDED].

Different National Doctrines. Each nation develops its own military doctrine based on its unique strategic culture, historical experience, and operational requirements. While allied nations often align their doctrines through exercises and agreements, significant differences remain. These differences affect how units communicate, how they interpret orders, and how they expect to operate in conjunction with partner forces [CITATION NEEDED].

Information Sharing Agreements. Coalition operations require extensive information sharing, but each nation maintains policies governing what information can be shared with whom. These policies, often referred to as “caveats,” restrict the dissemination of sensitive information to protect national sources and methods. While necessary for security, these caveats can create information gaps that impede coordination [CITATION NEEDED].

Release Authority Requirements. Before information can be shared with coalition partners, it typically must be reviewed and released by appropriate national authorities. This process, known as Foreign Disclosure, ensures that sensitive information is protected but can introduce delays that are incompatible with the tempo of modern operations. In rapidly evolving situations, the time required to obtain release authority may exceed the information’s useful lifespan [CITATION NEEDED].

NATO and Allied Coordination. The North Atlantic Treaty Organization has developed extensive procedures for coalition operations, including the Federated Mission Networking (FMN) framework for information sharing. FMN provides standards and processes for connecting coalition networks, but implementation remains uneven across member nations. Even with these frameworks, coalition coordination requires significant effort to overcome differences in systems, procedures, and authorities [CITATION NEEDED].

The Operational Gap

The proposal’s problem statement identifies the operational level as where the gap between strategic intent and tactical execution is most acute. This gap manifests in several ways.

Strategic Intent vs. Tactical Execution Disconnect. Strategic commanders establish objectives and allocate resources, but translating these broad directives into specific tactical actions requires interpretation and judgment at the operational level. Miscommunication or misunderstanding at this interface can result in tactical actions that fail to achieve strategic objectives or that create unintended consequences [CITATION NEEDED].

Siloed Decision-Making. As noted in the research problem statement, fragmented and stove-piped decision cycles, where branches, nations, or tactical units operate independently without cohesive governance, create costly delays. Each organization may optimize for its own objectives without adequate visibility into the broader operational picture. This siloed approach leads to duplication of effort, missed opportunities for synergy, and potential conflicts between friendly forces [CITATION NEEDED].

National Caveats Constraining Intelligence. The proposal specifically identifies national policies and caveats as constraining intelligence sharing, leading to critical gaps in situational awareness. When coalition partners cannot share relevant intelligence due to national restrictions, commanders may lack the information needed to make effective decisions. This challenge is particularly acute when time-sensitive intelligence cannot be shared quickly enough to be actionable [CITATION NEEDED].

Speed, Security, and Trust Tradeoffs. Coalition operations require balancing competing priorities. Faster information sharing improves coordination but increases security risks. More stringent security measures protect sensitive information but slow down decision cycles. Building trust among coalition partners takes time but enables more effective cooperation. These tradeoffs create friction that traditional C2 systems struggle to address [CITATION NEEDED].

Contemporary Example. The Russia-Ukraine conflict, which began in 2022 and continues through the present, has highlighted coalition coordination challenges in supporting a partner nation. Western nations have coordinated to provide military assistance, but this effort has required extensive coordination across dozens of nations with different capabilities, policies, and industrial bases. Challenges have included tracking donated equipment, coordinating training programs, and managing logistics across multiple national systems [CITATION NEEDED].

Current Limitations

Existing approaches to coalition coordination face several fundamental limitations.

Centralized Systems Vulnerable to Disruption. Current coalition C2 systems typically rely on centralized architectures that are vulnerable to the same disruption threats facing single-nation systems. If a coalition headquarters is degraded or destroyed, subordinate forces may lose the coordination mechanisms they depend on [CITATION NEEDED].

Poor Visibility into Partner Assets. Coalition commanders often lack complete visibility into the capabilities, status, and availability of partner nation forces. National restrictions may limit what information can be shared about force posture, while incompatible systems may prevent automatic sharing of status information. This lack of visibility impedes effective resource allocation and coordination [CITATION NEEDED].

Slow Consensus-Building Processes. Coalition decisions often require consensus or at least coordination among multiple nations. Traditional processes for building this consensus involve multiple rounds of communication, staffing through national chains of command, and negotiation of divergent positions. While thorough, these processes may be too slow for the tempo of modern operations [CITATION NEEDED].

No Decentralized Governance Framework. Perhaps most significantly, current coalition C2 approaches lack a framework for decentralized governance that could enable effective coordination even when centralized systems are degraded. There is no mechanism for encoding policies in a way that allows autonomous execution while maintaining human authority over critical decisions [CITATION NEEDED]. Section 2.10 examines this governance gap in the context of AI applications and identifies how BASTION addresses this limitation.

2.7 Frameworks and Doctrine

Military organizations have developed various frameworks and doctrinal approaches to address coordination challenges. Understanding these frameworks provides context for how BASTION's approach fits within existing military thinking.

DOTMLPF-P Model

The DOTMLPF-P model provides a framework for analyzing and developing military capabilities. The acronym stands for Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, Facilities, and Policy [CITATION NEEDED].

Framework Components. Each element of the DOTMLPF-P framework addresses a distinct aspect of capability development:

- **Doctrine:** The fundamental principles that guide military forces in support of objectives. Doctrine provides the common language and shared understanding that enable coordination.
- **Organization:** How forces are structured, including command relationships, staff organizations, and unit compositions. See Section 2.7.1 for detailed treatment of command relationships.
- **Training:** The preparation of individuals and units to perform their assigned tasks and missions.
- **Materiel:** The equipment, systems, and supplies needed to equip forces. This includes weapons, vehicles, communications systems, and logistics items.
- **Leadership and Education:** The development of leaders and the broader education of the force.
- **Personnel:** The human resources required, including skills, numbers, and career management.
- **Facilities:** The infrastructure needed to support forces, including bases, training areas, and maintenance facilities.
- **Policy:** The guidance and rules that govern how capabilities are employed and how forces operate.

Framework Purpose. When the military identifies a capability gap, the DOTMLPF-P framework ensures that all aspects of developing that capability are considered. A new capability may require not just new equipment (Materiel) but also new operating procedures (Doctrine), organizational structures (Organization), training programs (Training), leader development (Leadership and Education), personnel with specific skills (Personnel), infrastructure (Facilities), and supporting policies (Policy).

Coordination Improvements. Improvements to military coordination must be considered across all DOTMLPF-P dimensions. New technology alone is insufficient; it must be accompanied by doctrine that describes how to employ it, training that prepares personnel to use it effectively, and policies that govern its use. BASTION's approach addresses multiple DOTMLPF-P dimensions by providing not just technology but also a governance framework that can inform doctrine and policy.

Command Relationships and Force Structure

Effective coalition coordination requires precise understanding of how authority flows between organizations. Military doctrine defines specific command relationships that govern the degree of authority a commander exercises over assigned or attached forces [CITATION NEEDED].

Command Relationship Types

Combatant Command (COCOM). The highest level of command authority, COCOM provides full authority to organize and employ commands and forces, assign tasks, designate objectives, and give authoritative direction over all aspects of military operations. COCOM cannot be delegated or transferred and applies only to combatant commanders [CITATION NEEDED].

Operational Control (OPCON). OPCON grants authority to perform those functions of command involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction necessary to accomplish the mission. OPCON may be transferred between commanders as operational requirements dictate [CITATION NEEDED].

Tactical Control (TACON). TACON provides authority over assigned or attached forces limited to the detailed direction and control of movements or maneuvers within the operational area necessary to accomplish assigned missions or tasks. TACON represents a more limited form of authority than OPCON, typically used for specific tactical operations [CITATION NEEDED].

Administrative Control (ADCON). ADCON involves the direction or exercise of authority over subordinate organizations with respect to administration and support, including organization of units, control of resources and equipment, personnel management, and individual training. ADCON represents the administrative rather than operational chain of command [CITATION NEEDED].

Support Relationships

When forces are not placed under a commander’s direct authority but must provide assistance, support relationships define the priority and degree of that support [CITATION NEEDED].

Direct Support (DS). A mission requiring a force to support another specific force and authorizing it to answer directly to the supported force’s request for assistance. DS provides the highest priority of support.

General Support (GS). A mission requiring a force to support the force as a whole rather than any particular subdivision. GS assets are centrally controlled and allocated based on overall priorities.

General Support-Reinforcing (GSR). A mission requiring a force to provide support to the force as a whole and to reinforce another similar-type unit. GSR represents a combination of GS and reinforcing missions.

Reinforcing (R). A mission requiring a force to augment another force. The reinforcing unit remains under its parent command while prioritizing support to the reinforced unit.

Task Organization

The Task Organization, documented in Annex A of an Operations Order (OPORD), specifies exactly how forces are organized for a particular operation. It identifies which units are assigned, attached, or in support relationships, and to whom they report [CITATION NEEDED].

Mission-Tailored Structure. Task organization creates mission-tailored force packages by combining elements from different parent organizations. A task force might include infantry from one brigade, armor from another, and aviation from a third, all organized under a single task force commander for a specific mission.

Dynamic Reconfiguration. Unlike peacetime organizational structures, task organizations change frequently based on mission requirements. Forces may be attached for specific phases of an operation and then return to parent control. This dynamic reconfiguration enables commanders to mass combat power where needed while maintaining overall force structure.

Coalition Complexity. In coalition operations, task organization becomes more complex due to national caveats and restrictions. Forces contributed by partner nations may have limitations on how they can be employed, where they can operate, or what missions they can perform. The task organization must account for these restrictions while still creating an effective force structure.

Force Structure Data

Managing force structure requires tracking detailed information about participating units, personnel, and resources [CITATION NEEDED].

Unit Identification. Each military unit has a unique identification hierarchy: nation, service, major command, and subordinate unit designations. Standard unit identification codes (UICs) enable tracking across systems and time.

Equipment and Capabilities. Units bring specific equipment and capabilities to an operation. Force structure data tracks equipment types, quantities, and readiness states. Capability data indicates what missions a unit can perform based on its equipment, training, and current status.

Readiness Status. Military readiness uses standard indicators to communicate unit status. Equipment readiness typically uses FMC (Fully Mission Capable), PMC (Partially Mission Capable), and NMC (Not Mission Capable) categories with reason codes explaining degraded states. Personnel readiness tracks training currency, medical fitness, and specialty qualifications [CITATION NEEDED].

Resource Assignment Model. U.S. doctrine distinguishes between resources that are apportioned (identified for planning), allocated (made available for use), and assigned (placed under command). This progression enables planning to proceed before final resource commitments while maintaining visibility into expected versus actual resource availability [CITATION NEEDED].

Joint Planning Process

Joint Publication 5-0, Joint Planning, establishes the doctrine for how joint forces plan and execute operations [CITATION NEEDED].

Planning as Coordination Mechanism. The Joint Planning Process (JPP) serves as a primary coordination mechanism for military operations. Through iterative planning cycles, commanders and their staffs develop shared understanding of the operational environment, establish objectives, and develop courses of action. The process explicitly incorporates input from subordinate units, adjacent units, and coalition partners [CITATION NEEDED].

Where Automation Can Accelerate. While the JPP is comprehensive, it is also time-intensive. Many steps in the process involve gathering information, analyzing options, and coordinating among stakeholders. These activities present opportunities for AI-assisted acceleration without eliminating the human judgment that remains essential for military decision-making. Section 2.9 discusses the human-machine teaming concepts that inform how AI can augment human decision-making while maintaining appropriate authority positions. The methodology section will describe how BASTION’s AI agents can support various phases of the planning process while maintaining appropriate human authority.

Key Terms Summary

For reference, the following terms defined in this section are used throughout this paper:

Term	Definition
ADCON	Administrative Control - authority over administration and support functions
C2	Command and Control - the exercise of authority and direction over forces
COCOM	Combatant Command - highest level of command authority
DDIL	Disconnected, Degraded, Intermittent, Limited-bandwidth environments
DOTMLPF-P	Framework for capability development covering Doctrine, Organization, Training, Materiel, Leadership, Personnel, Facilities, Policy
DS/GS/GSR/R	Support relationships: Direct Support, General Support, General Support-Reinforcing, Reinforcing
FMC/PMC/NMC	Equipment readiness states: Fully Mission Capable, Partially Mission Capable, Not Mission Capable
FMN	Federated Mission Networking - NATO framework for coalition information sharing
JADC2	Joint All-Domain Command and Control - DOD concept for connecting sensors and shooters
JPP	Joint Planning Process - doctrinal approach to military planning
OPCON	Operational Control - authority to organize and employ forces for assigned missions
TACON	Tactical Control - limited authority for detailed direction of movements or maneuvers
Task Organization	Mission-tailored force structure documented in OPORD Annex A

The challenges described in this section create an environment where traditional approaches to military coordination are increasingly inadequate. The following section examines how artificial intelligence is currently applied in defense contexts and *introduces the human-machine teaming concepts that inform BASTION's approach to maintaining appropriate human authority over AI-augmented decision processes.*

2.8 AI Applications in Military Operations

Artificial intelligence has become increasingly central to military operations, offering capabilities that can enhance decision-making, improve situational awareness, and enable new operational concepts. Understanding current AI applications in defense provides essential context for BASTION's approach to AI-augmented governance. This section builds upon the military coordination challenges described in Sections 2.5-2.7, examining how AI can address those challenges while introducing its own governance requirements.

Current AI Uses

Military organizations worldwide are investing heavily in AI capabilities across multiple domains of application [CITATION NEEDED].

Sensor Fusion and Intelligence Analysis. One of the most mature applications of AI in defense involves processing and analyzing data from multiple sensors to create coherent intelligence products. Modern military operations generate enormous volumes of data from satellites, reconnaissance aircraft, unmanned systems, signals intelligence, and human sources. AI systems help analysts process this data, identify patterns, correlate information across sources, and generate actionable intelligence. Machine learning algorithms can detect objects of interest in imagery, identify patterns in signals, and flag anomalies for human review [CITATION NEEDED].

Decision Support Systems. AI-powered decision support systems assist military commanders and their staffs in understanding complex operational environments and evaluating courses of action. These systems can model potential outcomes, identify risks, and present options for human decision-makers. Decision support AI does not replace human judgment but rather augments it by processing more information than humans could review manually and presenting that information in actionable formats [CITATION NEEDED].

Autonomous and Semi-Autonomous Systems. Unmanned Aerial Vehicles (UAVs), Unmanned Ground Vehicles (UGVs), and other robotic systems increasingly incorporate AI for navigation, object recognition, and mission execution. These systems range from fully human-controlled (where AI assists with stability and navigation) to highly autonomous (where AI makes real-time decisions within programmed constraints). The degree of autonomy varies based on the mission, the operational environment, and policy constraints [CITATION NEEDED].

Resource Optimization and Logistics. AI applications in military logistics help optimize supply chains, predict maintenance requirements, and allocate resources efficiently. Machine learning models can forecast demand for supplies, identify optimal routing for logistics convoys, and predict when equipment will require maintenance before failures occur. These applications improve efficiency and readiness while reducing costs [CITATION NEEDED].

Cyber Operations. Both offensive and defensive cyber operations increasingly rely on AI. Defensive AI systems monitor networks for anomalies, detect intrusion attempts, and respond to threats faster than human operators could react. AI also supports offensive cyber operations by identifying vulnerabilities and optimizing attack vectors [CITATION NEEDED].

AI Governance Challenges

The integration of AI into military operations raises significant governance challenges that must be addressed to ensure responsible and effective use.

Trust and Reliability. Military commanders must trust the systems they rely on for critical decisions. AI systems, however, can produce unexpected outputs, fail in ways that are difficult to predict, and be vulnerable to adversarial manipulation. Building justified trust in AI systems requires extensive testing, validation, and operational experience [CITATION NEEDED].

Explainability Requirements. Military decision-makers need to understand why AI systems make specific recommendations or take particular actions. “Black box” AI systems that produce outputs without explainable reasoning are problematic in military contexts where commanders must justify their decisions and maintain accountability. The field of Explainable AI (XAI) aims to address this challenge, but practical solutions remain limited for complex AI systems [CITATION NEEDED].

Accountability for AI-Influenced Decisions. When AI systems contribute to military decisions, questions arise about accountability. If an AI-influenced decision leads to unintended consequences, determining responsibility requires understanding the role the AI played and whether human decision-makers appropriately evaluated its recommendations. Current military doctrine generally holds humans responsible for decisions, but the increasing speed and complexity of AI-augmented operations complicates this principle [CITATION NEEDED].

Ethical Considerations. The use of AI in military operations, particularly in lethal decision-making, raises profound ethical questions. International discussions continue regarding the appropriate role of AI in targeting decisions and the degree of human control required for lethal autonomous weapons systems. These debates inform policy constraints that shape how AI can be employed [CITATION NEEDED].

2.9 Human-Machine Teaming

The concept of human-machine teaming describes how humans and AI systems work together to accomplish tasks that neither could achieve as effectively alone. In military contexts, human-machine teaming aims to combine human judgment, creativity, and ethical reasoning with AI's speed, consistency, and data processing capabilities.

Authority Positions

A critical framework for understanding human-machine teaming involves the concept of human authority positions. These positions describe the relationship between human decision-makers and AI systems in terms of control and oversight. Three primary authority positions are commonly recognized [CITATION NEEDED].

Human-in-the-Loop (HITL). In the human-in-the-loop model, a human must approve each AI action before it executes. The AI system may analyze information, identify options, and make recommendations, but no action occurs without explicit human authorization. This approach provides maximum human control and is appropriate for high-consequence decisions where errors could have severe or irreversible effects.

The human-in-the-loop approach ensures that human judgment is applied to every decision point. It allows commanders to maintain direct control and intervene based on factors that the AI may not fully understand or weigh appropriately. However, this approach limits operational tempo because the speed of action is constrained by human decision-making speed. For time-critical situations, the delays inherent in human-in-the-loop operation may be unacceptable.

Human-on-the-Loop (HOTL). In the human-on-the-loop model, AI systems act autonomously within defined parameters while humans monitor operations and retain the ability to intervene. The human observer can override AI actions, halt operations, or adjust parameters as needed. This approach balances control with speed, allowing faster action while maintaining human oversight.

Human-on-the-loop operation enables AI systems to respond to situations at machine speed while ensuring that humans can intervene when AI behavior is inappropriate or when circumstances change. The effectiveness of this approach depends on humans maintaining sufficient situational awareness to recognize when intervention is needed and having the ability to intervene in time. This model is appropriate for moderate-consequence decisions where speed is important but errors can be corrected.

Human-out-of-the-Loop (HOOTL). In the human-out-of-the-loop model, AI systems operate fully autonomously within policy constraints, without real-time human involvement in individual decisions. Humans establish the policies, constraints, and objectives that guide AI behavior, but do not supervise or approve individual actions.

This approach enables maximum speed and is appropriate only for well-constrained, low-risk tasks where the consequences of errors are acceptable and the operational environment is well-understood. Human-out-of-the-loop operation is controversial for military applications, particularly for lethal operations, and most nations maintain policies requiring some degree of human control over weapons employment [CITATION NEEDED].

Trust and Autonomy Calibration

Effective human-machine teaming requires calibrating the appropriate level of autonomy for each situation. This calibration considers multiple factors.

Trust Earned Through Performance. Trust in AI systems should be based on demonstrated performance in relevant conditions. Systems that have proven reliable in testing and operational experience warrant greater trust than novel or untested systems. Trust should be calibrated to actual performance rather than assumed capabilities [CITATION NEEDED].

Graduated Autonomy. The appropriate level of autonomy may vary based on mission phase, operational conditions, and specific tasks. A single system might operate with different authority positions at different times, with more autonomy for routine tasks and greater human involvement for critical decisions. This graduated approach allows flexibility while maintaining appropriate control [CITATION NEEDED]. This concept directly addresses the operational gap described in Section 2.6, where the speed of tactical decisions may require different authority positions than strategic resource allocation.

Consequence Assessment. Higher-consequence decisions warrant more human involvement. The potential impact of errors or unexpected outcomes should inform the degree of human control required. Lethal decisions, in particular, typically require human-in-the-loop or at minimum human-on-the-loop operation [CITATION NEEDED].

Mission-Specific and Agent-Specific Levels. Different AI agents within a system may warrant different levels of autonomy based on their capabilities, track record, and the sensitivity of their tasks. Similarly, different missions may require adjusting autonomy levels based on the operational environment and acceptable risk.

2.10 Gap: AI Governance Frameworks

Despite significant progress in AI applications for defense, a critical gap remains in frameworks for governing AI-augmented multi-stakeholder decision-making.

Missing Integration

No Decentralized AI Governance. Current approaches to AI in defense focus primarily on single-organization implementations. Governance frameworks address how one nation or organization manages its AI systems, but do not provide mechanisms for coordinating AI-augmented decisions across organizational boundaries. Coalition operations involving multiple nations with their own AI systems lack frameworks for integrating these systems under coherent governance [CITATION NEEDED].

No Multi-Stakeholder AI Authority Framework. When multiple stakeholders (nations, organizations, or individuals) must coordinate AI-influenced decisions, existing frameworks do not define how authority should be allocated, how disagreements should be resolved, or how accountability should be assigned. Each stakeholder maintains its own AI governance approach, but no overarching framework coordinates these approaches [CITATION NEEDED].

Coalition AI Coordination Not Addressed. The challenges of coalition coordination described in Section 2.6 are compounded when AI systems are involved. Information sharing caveats, release authority requirements, and different national policies all affect how AI systems can be integrated across coalition boundaries. Current approaches do not address these AI-specific coalition challenges [CITATION NEEDED].

Policy Compliance Not Automated. AI systems must operate within policy constraints, but current approaches rely on programming these constraints directly into individual systems. There is no mechanism for encoding policies in a way that allows verification of compliance, enables automatic enforcement across systems, or provides transparency to stakeholders about what constraints are in effect [CITATION NEEDED].

Research Opportunity

The identified gaps create a research opportunity at the intersection of decentralized governance and AI agent systems.

DAO Governance for AI Coordination. Decentralized Autonomous Organizations provide governance mechanisms that could address multi-stakeholder AI coordination. Smart contracts can encode policies that constrain AI behavior, voting mechanisms can resolve disagreements, and blockchain provides transparency and auditability for AI-influenced decisions [CITATION NEEDED].

Smart Contract Enforcement of Autonomy Constraints. Rather than programming autonomy constraints directly into AI systems, these constraints can be encoded in smart contracts that govern AI agent behavior. This approach enables transparent, verifiable, and uniform enforcement of policies across multiple AI systems and organizational boundaries.

BASTION's Novel Contribution. BASTION addresses the identified gap by integrating DAO governance with AI agent systems. The platform provides mechanisms for multi-stakeholder coordination, encodes policies in smart contracts, and maintains appropriate human authority positions for different decision types. This integration enables coalition coordination with AI augmentation while maintaining the transparency, accountability, and human control that military operations require.

The background sections have established the technological, organizational, and doctrinal context for BASTION. The following methodology section describes how BASTION integrates these elements into a coherent architecture that addresses the identified gaps.

Gap Analysis Summary

The following table summarizes the gaps identified in existing approaches and how BASTION addresses each:

Approach	**Strength**	**Gap**	**BASTION Solution**
JADC2	Sensor connectivity across domains	No decentralized governance framework	DAO-based multi-stakeholder coordination
NATO FMN	Federation standards for information sharing	Centralized trust model	Blockchain immutability and transparency
Military AI	Decision support and autonomous systems	No policy compliance framework for coalitions	Smart contract encoded constraints
Commercial DAOs	Decentralized governance mechanisms	Not designed for C2 tempo or lethal decisions	AI agent augmentation with human authority positions

See Section 3 (Methodology) for how BASTION implements these solutions. See Section 4 (Results) for demonstration of AI governance in action.

2.4 Edge Computing and Military Robotics

The integration of autonomous robotic systems into military operations has accelerated significantly over the past decade, driven by advances in edge computing hardware, neural inference acceleration, and wireless mesh networking. BASTION’s physical demonstration layer engages this domain directly, deploying commercial edge computing platforms and mobile robotic vehicles within a governance-bound autonomous system. This section surveys the academic and doctrinal foundations underpinning that integration.

2.4.1 Edge Computing in Military Contexts

Edge computing relocates computational workloads from centralized data centers to nodes proximate to sensors, actuators, and operators.[15] In military contexts, this shift is motivated by the fragility of long-haul communications in contested environments and the latency requirements of autonomous systems that must make sub-second decisions. The NVIDIA Jetson platform — specifically the Jetson Orin Nano used in BASTION’s demonstration — provides up to 40 TOPS (trillion operations per second) of AI inference compute in a form factor appropriate for dismounted or vehicle-mounted edge deployment.[16] At this compute density, deep learning inference tasks such as object detection, scene classification, and pose estimation can execute locally without round-trip latency to cloud or datacenter infrastructure.

Military applications of edge computing extend beyond robotics to include forward-deployed sensor fusion, encrypted radio relay with local AI processing, and autonomous target recognition at the tactical edge. The U.S. Army’s Project Convergence and the Joint All-Domain Command and Control (JADC2) initiative both emphasize moving AI inference capabilities forward to reduce the sensor-to-decision cycle time.[17] BASTION’s architecture operationalizes this principle by running the robot vision pipeline entirely on the Jetson device, forwarding only high-level intent signals (threat type, position estimate, confidence score) to the command-and-control layer rather than raw video streams.

2.4.2 Autonomous Ground Vehicles in Military Applications

Unmanned Ground Vehicles (UGVs) have transitioned from experimental platforms to accepted elements of joint doctrine. The Department of Defense’s Unmanned Systems Integrated Roadmap identifies UGVs as essential contributors to reconnaissance, logistics, and counter-IED operations.[18] Contemporary doctrine addresses human-machine teaming in the ground domain, distinguishing between teleoperated systems (human controlled at all times), supervised autonomous systems (human-on-the-loop), and fully autonomous systems (human-out-of-the-loop for bounded task execution).[19]

BASTION’s robotic demonstration uses Sphero RVR+ platforms — commercially available differential-drive mobile robots with open SDK access — to represent tactical autonomous platforms. While not military-grade systems, this choice reflects a principled decision to validate the governance and coordination architecture using available hardware before scaling to operational platforms.

The architecture's DAO-governed mission assignment, intent translation, and authority position enforcement are hardware-agnostic and transfer directly to military UGV platforms.[20]

Human-machine teaming in ground maneuver contexts requires governance frameworks that specify: which decisions humans must authorize before robotic action, which actions robots may execute autonomously within defined parameters, and how humans can override or abort autonomous execution at any point. These governance requirements map precisely to BASTION's three authority positions (human-in-the-loop, on-the-loop, out-of-the-loop) described in Section 3.4.

2.4.3 DDIL Environments and Edge AI Resilience

Denied, Degraded, Intermittent, and Limited (DDIL) communications environments represent the baseline assumption for contested military operations.[21] Electronic warfare, terrain masking, adversary jamming, and network saturation all degrade or eliminate connectivity between robotic platforms and command elements. Effective autonomous systems must therefore be capable of operating under disconnected conditions while remaining bound by pre-authorized policy constraints.

Edge AI resilience in DDIL environments requires local inference capability (addressed by the Jetson's onboard neural engine), local mission state persistence (so robots retain their assignment when connectivity is lost), and conservative fallback behaviors that engage when communication is interrupted. BASTION's robot bridge architecture implements this pattern: the Python robot agent maintains mission state locally, executes within pre-authorized parameters when the central API is unreachable, and attempts reconnection on a configurable interval. This design reflects the broader principle that governance constraints must be enforced at the edge rather than relying on persistent uplink to a central authority.[22]

Research on military AI resilience emphasizes that autonomous systems should fail toward safety — defaulting to halt, hold position, or return-to-base behaviors — when operating outside the parameters of their mission authorization.[23] BASTION enforces this through the mission intent translation layer, which explicitly bounds robot behavior to the authorized mission type before command execution begins. Strike missions require explicit human approval regardless of connectivity status; the approval bit must be set before mission execution commences.

2.4.4 Swarm Robotics and Doctrinal Formation Concepts

Swarm robotics applies principles from biological collective behavior (flocking, stigmergy, distributed task allocation) to multi-robot coordination.[24] Military swarm applications include coordinated reconnaissance, distributed target acquisition, and mass saturation of adversary defensive systems. Academic literature distinguishes between reactive swarms (each robot responds to local sensor input without communication) and communicating swarms (robots share state over a mesh network and coordinate through explicit protocols).[25]

BASTION's swarm implementation belongs to the communicating swarm category. A designated swarm leader robot receives mission assignments from the DAO and translates them into formation commands distributed over a UDP peer mesh. Follower robots receive formation type, reference position, and interval parameters, then compute their individual positions relative to the leader using geometric formation models. This approach mirrors military doctrine's concept of movement formations — standardized spatial arrangements that provide specific tactical advantages.

Joint doctrine recognizes six primary ground movement formations: column (depth for road movement), line (maximum firepower to the front), wedge (all-around security with firepower focused forward), echelon left/right (oblique movement with protection on one flank), vee (observation and firepower to front and flanks), and staggered column (depth with mutual flank protection).[26] BASTION implements all six formations as first-class swarm behaviors, allowing the DAO to select the doctrinally appropriate formation based on mission type, threat axis, and terrain conditions expressed in the planning documents.

2.4.5 Self-Registration and Mesh Networking for Autonomous Platforms

Autonomous platforms that enter a network must be discoverable, authenticated, and authorized before they can receive mission assignments or contribute sensor data. This onboarding challenge mirrors the broader military resource management problem: how does a command element know what assets are available, where they are, and whether they are authorized for a given mission?[27]

BASTION addresses robot onboarding through mDNS-based self-registration. When a robot agent initializes, it broadcasts a service announcement on the local network containing its capability manifest (sensors, mobility type, communication range, current readiness). The robot bridge service discovers this announcement, validates the robot against the resource registry, and presents it to the DAO for authorization. This self-registration pattern extends the resource DID architecture (described in Section 3.10) to physical platforms, treating robots as blockchain-anchored resource entities with verifiable identity and capability claims.

Mesh networking enables robots to relay commands and sensor data through peer nodes when direct uplink to the control system is unavailable. BASTION's UDP peer mesh provides direct robot-to-robot communication for formation coordination, independent of the central API. This architecture ensures that formation integrity is maintained during periods of partial connectivity — the swarm can continue coordinated movement even when individual robots temporarily lose contact with the command element.[28]

2.4.6 Neural Vision Pipelines at the Tactical Edge

Computer vision systems based on convolutional neural networks (CNNs) have achieved human-level performance on object detection and classification benchmarks, enabling autonomous threat recognition without continuous human supervision.[29] NVIDIA's detectNet architecture, used in BASTION's robot vision pipeline, runs inference on a live camera feed to detect and classify objects against a trained label set, returning bounding box coordinates and confidence scores in real time. The Jetson Orin Nano's dedicated Deep Learning Accelerator (DLA) executes these inference workloads in parallel with the main CPU, enabling concurrent vision processing and communication without performance degradation.

Military applications of neural vision at the edge span perimeter security, route clearance, counter-drone, and dismounted operations.[30] The challenge of deploying neural vision in operational contexts involves not only model accuracy but also adversarial robustness — the susceptibility of deep learning models to adversarial perturbations that cause misclassification. BASTION's implementation addresses this by treating vision outputs as intelligence estimates rather than authoritative decisions: confidence scores from the vision pipeline feed into the knowledge graph as entity attributes subject to source reliability assessment (the same NATO Admiralty Code framework applied to OSINT). This framing correctly places the human authority question at the mission authorization layer rather than at the sensor layer.

2.5 Knowledge Graphs and Semantic Intelligence

The application of knowledge graph technologies to intelligence analysis has emerged as a significant research and operational priority as defense organizations confront the challenge of integrating heterogeneous information sources into coherent, queryable semantic models. BASTION's brain visualization and entity resolution capabilities are grounded in this research tradition. This section surveys the foundational concepts and relevant literature.

2.5.1 Knowledge Graphs for Intelligence Analysis and Sense-Making

A knowledge graph is a structured representation of entities (persons, places, organizations, capabilities, events) and the relationships between them, encoded as a graph of nodes and edges with typed predicates.[31] In intelligence analysis, knowledge graphs serve as a substrate for sense-making — the process by which analysts construct coherent narratives from fragmentary, often contradictory, evidence. Where traditional databases store facts in tabular structures optimized for known queries, knowledge graphs allow analysts to traverse relationship chains, identify indirect connections, and discover structural patterns that would be invisible in flat representations.[32]

Military intelligence agencies have invested in knowledge graph infrastructure as a component of their all-source fusion capabilities. Projects such as IARPA's CAUSE (Causal, Counterfactual, and Contrastive Explanations) and DARPA's Big Mechanism program have explored graph-based approaches to causal reasoning over intelligence text.[33] BASTION's RAFT (Retrieval-Augmented Fusion with Typing) graph extends this approach to the operational planning domain, capturing entities extracted from planning documents and intelligence feeds as a live graph that evolves with the operational situation.

2.5.2 JSON-LD and Semantic Web Standards in Defense Contexts

JSON-LD (JavaScript Object Notation for Linked Data) is a W3C standard that enables structured data to carry semantic context through the use of linked namespaces and type definitions.[34] JSON-LD documents are simultaneously valid JSON (parseable by any JSON library) and valid RDF (Resource Description Framework) serializations, enabling integration with semantic reasoning tools. This dual nature makes JSON-LD attractive for defense information exchange: it is interoperable with web infrastructure and existing data pipelines while supporting the ontological richness required for cross-domain intelligence fusion.

Defense information architectures have historically relied on XML-based standards (UCORE, NIEM) for structured data exchange. The shift toward JSON-LD reflects broader trends in web-oriented architecture adoption within the defense enterprise and offers advantages in tooling maturity, developer accessibility, and integration with modern AI pipelines that process JSON natively.[35] BASTION's knowledge graph uses JSON-LD for entity serialization, enabling direct export to RDF triple stores and compatibility with SPARQL-based query tools used in intelligence analysis platforms.

2.5.3 Graph Visualization: Force-Directed Layouts and Neural Canvas Approaches

Visualizing large knowledge graphs presents a fundamental challenge: graph drawing algorithms must balance aesthetic criteria (minimizing edge crossings, distributing nodes evenly) with semantic criteria (placing related entities near each other, highlighting high-importance nodes) within interactive performance constraints.[36] Force-directed layout algorithms, pioneered by Fruchterman-Reingold and refined through numerous variants, simulate physical systems where nodes repel each other and edges act as springs, converging to layouts that reveal graph structure without requiring explicit hierarchical or temporal arrangement.[37]

BASTION's brain visualization extends force-directed layout with neural metaphor design: node importance (determined by connectivity degree and evidence weight) controls node size and glow intensity, while edge weight (derived from co-occurrence frequency and confidence scores) controls edge thickness and opacity. The result is a visualization that resembles a neural network or brain scan, making the semantic density of intelligence relationships visually interpretable to analysts without requiring graph theory expertise. This design choice reflects research on cognitive fit between visualization metaphor and analyst mental models — analysts who think about intelligence as a network of connections benefit from a visualization that renders those connections as spatial, visceral relationships.[38]

2.5.4 Entity Resolution and Confidence Scoring in Multi-Source Intelligence Fusion

Entity resolution — determining whether two references to a named entity (a person, organization, location, or military unit) in different documents refer to the same real-world entity — is a fundamental challenge in intelligence fusion.[39] Named entity references are noisy: transliterations vary, abbreviations are context-dependent, and adversaries deliberately introduce ambiguity. Entity resolution algorithms use string similarity, context embedding similarity, and structural graph evidence to make probabilistic merge decisions, producing entity clusters where each cluster represents a single real-world entity with multiple textual variants.

Confidence scoring in intelligence products follows NATO's Admiralty Code, which rates source reliability (A through F) independently of information quality (1 through 6).[40] BASTION applies this framework at the entity level: each entity in the knowledge graph carries source reliability ratings derived from the provenance of the documents that introduced it. Conflicting claims from different sources produce competing entity attributes with differential confidence, allowing analysts to inspect the evidentiary basis for each claim rather than receiving a single merged verdict. This approach reflects the intelligence community's emphasis on maintaining analytical uncertainty rather than prematurely collapsing ambiguity into false certainty.[41]

2.5.5 Ontology Alignment and Interoperability Standards

Defense knowledge graphs must interoperate with existing ontological frameworks to enable cross-system intelligence sharing. The Basic Formal Ontology (BFO) provides a foundational upper ontology that domain-specific defense ontologies extend.[42] The Common Core Ontologies (CCO), developed under the Information Warfare Cross-Functional Team, provide mid-level ontologies for military domains including geospatial, temporal, and organizational concepts. The DoDAF/DM2 (DoD Architecture Framework Data Model) provides data architecture standards for system interoperability documentation.

BASTION's knowledge graph uses a pragmatic approach: entities are typed using a vocabulary drawn from military planning doctrine (units, capabilities, objectives, lines of effort, tasks, constraints) rather than formalizing against BFO or CCO. This design decision prioritizes operational utility over formal ontological rigor — analysts and planners can contribute to and query the graph using familiar doctrinal terminology without ontology expertise. Future work (noted in Section 5) addresses alignment of BASTION's entity vocabulary with CCO for interoperability with other defense knowledge graph systems.[43]

---

3. Methodology

This section describes BASTION's architecture and the design principles that guide its implementation. The architectural decisions presented here directly address the gaps identified in Section 2: the lack of decentralized governance for military command and control, the absence of frameworks for multi-stakeholder AI coordination, and the need for policy-compliant autonomous execution with appropriate human authority positions. Each design choice is justified in terms of how it enables effective coalition coordination while maintaining the transparency, accountability, and human control that military operations require.

3.1 Design Principles

BASTION's architecture rests on four foundational principles derived from the challenges identified in the background sections. These principles inform every architectural decision and ensure that technical capabilities align with operational requirements.

Decentralization for Resilience

Modern military operations face adversaries who actively target command and control infrastructure. As described in Section 2.5, centralized C2 systems present attractive targets for electronic warfare, cyber attacks, and kinetic strikes. When a coalition's coordination depends on a single command post or network node, disrupting that node can paralyze operations across multiple nations' forces.

BASTION addresses this vulnerability through architectural decentralization. Rather than concentrating coordination logic in any single location, the system distributes governance across a network of nodes operated by coalition partners. Each partner maintains the capability to participate in coordination even if communication with other partners is degraded. The blockchain substrate ensures that when connectivity is restored, the system can reconcile state across previously disconnected nodes and continue operations from a consistent foundation.

This decentralized architecture directly addresses the DDIL (Disconnected, Degraded, Intermittent, Limited-bandwidth) challenges identified in Section 2.5. Forces operating in contested electromagnetic environments can continue to execute pre-approved missions using locally cached governance state. When they reconnect, their actions are recorded on the shared ledger, maintaining the audit trail and enabling reconciliation with broader coalition activities. No single point of failure exists that an adversary could target to disable coalition coordination.

Transparency for Trust

Coalition operations require trust among partners who may have competing national interests, different operational doctrines, and varying levels of information sharing authority. As described in Section 2.6, building this trust through traditional means is time-consuming and may never fully overcome suspicions that partner nations are not fully sharing information or are pursuing hidden agendas.

BASTION provides transparency through blockchain's immutable audit trail. Every proposal submitted to the governance system, every vote cast, every resource allocation approved, and every mission authorized is permanently recorded with cryptographic proof of its origin. Coalition partners can verify for themselves that the system is operating according to agreed-upon rules rather than relying on assurances from other partners.

This transparency operates across organizational boundaries. When multiple nations collaborate on a campaign, each nation's authorized representatives can observe how their partner nations are voting, what resources they are committing, and whether approved actions are being executed. This visibility enables accountability without requiring any nation to cede control to a centralized authority. The shared ledger serves as a neutral, verifiable record that all parties can trust.

The audit trail also supports after-action review and learning. When coalition operations produce unexpected outcomes, the complete record of decisions enables analysis of what went wrong and how procedures can be improved. Attribution in disputes is straightforward: the blockchain records precisely who approved what, when, and under what circumstances.

AI Augmentation for Speed

Section 2.7 identified the operational level as where the gap between strategic intent and tactical execution is most acute. Joint All-Domain Command and Control (JADC2) envisions accelerating decision cycles through sensor-to-shooter connectivity, but connectivity alone does not address the cognitive burden on human decision-makers who must process vast amounts of information and make timely decisions.

BASTION integrates AI agents at each level of the governance hierarchy to augment human decision-making. These agents analyze proposals, assess risks, and present recommendations in formats optimized for human comprehension. They can process intelligence from multiple sources, identify patterns that might escape human attention, and flag issues that require human judgment. By handling routine analysis and presentation, AI agents free human decision-makers to focus on the judgments that require human wisdom, creativity, and ethical reasoning.

Critically, AI augmentation in BASTION does not replace human authority. As described in Section 2.9, the appropriate relationship between humans and AI depends on the consequences of potential errors. BASTION implements graduated autonomy: AI agents may execute routine, low-consequence actions autonomously, while higher-consequence decisions require human approval at appropriate authority levels. The speed benefits of AI augmentation are realized within a framework that preserves human control over decisions with significant implications.

This approach addresses the tempo requirements of modern operations while maintaining the accountability that military command requires. Commanders retain authority over critical decisions but are not overwhelmed by routine matters that AI agents can handle within established policy constraints.

Policy Compliance by Design

Section 2.6 identified national caveats and information sharing restrictions as significant constraints on coalition coordination. Each nation participates in coalitions subject to its own national policies regarding what information can be shared, what actions can be authorized, and what resources can be committed. These policies are not merely preferences; they are legal requirements that coalition systems must respect.

Traditional approaches to policy compliance rely on human review at each decision point to ensure that proposed actions comply with applicable policies. This approach introduces delays and places the burden of compliance on individuals who may not fully understand all applicable constraints. Errors in compliance can have serious diplomatic and legal consequences.

BASTION addresses this challenge by encoding policies directly in smart contracts that govern the system’s operation. National caveats, information classification requirements, release authority rules, and operational constraints are expressed as executable code that the governance system enforces automatically.

Proposals that violate encoded policies cannot be approved; resources subject to restrictions cannot be allocated outside their permitted uses; information can only flow to recipients authorized to receive it.

This approach provides several advantages. First, compliance is consistent: the same rules apply every time, without variation based on individual interpretation or oversight. Second, compliance is verifiable: coalition partners can examine the smart contract code to confirm that their policies are correctly encoded. Third, compliance is auditable: when questions arise about whether actions complied with policy, the blockchain record shows exactly what rules were in effect and how they were applied.

Smart contract policy enforcement is particularly valuable for bounded autonomy. AI agents can be permitted to act within the bounds established by policy without requiring case-by-case human approval, because the policies themselves constrain what actions are possible. This enables speed where policy permits while ensuring that AI agents cannot exceed their authorized scope.

3.2 Architecture Overview

BASTION implements a layered architecture that maps to the military levels of warfare described in Section 2.5. This mapping ensures that the governance structure aligns with existing military organizational concepts while enabling novel coordination capabilities through blockchain and AI integration.

Three-Tier DAO Structure

At the core of BASTION's architecture are three interconnected Decentralized Autonomous Organizations, each corresponding to a level of warfare. This structure ensures that governance mechanisms match the decision-making requirements at each level while enabling coordination across levels through well-defined interfaces.

Strategic-Level DAO. The Strategic DAO governs long-term objectives, resource allocation, and coalition membership. Decisions at this level correspond to the strategic level of warfare: determining which military objectives will achieve political ends, allocating resources among theaters, and establishing overall priorities. Coalition membership is managed through the Strategic DAO, with voting weights reflecting each nation's commitment and contribution to the coalition effort.

The Strategic DAO operates on longer time horizons, with proposals subject to extended deliberation periods that allow for thorough review and consensus-building among coalition partners. Supermajority thresholds ensure that strategic decisions reflect broad agreement rather than the preferences of a narrow majority. The immutable record of strategic decisions provides the foundation against which lower-level actions are evaluated for alignment with coalition intent.

Operational-Level DAO. The Operational DAO bridges strategic intent and tactical execution, coordinating campaigns and major operations across multiple tactical units. This level manages the AI agent orchestration that enables rapid analysis and recommendation generation, ensuring that agents operate within strategically approved parameters while supporting tactical decision-making.

Resource-to-objective mapping occurs at the operational level, translating strategic resource allocations into specific commitments to campaigns and operations. The Operational DAO monitors the state of resources and objectives, automatically generating proposals to the Strategic DAO when resource states require attention or when tactical results suggest strategic reassessment. This automated proposal generation accelerates the feedback loop from tactical execution to strategic planning.

AI agents at the operational level fuse intelligence from multiple sources, assess progress toward objectives, and identify risks or opportunities that may not be apparent at either the strategic or tactical level. Human operational commanders retain authority over significant decisions while benefiting from AI-augmented situational awareness.

Tactical-Level DAO. The Tactical DAO governs mission-specific decisions, asset coordination, and target identification within policy constraints. Decisions at this level must often occur in real-time or near-real-time, requiring different governance mechanisms than higher levels.

Tactical proposals may be approved through streamlined processes that enable rapid response while maintaining accountability. Pre-approved mission types can be executed by AI agents within the bounds established by higher-level policy, with human authority required only for exceptions or high-consequence actions. The Tactical DAO records all actions for later review, ensuring that even rapid tactical execution leaves an auditable trail.

The interaction between human decision-makers and AI agents at the tactical level is calibrated to mission requirements. Routine coordination actions may proceed autonomously, while actions with significant risk or policy implications require human approval. The authority position for each decision type is explicitly defined in smart contract code, ensuring consistent application across the coalition.

Inter-DAO Communication

The three DAOs are not isolated; they communicate through well-defined interfaces that enable information flow in both directions. Strategic guidance flows downward as policy constraints and resource allocations that bound lower-level decision-making. Tactical and operational information flows upward as status reports and automatic proposals that inform higher-level planning.

Automatic proposal generation enables tactical and operational experience to influence strategic decisions without requiring manual escalation. When tactical resource expenditure reaches defined thresholds, the system automatically generates proposals to the Operational or Strategic DAO requesting replenishment or reallocation. When tactical results suggest that strategic objectives may not be achievable through current approaches, the system can flag the situation for strategic review.

Resource state monitoring provides real-time visibility into coalition capabilities across all levels. Strategic planners can observe the aggregate state of resources committed to campaigns; operational commanders can track resource availability for upcoming operations; tactical units can verify resource status before commencing missions. This shared visibility reduces the information gaps that impede effective coordination in traditional coalition operations.

Figure 1 illustrates the three-tier architecture, showing the relationships between DAOs, the AI agent layer, and the blockchain infrastructure that secures all transactions. The diagram indicates human authority positions at each level and the decision flow patterns that connect strategic intent to tactical execution.

Mission Context Establishment

Before operational planning can begin, BASTION requires establishment of a mission context that defines the participants, command relationships, available resources, and operational boundaries. This setup phase translates the abstract governance architecture into a concrete operational configuration.

Workspace Creation. Each mission operates within an isolated workspace containing its own participant roster, command hierarchy, resource inventory, and geographic boundaries. Workspace isolation ensures that coalition members participating in multiple missions maintain appropriate separation of concerns, while cross-references enable coordination when missions overlap. Any authenticated organization member can initiate workspace creation, democratizing mission establishment while maintaining accountability through the blockchain audit trail.

Participant Onboarding. Mission participants join through two mechanisms: direct invitation from the organization roster for known members, and invitation links or codes for external participants. Both methods ultimately result in cryptographic identity verification through the NEAR Protocol identity layer. Upon joining, participants receive role assignments that govern their authorities within the mission context.

Command Relationship Configuration. BASTION implements the full spectrum of military command relationships described in Section 2.7.1: OPCON, TACON, ADCON, COCOM, and support relationships (DS, GS, GSR, R). Commanders configure these relationships through a visual editor that presents both organizational chart and matrix views. The system validates configurations, warning of circular command relationships or orphaned units while permitting overrides for non-standard situations. Critically, command relationships drive permission cascades: a commander automatically gains visibility into subordinate resources and activities based on the configured relationship type.

Mission Lifecycle States. Missions progress through defined lifecycle states: Planning, Active, Complete, and Archived. These states govern what actions are permitted and what visibility coalition members have. Planning state enables configuration changes; Active state locks critical settings while enabling operational execution; Complete state preserves the mission record while preventing further actions; Archived state removes the mission from active views while maintaining the immutable blockchain record for historical reference and accountability.

Resource Management Architecture

Effective coalition coordination requires precise tracking of resources from initial contribution through operational employment to consumption and replenishment. BASTION’s resource management architecture addresses the full resource lifecycle within DAO governance.

Resource Catalog. The resource catalog maintains a multi-tier taxonomy aligned with military logistics standards. Pre-built equipment databases provide common starting points for standard military equipment (vehicles, weapons systems, communications equipment), while organizations can extend the catalog with custom resource types. Each resource entry captures technical specifications, capabilities, certifications, and maintenance history, enabling AI agents to match resources to mission requirements automatically.

Availability Tracking. Resource readiness uses the standard FMC/PMC/NMC (Fully Mission Capable, Partially Mission Capable, Not Mission Capable) framework with reason codes that explain degraded states. This granular tracking enables planners to distinguish between temporarily degraded resources that may become available and resources requiring significant repair or replacement.

Assignment Model. BASTION implements the doctrinal resource assignment progression: Apportioned (identified for planning), Allocated (made available for use), and Assigned (placed under command). This progression enables planning to proceed before final resource commitments while maintaining visibility into expected versus actual resource availability. Smart contracts enforce transitions between assignment states, ensuring that resources cannot be double-committed and that only authorized commanders can effect assignment changes.

Location Tracking. Resources track their physical or logical locations through multiple data sources: device GPS feeds for mobile assets, last known positions from situation reports, and sensor fusion for best estimates when primary data is unavailable. Pipeline tracking enables visibility into resources in procurement or transit, with expected availability dates informing future planning.

Consumables and Supply Points. Expendable resources (ammunition, fuel, supplies) are tracked by supply point rather than individual item. Each supply point maintains quantity on hand, consumption rates, and replenishment status. When consumption crosses defined thresholds, the system automatically generates replenishment proposals through the inter-DAO communication mechanisms described above.

Personnel Management. Personnel form a distinct resource type with their own tracking requirements: qualifications, training currency, medical fitness, and duty status. Personnel are assignable to units and linkable to equipment resources (e.g., crew assignments). Readiness definitions are configurable per mission context, enabling commanders to specify what qualifications and currency levels constitute “ready” for their specific operational requirements.

3.3 Key Design Decisions

BASTION's architecture required numerous design decisions, each with alternatives that were considered and evaluated. This section documents the major decisions, the alternatives considered, and the rationale for the choices made. These decisions establish patterns that future development must maintain and provide context for understanding why the system operates as it does.

Decision	Alternatives Considered	Rationale
NEAR Protocol for blockchain	Ethereum, Solana, private blockchain	Developer-friendly contracts, high performance, native DID support
DAO-based governance	Traditional RBAC, centralized authority	Decentralized trust compatible with coalition structure
AI agent augmentation	Full automation, human-only processes	Balances operational speed with accountability
Smart contract policy encoding	Runtime policy evaluation	Immutable enforcement with complete audit trail
Three-tier architecture	Flat DAO, traditional hierarchical C2	Maps directly to military levels of warfare

NEAR Protocol Selection

The choice of blockchain platform is fundamental to BASTION's capabilities. Three primary alternatives were considered: Ethereum, the most established smart contract platform; Solana, known for high transaction throughput; and private blockchain implementations that would restrict network participation to coalition members.

NEAR Protocol was selected for several conceptual advantages that align with BASTION's requirements. First, NEAR's sharded architecture provides the scalability necessary for military operations that may involve thousands of transactions representing tactical actions, intelligence updates, and resource state changes. The protocol can process transactions with finality in approximately two seconds, enabling near-real-time coordination without sacrificing the security guarantees that blockchain provides [CITATION NEEDED: NEAR Protocol documentation].

Second, NEAR provides native support for decentralized identifiers (DIDs) and account abstraction, enabling the identity management capabilities essential for coalition operations. Coalition partners can maintain cryptographically verified identities without relying on centralized identity providers, addressing the multi-stakeholder coordination requirements identified in Section 2.6.

Third, NEAR's developer-friendly smart contract environment, using Rust and JavaScript/TypeScript, reduces the barriers to developing and maintaining governance contracts. This accessibility is important for military organizations that must maintain and audit critical systems over extended periods.

The decision to use a public blockchain rather than a private blockchain reflects a deliberate tradeoff. Private blockchains offer more control but sacrifice the resilience benefits of a globally distributed network. BASTION addresses classification and access control through encryption and access control layers rather than network isolation, maintaining the resilience of the public blockchain while protecting sensitive information.

AI Agent Architecture

BASTION's AI agents follow a single-responsibility design pattern, where each agent performs one well-defined function rather than attempting to be a general-purpose assistant. This architecture was chosen over two alternatives: a monolithic AI system that handles all decision support functions, and a fully automated approach that removes humans from routine decisions entirely.

The single-responsibility pattern provides several advantages for military applications. First, it enables graduated trust: each agent can be evaluated, tested, and trusted independently based on its specific function. An agent that has proven reliable for logistics coordination can be granted greater autonomy in that domain without implying equivalent trust for intelligence analysis.

Second, single-responsibility agents are more explainable. When an agent makes a recommendation, understanding why requires understanding only that agent's limited scope rather than tracing reasoning through a complex multi-function system. This explainability supports the accountability requirements of military command.

Third, the pattern supports defense in depth. Multiple agents with different functions can cross-check each other's outputs, reducing the risk that a single point of AI failure leads to incorrect decisions.

Agent types in BASTION include facilitators that coordinate workflows and present information, analysts that process data and generate assessments, and executors that carry out approved actions within defined constraints. Each agent type has its own trust calibration, with executors subject to the strictest constraints and facilitators permitted the greatest autonomy for their coordination functions.

The NEAR AI Governance Framework, which defines phases of AI capability maturation (Support, Represent, Organize), informs how agent autonomy evolves over time [CITATION NEEDED: NEAR AI documentation]. New agents begin in the Support phase, where they can only provide information and recommendations. As they demonstrate reliability, they may advance to Represent (acting on behalf of humans within constraints) and eventually Organize (coordinating other agents) phases.

Security Architecture

BASTION's security architecture follows a zero-trust model appropriate for multi-stakeholder environments where no single organization controls all participants. Rather than assuming that participants within the coalition network are trusted, every access request is verified against the requestor's attributes, credentials, and the policies applicable to the requested resource.

Attribute-Based Access Control (ABAC) provides the flexibility necessary for coalition environments with complex, overlapping authorization requirements. Unlike Role-Based Access Control (RBAC), which assigns permissions based on predefined roles, ABAC evaluates access decisions based on attributes of the subject, resource, action, and environment. This approach can express policies such as "NATO SECRET information may be accessed by personnel with SECRET clearance from nations that are NATO members and have signed the applicable information sharing agreement."

The ABAC implementation encodes national caveats directly, enabling automatic enforcement of the information sharing restrictions that impede traditional coalition coordination. When a coalition partner specifies that certain information is "REL TO FVEY" (releasable to Five Eyes nations), the access control system automatically restricts access to personnel from the United

States, United Kingdom, Canada, Australia, and New Zealand. These restrictions are enforced consistently across all access paths, eliminating the risk of inadvertent disclosure through unchecked channels.

Post-quantum cryptography considerations inform BASTION's cryptographic choices, recognizing that data protected today may face quantum computing threats during its classification lifetime. While current quantum computers cannot break modern encryption, the development of cryptographically relevant quantum computers could compromise data encrypted with today's algorithms. BASTION's architecture supports cryptographic agility, enabling transition to post-quantum algorithms as standards mature and implementations become available [CITATION NEEDED: NIST post-quantum cryptography standardization].

The security architecture treats classification as a first-class concern. Information objects carry classification metadata that travels with them through the system, enabling consistent enforcement of access controls regardless of where data is processed or stored. Sensitive data is encrypted at rest and in transit, with keys managed through a distributed key management system that does not rely on any single trusted authority.

3.4 Human Authority Integration

Maintaining appropriate human authority over AI-augmented decision-making is central to BASTION's design. The system implements the authority positions described in Section 2.9 (Human-in-the-Loop, Human-on-the-Loop, Human-out-of-the-Loop) in a structured framework that matches authority requirements to decision consequences.

Autonomy Levels by Decision Type

BASTION categorizes decisions according to their consequences and assigns appropriate autonomy levels. This categorization is encoded in smart contracts, ensuring consistent application across the coalition regardless of which nation's forces or systems are involved.

High-Consequence Decisions require Human-in-the-Loop (HITL) authority. These include strategic resource allocations, changes to coalition membership, modifications to governance rules, and any decision that could result in lethal effects. For HITL decisions, AI agents may analyze options, present recommendations, and prepare proposals, but no action occurs until a qualified human explicitly approves.

Medium-Consequence Decisions may operate with Human-on-the-Loop (HOTL) authority. The system can execute these decisions autonomously, but human operators monitor execution and can intervene to halt or modify actions. Examples include routine resource transfers within pre-approved allocations, standard operational coordination, and information sharing within established classification boundaries. HOTL authority enables operational tempo while maintaining human oversight.

Low-Consequence Decisions may operate with Human-out-of-the-Loop (HOOTL) authority for well-constrained scenarios. These include status reporting, routine logistics coordination, and execution of pre-approved tactical actions. Even for HOOTL decisions, all actions are recorded on the blockchain for later audit, and humans can retrospectively review agent actions and adjust autonomy levels if agents demonstrate unreliable behavior.

The categorization of specific decision types is itself a governance decision that can be adjusted through DAO proposals. If coalition partners determine that a certain class of decisions requires more human involvement, they can vote to change the autonomy level for that decision type. This flexibility enables the governance framework to evolve with operational experience.

Strike Authorization as Special Case

One category of decision is treated as an inviolable special case: strike authorization involving lethal effects always requires Human-in-the-Loop approval, regardless of any other settings or configurations. This constraint is hardcoded in the system architecture, not merely a policy setting that could be changed through governance processes.

The rationale for this constraint reflects both ethical principles and legal requirements. International discussions on autonomous weapons systems consistently emphasize the need for meaningful human control over lethal decisions [CITATION NEEDED: International discussions on LAWS]. Military doctrine in most nations requires human authorization for weapons employment. Public trust in AI-augmented military systems depends on assurance that machines do not make life-or-death decisions autonomously.

In BASTION’s implementation, strike authorization proposals are routed through a separate approval pathway that requires explicit human votes. AI agents may prepare targeting recommendations, assess collateral damage estimates, and present relevant intelligence, but they cannot approve strikes and cannot be configured to do so. The smart contract logic enforces this constraint at the protocol level, making it impossible to bypass through configuration changes or software updates.

Furthermore, strike authorization requires supermajority approval and extended deliberation periods, providing multiple opportunities for human judgment to intervene. Every strike authorization decision is logged with special audit markers that enable retrospective review and accountability. This approach ensures that BASTION's AI augmentation enhances decision quality without eroding human control over the most consequential military decisions.

Trust Earned Through Performance

AI agent autonomy is not static; it evolves based on demonstrated performance. Agents that consistently provide accurate analysis, make reliable recommendations, and execute actions correctly can be granted increased autonomy over time. Agents that produce errors, demonstrate bias, or behave unexpectedly have their autonomy reduced.

This graduated trust approach mirrors how military organizations develop confidence in subordinate units and personnel. Just as a newly assigned officer might be given limited initial responsibility that expands as they demonstrate competence, AI agents begin with minimal autonomy and earn greater latitude through reliable performance.

The system maintains performance metrics for each agent, tracking accuracy of predictions, quality of recommendations, and reliability of execution. Human reviewers periodically assess agent performance and can adjust autonomy settings accordingly. Agents flagged for performance issues are automatically reduced to HITL authority until human operators explicitly restore higher autonomy levels.

3.5 MDMP Governance Integration

While Sections 3.2 through 3.4 describe BASTION's foundational architecture for DAO governance and human authority, operational military planning requires a more granular governance framework that maps to established doctrinal processes. The Military Decision Making Process (MDMP), as defined in Army doctrine for Theater Army (ASCC-level) planning, provides this structure. BASTION integrates MDMP governance directly into the DAO layer, extending the foundational architecture with formal phase progression enforcement, assumption lifecycle tracking, and a safety matrix that prevents authority violations at the smart contract level.

Five-Tier Authority Model

Section 3.4 described three human authority positions: Human-in-the-Loop, Human-on-the-Loop, and Human-out-of-the-Loop. Analysis of the 65 distinct activities within the MDMP process revealed that this three-level framework, while conceptually sound, lacks the granularity needed for operational planning governance. Some activities require AI to lead analysis with human oversight, while others require humans to lead with AI assistance. The distinction between these cases is operationally significant.

BASTION extends the authority framework to five tiers that map more precisely to MDMP activity requirements:

Authority Level	Description	Example Activities
AI_AUTONOMOUS	AI executes without human involvement	Data aggregation, format validation, monitoring
AI_PRIMARY	AI leads with human oversight	Intelligence preparation, trend analysis
HYBRID_AI_LED	AI generates, human reviews before action	COA development, risk assessment
HYBRID_HUMAN_LED	Human leads with AI assistance	Mission analysis, commander's guidance
HUMAN_ONLY	Human decides, AI may present information	Strike authorization, ethical judgments, risk acceptance

Three activity categories are permanently locked to HUMAN_ONLY regardless of any governance action: AUTHORITY_DECISION (decisions about who has authority to act), ETHICAL_LEGAL (judgments involving ethical or legal reasoning), and RISK_JUDGMENT (acceptance of risk to mission or force). These locks are immutable in the smart contract implementation, reinforcing the strike authorization invariant described in Section 3.4 with a broader class of decisions that require human judgment.

The five-tier model is enforced at the smart contract level. When an AI agent attempts to execute an activity, the contract verifies that the agent's authority level is permitted for that activity's category. Transactions that violate the safety matrix are rejected before execution, providing defense in depth against both misconfiguration and adversarial manipulation of agent permissions.

Governance Gate System

MDMP divides planning into nine phases, from Receipt of Mission through Transition. BASTION enforces phase progression through a governance gate system that prevents advancement until prerequisite conditions are satisfied. This enforcement addresses a common failure mode in military planning: pressure to skip steps or abbreviate analysis under time constraints.

Eighteen governance gates span the nine MDMP phases, organized into six gate types:

- **PhaseTransition gates** require completion of all mandatory products and analyses before advancing to the next MDMP phase.
- **ProductApproval gates** require human approval of key planning products (mission statement, commander’s guidance, COA sketches, OPORD).
- **AuthorityCheckpoint gates** enforce human decision authority at points where AI analysis must yield to human judgment.
- **RedTeamGate gates** require that red team challenges achieve minimum completeness thresholds before COA approval.
- **CoalitionGate gates** require multi-party consensus from coalition partners at key decision points.
- **AssumptionGate gates** require explicit human acceptance of planning assumptions before they can inform subsequent analysis.

Each gate is implemented as a DAO proposal that must achieve the required approval threshold before the workflow engine permits phase advancement. This mechanism extends BASTION’s existing proposal and voting infrastructure rather than introducing parallel governance mechanisms, maintaining architectural consistency while adding MDMP-specific enforcement.

Assumption Lifecycle Management

Military planning relies on assumptions that may prove invalid as operations unfold. Traditional planning processes track assumptions informally, creating risk that invalidated assumptions continue to inform decisions. BASTION implements formal assumption lifecycle management through a dedicated smart contract.

Assumptions progress through three states: Pending (identified but not yet accepted), Accepted (explicitly approved by a human decision-maker through a DAO proposal), and Invalidated (contradicted by new information or events). Two governance invariants enforce assumption discipline: Invariant 3 requires explicit human acceptance for all planning assumptions, ensuring that no assumption enters the planning basis without accountability; Invariant 6 triggers automatic replanning workflows when an accepted assumption is invalidated, ensuring that plans built on false premises are flagged for review.

Each assumption carries sensitivity analysis metadata indicating which planning products and decisions depend on it. When an assumption is invalidated, the system identifies all downstream products that may require revision, providing decision-makers with immediate visibility into the blast radius of changed circumstances. This capability directly addresses the challenge identified in Section 2.7: the gap between strategic intent and tactical execution widens when planning assumptions fail silently.

MDMP-Specific AI Agents

Six AI agents support the MDMP governance framework, each following the single-responsibility pattern described in Section 3.3:

The **Assumption Auditor** continuously monitors the planning environment for indicators that accepted assumptions may be invalid. It surfaces new assumptions implicit in planning products and tracks assumption sensitivity across the planning timeline. The **Orders Validator** performs format and consistency validation on OPORD products, simulating degraded execution scenarios to identify orders that may fail under realistic conditions. The **Uncertainty Quantifier** attaches calibrated confidence intervals to all AI-generated analyses, detecting false precision where AI outputs imply greater certainty than evidence supports. The **Data Bias Detector** identifies statistical bias, coverage gaps, and staleness in intelligence inputs that could skew planning analysis. The **Problem Framing** agent generates alternative problem perspectives from multiple viewpoints, countering the cognitive trap of premature closure on a single problem definition. The **ROE Compliance** agent parses rules of engagement, maps authorities to specific tasks, and validates that planned actions comply with applicable constraints.

These agents operate within the authority levels defined by the safety matrix. None can override governance gates or approve proposals autonomously. Their outputs feed into the governance workflow, where human decision-makers evaluate AI analysis within the structured MDMP process.

3.6 Escalation & Competition Modeling

Military planning must account for adversary behavior, escalation dynamics, and cascading effects across multiple domains. While Section 3.5 describes the governance framework that structures the planning process, this section describes the analytical capabilities that inform planning decisions within that framework. BASTION integrates adversary modeling, escalation simulation, and effects analysis as first-class capabilities rather than ad hoc staff processes.

Adversary Modeling

Effective COA development requires understanding adversary capabilities and likely responses. BASTION implements adversary modeling following ATP 2-01.3 doctrine, with an AI agent that synthesizes adversary capability models from available intelligence inputs and generates two standard adversary products: the Most Likely Course of Action (MLCOA), representing the adversary response with highest probability given observed indicators, and the Most Dangerous Course of Action (MDCOA), representing the adversary response that poses greatest risk to friendly operations regardless of probability.

The adversary modeler operates at SemiAutonomous authority, generating adversary COAs that human analysts review before incorporation into wargaming. This authority level reflects the analytical nature of adversary modeling: the AI performs pattern synthesis and doctrinal reasoning that benefits from machine speed, while human analysts provide the contextual judgment and cultural understanding that current AI systems cannot reliably provide.

Escalation Dynamics and Effects Analysis

Military operations produce effects that extend beyond immediate tactical outcomes. A tactical action in one domain may trigger diplomatic consequences, information effects, economic disruptions, or further military responses. BASTION models these dynamics through two complementary capabilities.

The **Escalation Modeler** simulates escalation dynamics using configurable theoretical frameworks. Each scenario is represented as an escalation ladder with discrete rungs representing levels of conflict intensity, trigger conditions that cause transitions between rungs, thresholds that define escalation boundaries, and de-escalation options available at each level. The modeler identifies escalation pathways that planned actions might trigger and highlights off-ramps that preserve strategic flexibility.

This capability directly addresses the risk that tactical decisions made under time pressure may inadvertently cross escalation thresholds with strategic consequences.

The **Effect Cascader** maps second and third-order effects of each COA across the four DIME domains: Diplomatic, Information, Military, and Economic. For each planned action, the cascader traces likely consequences through interconnected domains, identifying effects that planners might not anticipate from a single-domain perspective. The visualization presents these cascading effects as directed flows across DIME swim lanes, enabling planners to trace cause-and-effect chains and identify unintended consequences before committing to a course of action.

A **Deception Detector** complements these analytical agents by identifying inconsistencies between adversary stated intent and observed behavior. This capability supports the Intelligence Preparation of the Battlefield (IPB) process by flagging indicators that adversary actions may not align with their declared posture, alerting analysts to potential deception operations.

Enhanced Wargaming

BASTION extends its red team simulation capability into a full wargaming framework that supports the action-reaction-counteraction methodology prescribed by military doctrine. The wargaming engine operates in a hybrid mode: AI agents first execute automated scenario runs that explore the decision space rapidly, identifying critical decision points and high-impact variables. Commanders then use interactive “what-if” exploration to test specific scenarios, modify assumptions, and probe edge cases that automated analysis may not cover.

Each wargaming session produces a complete move log with reasoning audit trail, preserving the analytical basis for COA selection. This audit trail integrates with the governance gate system described in Section 3.5, providing the evidence base that RedTeamGate checks require before COA approval can proceed.

Quantitative Planning Support

Three additional capabilities provide quantitative rigor to COA comparison and selection:

Force Ratio Analysis implements the Correlation of Forces Methodology (COFM) with seven combat power modifiers and doctrinal thresholds (3:1 for deliberate attack, 1:1 for defense). This quantitative assessment complements the qualitative COA comparison performed by the COA Comparator agent, providing numerical basis for feasibility judgments.

COA Sketch Generation produces visual representations of each COA using MIL-STD-2525D military symbology overlaid on geographic map displays. Phased timelines with animation enable commanders to visualize the temporal progression of operations, while affiliation filtering isolates friendly, hostile, and neutral force dispositions.

Sustainment Modeling assesses logistics feasibility for each COA through resource burndown analysis aligned with ADP 4-0 resource categories. Phase-level risk flags (green, amber, red) provide rapid feasibility assessment, while detailed burndown charts enable planners to identify specific resource constraints that may limit operational duration or tempo.

Branch and Sequel Planning supports contingency development by identifying decision points where planned operations may need to diverge based on battlefield conditions. Decision points are modeled with trigger conditions categorized across multiple domains, enabling planners to define what observable conditions would require transitioning to alternative plans. This capability ensures that operational planning accounts for uncertainty rather than assuming a single predicted outcome.

3.7 Doctrinal Lifecycle Interface

BASTION's user interface evolved from a functional grouping of capabilities (Decide, Design, Campaign, Monitor) to a doctrine-aligned lifecycle that mirrors the Joint Planning Process as described in JP 5-0. This restructuring ensures that the system's workflow organization reinforces doctrinal thinking rather than imposing an arbitrary software workflow on military planners.

Six-Tab Doctrinal Flow

The interface implements six tabs corresponding to the phases of joint operational planning:

Understand. The first tab consolidates strategic environment analysis. Commanders and staff upload strategic guidance documents, scenario training packages, and intelligence reports. The system provides PMESII-PT (Political, Military, Economic, Social, Information, Infrastructure, Physical Environment, Time) operational environment analysis through AI agents. RAFT graph visualization displays actor relationships, tensions, and functions extracted from uploaded documents. This tab absorbs capabilities previously distributed across separate strategic planning and intelligence views, presenting them as a unified understanding of the operational environment. A “Create from Scenario” wizard enables rapid exercise setup by ingesting multi-file training packages with AI-driven tag inference for document type, team assignment, and exercise phase classification.

Design. The second tab implements operational design as described in JP 5-0 Chapter III. A problem framing canvas enables commanders to define the problem from multiple perspectives, with AI agents identifying key tensions and suggesting alternative framings. Center of gravity analysis follows Strange’s CG-CC-CR-CV (Center of Gravity, Critical Capabilities, Critical Requirements, Critical Vulnerabilities) framework for both friendly and adversary forces. Lines of effort and lines of operation are defined visually with explicit linkages to strategic objectives and decisive points. An operational approach builder synthesizes the center of gravity analysis and lines of effort into a coherent phased approach. The design-to-plan handoff exports operational design outputs directly as inputs to mission analysis in the Plan tab. Real-time collaborative editing through Yjs CRDTs enables multi-staff contribution to the operational design.

Plan. The third tab implements the JP 5-0 Joint Planning Process and MDMP workflow described in Section 3.5. Mission analysis, course of action development, wargaming, COA comparison, and plan development proceed through governance-gated phases. The escalation modeling and competition analysis capabilities described in Section 3.6 integrate here as analytical tools within the planning workflow.

Decide. The fourth tab manages the transition from planning to execution and houses the decision dashboard that consolidates all pending governance decisions requiring human action. WARNORD, OPORD, and FRAGO generation produces formatted orders with classification banners and handling instructions. Task organization displays show force assignment by unit and function. Resource allocation from the registry (Section 3.10) is managed through DAO proposals at decision gates. A RACI-filtered decision queue presents pending approvals, rejections, and deferrals through PendingDecisionModal workflows. Ironclaw—the AI Chief of Staff agent—polls the decision queue every sixty seconds, proactively surfacing overdue decisions and surfacing context to inform the commander’s choice. Per-team information barriers ensure that each staff role and coalition partner sees only the orders and directives relevant to their function and clearance level.

COP (Common Operating Picture). The fifth tab displays the operational picture generated by autonomous AI agent teams (Section 3.9). Multiple overlay layers render MIL-STD-2525D military symbology as interactive SVG graphics. Resources from the registry appear as standard military symbols alongside AI-generated entity positions. Friendly and adversary perspective toggles, phase sliders, and playback controls enable temporal and perspective-based COP exploration.

Assess. The sixth tab supports running assessment against measures of effectiveness (MOE) and measures of performance (MOP). Running estimates from each staff role feed into assessment dashboards. The assessment tab closes the doctrinal loop: when assessment reveals that the operational approach is no longer achieving desired effects, the commander can reframe the problem and cycle back to the Understand or Design tabs.

Role-Based Tab Visibility

Not all roles require access to all tabs. The interface implements role-based tab visibility: a J2 Intelligence officer may see Understand, COP, and Assess but not Decide. A Commander sees all tabs. This visibility control reduces cognitive load by presenting only the workflow phases relevant to each user's function while maintaining a single coherent interface for the entire planning lifecycle.

Iterative Navigation

The tab structure supports the iterative nature of operational planning described in JP 5-0. Planners are not locked into a linear progression; they can revisit earlier tabs as understanding evolves. Assessment findings may trigger a return to Design for operational approach adjustment. New intelligence in Understand may invalidate assumptions tracked in Plan. This iterative capability mirrors doctrinal guidance that planning is continuous and cyclical rather than linear.

3.8 Intelligence Preparation of the Battlefield

BASTION implements a complete Intelligence Preparation of the Battlefield (IPB) cycle supporting both friendly (Blue) and adversary (Red) perspectives with strict information isolation between the two.

Dual-Perspective Analysis

The IPB system provides a Blue/Red toggle that switches the analysis context between friendly and adversary perspectives. Information isolation ensures that Red team analysis of adversary capabilities and intentions does not leak into Blue team planning products, and vice versa. This separation supports realistic exercise scenarios where opposing teams develop independent assessments.

Scenario-Driven IPB

Exercise scenarios drive the IPB process. Scenario package upload accepts multi-file training packages with AI-driven tag inference that automatically categorizes documents by type (intelligence report, operations order, logistics estimate), team assignment (Blue, Red, White), and exercise phase (Competition, Crisis, Conflict). Async LLM extraction processes uploaded documents and surfaces extracted intelligence in the Understand tab.

COA Scoring and Commander Decision Support

Courses of action are scored against five doctrinal criteria with wargame evidence integration. The COA scoring system draws on the wargaming framework described in Section 3.6 to incorporate action-reaction-counteraction outcomes as evidence supporting or undermining each COA’s viability. A commander decision matrix presents scored COAs with blockchain-anchored decision records, ensuring that the rationale for COA selection is permanently recorded.

Order Generation with Information Barriers

The IPB cycle produces WARNORD, OPORD, and FRAGO documents with per-team information barriers. Each team receives only the products and information appropriate to their role and perspective. A Kanban task board manages exercise phase gate progression, providing visual workflow management for multi-phase exercises.

3.9 AI Common Operating Picture Generation

A distinctive capability of BASTION is the autonomous generation of Common Operating Picture (COP) layers by AI agent teams. Rather than requiring manual placement of symbols on a map, AI agents parse planning documents, orders, and intelligence reports to derive location, resource, and intent data, then generate MIL-STD-2525D-compliant military symbology overlays.

Agent Team Architecture

Each workspace section maintains an autonomous agent pool that monitors document and plan changes. When source material is created or updated, agents process the content through an extraction pipeline that identifies:

- **Geographic references** that can be resolved to map coordinates
- **Military units and resources** that can be represented with standard symbology
- **Temporal phasing** that enables phase-based COP animation
- **Intent and objectives** that can be visualized as operational graphics

Symbol Generation

A SIDC (Symbol Identification Coding) builder constructs standard MIL-STD-2525D symbol codes from extracted entity attributes. An SVG specification generator produces interactive vector graphics for each symbol, supporting hover/click detail panels that link symbols back to their source documents. An entity linker resolves document mentions (unit names, location references, resource identifiers) to specific map positions using a combination of geocoding and context-aware disambiguation.

Layer Management and Governance

Generated COP layers undergo a publish review cycle before promotion to the top-level COP visible to all workspace participants. This governance integration ensures that AI-generated operational picture content receives human review before influencing command decisions. Layer version snapshots support rollback, and conflict detection identifies when multiple agents produce contradictory symbol placements. Friendly and adversary perspective toggles enable separate COP views, and a phase slider with playback controls enables temporal COP exploration showing planned force movements and phasing.

3.10 Resource Registry and DID Plugin Architecture

BASTION elevates military resources from simple inventory entries to first-class entities with blockchain-anchored Decentralized Identifiers and a plugin architecture for extensible resource type management.

Resource Identity

Each resource receives a DID in the form `did:near:resource-{id}`, anchored on the NEAR blockchain with blinded and public key pairs. This identity model extends the DID framework established in Phase 2 (Section 3.2) to military assets, enabling cryptographic verification of resource provenance, assignment history, and capability claims. Resources are registered in a singleton registry that supports DID lookup, capability queries, and geographic area queries.

Plugin Interface

Resource types are managed through a plugin interface that defines five extension points for each resource category: schema (data model and validation), state machine (lifecycle states and transitions), capabilities (what the resource can do), data handler (telemetry and status ingestion), and COP renderer (how the resource appears on the Common Operating Picture). This architecture enables new resource types to be added without modifying core platform code.

Built-in Plugins

Five built-in plugins cover the primary military resource categories:

- **AutonomousVehiclePlugin**: Manages unmanned systems with navigation state, mission assignment, and sensor payload tracking
- **SensorPlatformPlugin**: Handles sensors across domains (airborne, ground, maritime, space) with coverage areas and detection capabilities
- **WeaponSystemPlugin**: Tracks weapon systems with ammunition states, engagement envelopes, and employment restrictions
- **CommsPlugin**: Manages communication assets with frequency assignments, encryption status, and network topology
- **LogisticsPlugin**: Tracks supply items with consumption rates, distribution status, and replenishment thresholds

COP Integration and DAO-Governed Allocation

Resources render on the COP as MIL-STD-2525D symbols alongside AI-generated layers. Real-time readiness tracking reports Full Mission Capable (FMC), Partially Mission Capable (PMC), or Not Mission Capable (NMC) status with location, degradation factors, and maintenance scheduling. Telemetry ingestion from sensors and vehicles feeds AI agent context for situational awareness. Resource allocation flows through DAO proposals at decision gates, ensuring that resource assignment decisions receive appropriate governance oversight.

3.11 Training and Operational Mode Architecture

BASTION implements a global application mode toggle that switches between training (exercise) and operational contexts, embodying the military doctrine of “train as you fight.”

Mode Design

The training mode activates a persistent amber “EXERCISE - EXERCISE - EXERCISE” banner across the application, providing unambiguous visual distinction from operational mode. All documents generated in training mode receive automatic EXERCISE watermarks. The operational mode presents a clean interface where the absence of the training banner serves as the mode indicator.

Data Isolation

Training and operational modes maintain complete data isolation. Exercise scenarios, planning products, and governance decisions in training mode exist in a separate context from operational data. This isolation prevents exercise data from contaminating operational systems while allowing exercise scenarios to use realistic data structures and workflows.

Governance Parity

A core design principle is that DAO governance operates identically in both modes. Voting thresholds, authority models, governance gates, and safety matrix enforcement apply equally in training and operational contexts. This parity ensures that training exercises exercise the actual governance mechanisms that would be used in operations, avoiding the common problem of training systems that simplify governance for convenience and thereby fail to prepare users for operational governance complexity.

Exercise Management

Training mode provides reset and checkpoint capabilities for exercise iteration. After-action review capture enables structured debriefing by recording exercise events, decisions, and outcomes for post-exercise analysis. Exercises can be reset to specific checkpoints to replay scenarios with different decisions, supporting iterative learning.

3.12 JPP Staff Organization

BASTION organizes exercise participation around the Joint Planning Process staff structure, providing per-role workspaces that mirror the organizational structure of a joint staff.

Role-Based Workspaces

Each JPP staff role—Commander, Deputy Commander, Chief of Staff, J1 through J9, and specialized roles including Staff Judge Advocate, Political Advisor, Public Affairs Officer, and component commanders—receives a dedicated workspace with templated doctrinal products appropriate to their function. Templates follow service-specific formats and include the required sections, coordination fields, and approval workflows that each staff role is responsible for producing.

Cross-Staff Coordination

Real-time notifications delivered via WebSocket alert staff members when products they depend on are updated, when coordination is requested, or when governance gates require their input. A hybrid editor combining structured fields with narrative text enables both standardized data entry and free-form analysis within each doctrinal product. Real-time product merging allows multiple staff roles to contribute to shared products (such as running estimates) simultaneously through CRDT-based conflict resolution.

AI Staff Agent Integration

Each staff role has access to AI agent teams that automate routine doctrinal tasks within the role's functional area. Rather than dedicating one AI agent per staff role, Ironclaw — BASTION's Chief of Staff agent — provides role-aware coordination across all staff functions, drawing on doctrinal publications, product formats, and coordination requirements as needed. Strategic direction import from the Design tab ensures that staff workspaces receive operational design outputs as context for their planning products, maintaining alignment between design intent and staff products.

3.13 Component Integration

BASTION's contribution lies not in inventing new technologies but in integrating existing technologies in novel ways to address gaps in military coordination. This section highlights the integration points that represent BASTION's systems integration novelty, including the MDMP governance and escalation modeling capabilities described in Sections 3.5 and 3.6, and the doctrinal lifecycle, COP generation, resource registry, and training mode capabilities described in Sections 3.7–3.12.

Novel Integration Points

DAO Governance with AI Agents. While DAOs and AI agents each have extensive prior art, their integration for command and control applications has not been previously demonstrated. BASTION combines DAO voting and smart contract enforcement with AI agent analysis and execution, creating a system where AI augments human decision-making within a decentralized governance framework. This integration addresses the gap identified in Section 2.10: no existing systems provide frameworks for multi-stakeholder AI coordination with policy compliance and appropriate human authority.

Blockchain with Military C2. As documented in Section 2.4, existing blockchain applications in defense focus primarily on data integrity for supply chains and identity management. BASTION extends blockchain use to command and control governance, leveraging immutability and transparency for decision accountability rather than merely data verification. This application demonstrates that blockchain's governance capabilities, not just its data integrity properties, are valuable for military coordination.

Multi-Level Coordination. The three-tier DAO structure maps blockchain governance to the military levels of warfare: strategic, operational, and tactical. This mapping enables coordinated decision-making across levels while respecting the different tempo, scope, and authority requirements at each level. The inter-DAO communication mechanisms provide the strategic-to-tactical linkage that JADC2 envisions, but with decentralized governance rather than centralized control.

Policy Encoding with Autonomous Execution. Smart contract policy encoding enables bounded autonomy: AI agents can act quickly within policy constraints without case-by-case human approval, while policy violations are automatically prevented. This integration addresses the tradeoff between speed and compliance that characterizes traditional coalition coordination.

Doctrinal Process Governance with Blockchain Enforcement. The integration of MDMP doctrinal processes with DAO governance gates represents a novel application of blockchain to military planning process enforcement. While MDMP checklists and phase gates exist in doctrine, they are traditionally enforced through procedural discipline rather than technical controls. BASTION encodes these gates as smart contract conditions that must be satisfied through DAO proposals before workflow advancement, transforming doctrinal guidance into enforceable governance. The assumption lifecycle management system further demonstrates this integration: planning assumptions that would traditionally be tracked in spreadsheets or staff notes become blockchain-recorded artifacts with formal acceptance requirements, invalidation triggers, and automatic replanning workflows.

Adversary Modeling with Governance-Aware Wargaming. The integration of AI-driven adversary modeling, escalation simulation, and effects analysis with the MDMP governance framework ensures that analytical rigor is both enabled by AI speed and bounded by governance controls. Wargaming results feed directly into RedTeamGate checks that must be satisfied before COA approval, creating a closed loop between AI analysis and governance enforcement that prevents premature commitment to courses of action that have not been adequately challenged.

Doctrine-First Interface Design with Lifecycle Governance. The six-tab doctrinal lifecycle (Section 3.7) integrates UI structure with the governance framework: each tab corresponds to a planning phase with associated governance gates, and tab transitions mirror the iterative planning process described in JP 5-0. This integration ensures that the interface reinforces doctrinal thinking rather than allowing planners to shortcut governance requirements through arbitrary navigation.

Autonomous COP Generation with Governance Review. The AI COP layer generation system (Section 3.9) integrates document analysis, symbol generation, and governance review in a pipeline where AI agents produce operational picture content but human review gates control what reaches the shared COP. This integration demonstrates that AI can accelerate operational picture construction while governance ensures quality control over the information that informs command decisions.

Resource Identity with DAO Governance. The resource DID architecture (Section 3.10) integrates blockchain identity, plugin-based resource management, and DAO governance for allocation decisions. Resources are not merely tracked in a database; they are blockchain-verified entities whose assignment flows through the same governance mechanisms used for strategic decisions, providing end-to-end accountability from resource registration through employment.

Training-Operational Mode Parity. The global mode architecture (Section 3.11) integrates with every other subsystem to ensure that exercise environments are indistinguishable from operational environments in governance behavior. This integration validates that training exercises truly exercise the governance mechanisms, addressing the common problem of training systems that simplify governance for convenience.

Cross-Reference to Results

The following section (Section 4, Results) demonstrates these architectural components in an end-to-end scenario. The minimum viable product (MVP) system implements the three-tier DAO structure, deploys 16 AI agents across governance, planning, intelligence, and staff coordination functions, and demonstrates the human authority integration described here. The MDMP governance framework (Section 3.5), escalation modeling capabilities (Section 3.6), doctrinal lifecycle interface (Section 3.7), COP generation (Section 3.9), resource registry (Section 3.10), and training mode architecture (Section 3.11) collectively provide a comprehensive military planning and execution environment that extends well beyond the initial demonstration scope.

The Results section shows how the theoretical architecture described here translates into functioning software and hardware, providing evidence that BASTION's integration approach is not merely conceptual but practically achievable. Quantitative metrics from the implementation validate the system's ability to accelerate coordination while maintaining appropriate human control.

3.14 Robot Integration Architecture

Physical domain augmentation requires bridging the gap between cloud-resident governance infrastructure and edge-deployed autonomous systems operating in contested, bandwidth-limited environments. BASTION's robot integration architecture resolves this tension through a Docker-based local network bridge that mediates communication between the NEAR blockchain and physical robotic platforms, avoiding the latency, dependency, and security vulnerabilities associated with direct cloud-to-robot connectivity.

Bridge Architecture and Self-Registration

The Docker bridge runs as a containerized service on a local network accessible to both the BASTION application server and the robot's on-board Python agent. Rather than requiring the cloud platform to initiate contact with the robot—an approach that introduces firewall traversal complexity and creates a command dependency—the Python robot agent establishes an outbound WebSocket connection to the bridge at startup, self-registering with a cryptographically signed DID credential. This self-registration pattern eliminates the need for static IP configuration or VPN tunneling: the robot registers itself with the system rather than waiting to be discovered.

Multicast DNS (mDNS) auto-discovery enables the bridge to locate robot agents on the local network without manual configuration. When a Python robot agent starts, it advertises its presence via mDNS; the bridge discovers the advertisement and completes the WebSocket handshake. This zero-configuration approach is critical for field deployment where network administrators may not be available to configure static routes or DNS entries.

Mission Intent Translation

BASTION issues missions as structured intent objects specifying objective, parameters, and DID-encoded constraints. The Python robot agent receives these intent objects and translates them into platform-specific command sequences through an LLM-based translation layer with a deterministic template fallback. The LLM path handles novel mission specifications by reasoning about platform capabilities and composing appropriate command sequences; the template path ensures reliable execution for the four established vision-enabled mission types: `recon_area`, `visual_search`, `overwatch`, and `resupply_route`. Pre-flight DID constraint validation confirms that the robot's registered capabilities match the mission's requirements before execution begins, preventing mission assignment to platforms that cannot complete the specified task [CITATION NEEDED: autonomous mission planning literature].

Vision Pipeline

The robot vision pipeline runs on an NVIDIA Jetson Orin Nano edge computing module at the robot platform, providing onboard inference without dependence on cloud connectivity. A CSI camera feeds raw frames into NVIDIA detectNet for object detection and classification, producing bounding boxes with confidence scores for entities of tactical interest. ORB (Oriented FAST and Rotated BRIEF) feature matching performs re-identification across video frames, enabling the system to track previously detected entities across discontinuous observation windows. Detection results stream back through the bridge to the BASTION COP layer, where they appear as resource-linked intelligence contributions subject to the governance review cycle described in Section 3.9 [CITATION NEEDED: edge AI for military robotics].

3.15 Knowledge Graph Architecture

Operational intelligence exists as a distributed, heterogeneous collection of documents, orders, signals intelligence, and human reporting. Extracting actionable knowledge from this corpus requires a semantic layer that moves beyond keyword search to represent entities, relationships, and temporal context in a machine-queryable form. BASTION implements a knowledge graph—termed the “brain”—as the semantic substrate for intelligence integration across the platform.

JSON-LD Semantic Brain

The brain is stored as JSON-LD (JSON Linked Data), a lightweight serialization of RDF that enables semantic annotation of knowledge graph nodes without requiring a dedicated triple store. JSON-LD’s use of `@context` declarations links BASTION’s entity vocabulary to established upper ontologies: Basic Formal Ontology (BFO), the Common Core Ontologies (CCO), and the conceptual structures of DODAF and DNDAF architectural frameworks. This ontological alignment ensures that BASTION’s knowledge graph is interpretable against existing defense data standards and can exchange structured data with compliant systems [CITATION NEEDED: defense ontology standards].

Entity Resolution and Confidence Scoring

Multi-source intelligence produces redundant and sometimes contradictory entity references. A dedicated entity resolution subsystem identifies co-references across documents—recognizing, for example, that “the 3rd Brigade,” “3 BDE,” and “the Vanguard Brigade” all refer to the same tactical entity—and merges them into a single graph node with provenance pointers to each source. Confidence scoring follows NATO intelligence source reliability and information credibility ratings (the STANAG 2511 two-character code scheme), attaching reliability grades to each knowledge contribution. Graph nodes therefore carry not just their attribute values but calibrated uncertainty metadata that downstream analytical agents can use to weight evidence appropriately [CITATION NEEDED: intelligence confidence scoring methodology].

Brain Visualization and Temporal Reasoning

A force-directed neural canvas renders the knowledge graph as an interactive visual interface that presents nodes sized by relationship density and edges colored by relationship type. This visualization enables analysts to identify clusters of highly connected entities—potential centers of gravity or critical nodes—and trace relationship chains that might indicate adversary intent or capability. A brain timeline provides temporal reasoning capability: analysts can advance the timeline to observe how the graph state evolves as intelligence is incorporated over the course of an exercise or operation, identifying knowledge gaps and tracking adversary activity patterns across time.

3.16 Swarm Leadership and Doctrinal Formations

Heterogeneous robotic swarms operating in contested environments require coordinated behavior that degrades gracefully when communication is disrupted. BASTION's swarm architecture designates a single vision-equipped platform—the Sphero RVR+—as the swarm leader, leveraging its onboard NVIDIA processing and CSI camera to provide situational awareness for the entire formation while coordinating heterogeneous subordinate platforms.

Formation Control

Six doctrinal formations derived from Army tactical movement doctrine govern swarm geometry: wedge, line, column, echelon left, echelon right, vee, and staggered column. Four movement techniques—traveling, traveling overwatch, bounding overwatch, and successive bounds—define the temporal pattern of advance. Formation parameters (spacing, interval, lead platform designation) are encoded in the mission intent object and transmitted to swarm members through the bridge. The swarm leader broadcasts position updates and formation correction signals, and subordinate platforms adjust their positions to maintain prescribed spacing relative to the leader's position [CITATION NEEDED: autonomous swarm coordination].

UDP Broadcast Peer Mesh

In DDIL environments where bridge connectivity may be intermittent, swarm members maintain peer-to-peer coordination through a UDP broadcast mesh running on the local robot network. The mesh enables swarm members to share position data and formation status without routing through the cloud bridge, ensuring that formation cohesion and collision avoidance operate independently of external connectivity. When bridge connectivity is restored, the swarm leader synchronizes accumulated state back to BASTION through the WebSocket connection.

DAO-Driven Swarm Membership

Swarm composition is governed by the Tactical DAO. Adding a new platform to a swarm or removing a degraded platform requires a DAO proposal that records the membership change on the blockchain, associating each platform’s DID with the swarm mission. This governance approach ensures that swarm composition is tracked with the same accountability as human force assignment decisions. The leader shares its vision feed with swarm members that lack independent visual sensing capability, enabling collective situational awareness proportional to the leader’s detection range.

3.17 Document Intelligence Pipeline

Military planning generates and consumes large volumes of unstructured text. Intelligence reports, orders, doctrine publications, and exercise scenario packages arrive in diverse formats and must be integrated into a coherent operational picture. BASTION’s document intelligence pipeline applies a multi-agent processing team to each ingested document, extracting structured knowledge and inserting it into the brain graph in a form that downstream analytical agents can query.

Multi-Agent Processing Team

Each document is processed by a team of ten specialist agents operating in a coordinated pipeline: an orchestrator that manages pipeline state and resolves conflicts between agent outputs; a converter that transforms source documents from PDF, DOCX, and other formats to normalized text; a classifier that categorizes documents by type, team assignment, and exercise phase; three perspective analysts that extract content from Blue, Red, and White team perspectives with strict information isolation; a fact extractor that identifies discrete factual claims; an objective extractor that identifies stated goals and intentions; a linker that resolves entity mentions to existing brain graph nodes or creates new nodes for novel entities; a bias identifier that flags potential analytical bias or coverage gaps in the document; and a quality assessor that grades the document's reliability and completeness using NATO source reliability ratings [CITATION NEEDED: multi-agent information extraction].

Scoping Interview and Gap Identification

Before document upload, an AI-led scoping interview captures the boundaries of the current problem set: geographic area of interest, time period, key actors, and classification context. These boundaries constrain the linker and fact extractor, preventing false entity resolution and scope creep. Following extraction, the pipeline generates standing intelligence requirements that identify knowledge gaps—questions about entities or relationships that source documents do not address—and triggers autonomous web search and OSINT processes to fill identified gaps within the defined scope boundaries.

ExtractionTheater Visualization

An ExtractionTheater component presents live visualization of pipeline progress to planning staff, showing which agents are processing which documents, what entities have been extracted, and which gaps have been identified. This transparency enables analysts to monitor extraction quality in real time and intervene when the pipeline produces incorrect or incomplete outputs, maintaining human oversight over the automated intelligence processing workflow.

3.18 Hierarchical Problem Set Inheritance

Joint operations involve parallel and nested planning efforts: strategic campaigns that spawn operational missions, operational missions that spawn tactical subordinate actions, and FRAGO updates that must propagate across multiple levels simultaneously. BASTION's hierarchical problem set architecture formalizes these relationships, enabling structured context propagation from higher to lower echelons and upward reporting from tactical execution to campaign assessment.

Context Propagation

Strategic-level problem sets serve as context providers for child operational and tactical problem sets. When a strategic problem set is created, commanders designate which directives, policies, and intelligence products are eligible for inheritance by subordinate problem sets. Child problem sets inherit this context automatically at creation and receive push notifications when inherited content changes, enabling subordinate planners to identify when strategic guidance has been updated and assess the impact on their plans. Override tracking records when a subordinate problem set deviates from inherited guidance, providing parent echelon visibility into plan divergence with documented rationale [CITATION NEEDED: hierarchical planning systems].

OPORD-Triggered Mission Creation

When an OPORD is published in an operational problem set, the system identifies subordinate units and generates a tactical child problem set for each assigned unit, automatically populating the child problem set with the relevant portions of the parent OPORD as inherited context. This automation eliminates the manual transcription of orders across planning levels and ensures that tactical planners begin their work with correct, current guidance from the operational command. FRAGO propagation follows the same pattern: when a parent problem set issues a FRAGO, affected child problem sets receive notification and can apply the fragmentary order to their own plans.

Upward Reporting

Tactical execution generates assessment data—task completion status, resource consumption, engagement outcomes—that strategic and operational commanders require for running estimates. BASTION’s upward reporting mechanism aggregates tactical COP state and execution status from child problem sets, surfacing it in parent problem set assessment dashboards without requiring manual reporting from subordinate staffs. This automation closes the information loop from tactical action to strategic assessment, enabling the continuous operational assessment described in JP 5-0.

3.19 Operational Design Workspace

The transition from operational art to operational science—from understanding the problem to developing a plan to solve it—is the most demanding cognitive task in military planning. JP 5-0 defines operational design as the application of creative thinking and operational art to design an approach to solve the problem; BASTION provides dedicated tooling that structures this creative process without constraining it.

Center of Gravity Analysis

The operational design workspace implements Strange's CG-CC-CR-CV analytical framework for both friendly and adversary forces. Commanders and staff identify each force's Center of Gravity, decompose it into Critical Capabilities and Critical Requirements, and analyze Critical Vulnerabilities that opposing action could exploit. AI agents assist by cross-referencing identified critical capabilities against available intelligence in the brain graph, surfacing evidence that supports or challenges each analytical judgment and generating alternative decompositions for staff consideration. The analysis products feed directly into operational approach development and inform the targeting priorities encoded in planning products [CITATION NEEDED: center of gravity analysis methodology].

Lines of Effort and Operational Approach

Lines of effort and lines of operation are defined visually with explicit linkages to strategic objectives and decisive points. The operational approach builder synthesizes center of gravity analysis and lines of effort into a phased operational approach, with AI-assisted recommendations for sequencing and resource alignment. The design-to-plan handoff exports operational design outputs directly as structured inputs to mission analysis in the Plan tab, maintaining analytical continuity between the design and planning phases.

Fork-and-Merge Revision System

As planning proceeds, new intelligence or changed conditions may require operational design revision. BASTION implements a fork-and-merge revision system analogous to version control: a staff member proposes a design revision by forking the current approved design, developing the proposed change in isolation, and submitting a merge request that presents the diff between current and proposed designs. DAO governance gates require review and approval of design revisions before they replace the baseline, ensuring that approved designs are not modified unilaterally and that revision rationale is preserved in the governance record.

3.20 Training Assessment and Readiness

Military readiness is a continuous assessment process, not a binary state. BASTION's training assessment architecture provides structured mechanisms for capturing training event outcomes, tracking proficiency over time, and aggregating readiness data from unit level through campaign level to support commander decision-making about training investment priorities.

After-Action Review Capture

After-action review (AAR) capture at training events provides structured recording of observed outcomes, identified strengths, areas requiring improvement, and corrective action assignments. AAR records are linked to the training event, the units and personnel who participated, and the doctrinal tasks that were exercised, enabling longitudinal tracking of performance across multiple training iterations. AAR outputs feed into the proficiency tracking subsystem as assessment inputs.

METL Proficiency Tracking

Mission Essential Task List (METL) proficiency is tracked using the standard T/P/U (Trained, Practice, Untrained) assessment scale for each doctrinal task. Proficiency assessments from individual training events are aggregated by unit, task, and time period to produce proficiency trend lines that reveal whether unit capability is improving, degrading, or static. Commanders can drill down from aggregate readiness scores to individual task assessments to understand the specific training gaps driving readiness shortfalls.

Upward Aggregation and Training Readiness Dashboard

Training readiness data aggregates from tactical training events through exercises to strategic training programs using the same hierarchical problem set inheritance architecture described in Section 3.18. A training readiness dashboard presents proficiency trends, upcoming training milestones, and readiness projections for scheduled operational commitments. The Pacific Strategy AY26 scenario package serves as the operational demonstration data package for this capability, providing a complete multi-echelon exercise scenario—spanning six phases from Competition through Negotiation—that exercises the full training assessment pipeline from event capture through strategic readiness aggregation [CITATION NEEDED: military readiness assessment frameworks].

3.21 Ironclaw as Chief of Staff for Operational Design (Phase 55)

Operational design is among the most cognitively demanding tasks in military planning. Commanders and staff must engage deeply with problem framing, center of gravity analysis, lines of effort, and operational approach development under time pressure. BASTION extends the operational design workspace with Ironclaw functioning as a Chief of Staff for the design process — coordinating inputs across staff sections, obtaining the commander's design decisions through structured doctrinal questioning, and validating each input against JP 5-0 requirements before recording it as part of the plan's foundation.

Staff Coordination Architecture

The operational design coordination is implemented as a LangGraph StateGraph with interrupt-resume semantics, mirroring the architecture used for OSINT scoping. The graph progresses through four sequential doctrinal sections aligned to JP 5-0 Chapter III: problem framing (5 coverage criteria including environment characterization and tensions identification), center of gravity analysis (5 criteria with a minimum of two adversary critical capabilities for doctrinal completeness), lines of effort/ operation (3 criteria), and operational approach (3 criteria including phasing and end state articulation).^1

Each section carries explicit doctrinal coverage criteria derived from JP 5-0 and operational design doctrine. The interview graph uses structural inspection of the derived design object to assess coverage — a deterministic check against field completeness rather than a separate LLM evaluation. This design choice provides consistency and speed over qualitative assessment. Coverage assessment is transparent: if a field is empty or below the minimum count threshold, the section is not marked complete, and Ironclaw continues questioning.

Chief of Staff vs. Chatbot

The operational design capability positions Ironclaw as a *Chief of Staff* rather than a *chatbot*. The distinction is operationally significant. A chatbot responds to questions and presents information on request. A Chief of Staff actively coordinates the process — determining what inputs are needed, obtaining them from the right sources, validating them against doctrinal requirements, and challenging framings that appear incomplete through red-team follow-up questions embedded in the coordination.

Ironclaw's coordination style applies a challenge-then-validate pattern: for each section, it obtains the commander's reasoning through directed questioning, identifies apparent gaps or tensions in responses, and surfaces alternative framings that the commander may not have considered. Red-team probing questions are appended as AI messages in the same graph node as the primary question — a single LLM call that produces both the primary question and the follow-up challenge — rather than a separate graph node. This design minimizes latency while maintaining the adversarial questioning discipline.

Coordination Output and Workflow Integration

When a section achieves coverage, Ironclaw surfaces a section confirmation prompt. Upon confirmation, the section's derived design object is persisted via a dedicated store and the coordination advances to the next section. Confirmed sections cannot be automatically overwritten; they become locked design artifacts requiring explicit intent to revise.

Upon coordination completion, the derived design output integrates directly with the operational design workspace. The process captures design decisions in the same structured format as the manual workspace (problem framing canvas, CoG analysis object, LOE definitions, phased operational approach), enabling the design-to-plan handoff to proceed from coordinated artifacts through the same export path as manually constructed designs.

Knowledge Graph Gap Detection

The coordination process incorporates real-time knowledge graph gap detection. When the commander's inputs reference actors, relationships, or capabilities that have no corresponding entries in the problem set's brain graph, the system flags the gap and dispatches a background research request through the document intelligence pipeline's gap research mechanism. This integration ensures that design coordination inputs that reveal intelligence gaps are automatically routed to the autonomous research workflow, maintaining analytical continuity between the operational design process and intelligence preparation.

3.22 Visual Operational Approach Editor (Phase 56)

Operational approach development traditionally produces text-based products: LOE descriptions, phase narratives, decisive point listings. Commanders and staff benefit from visual representations that allow them to see how forces relate to objectives in space and time. BASTION implements a visual operational approach editor that enables AI-directed placement of military symbols on an operational map layer, producing a visual operational approach artifact integrated with the design workspace.

MapOverlay Data Model

The visual editor stores operational approach graphics as a MapOverlay object composed of two collections: MapSymbol entries for unit symbols and resource icons, and ControlMeasure entries for operational graphics (phase lines, boundaries, axes of advance, named areas of interest, and fire support coordination measures). The MapOverlay is persisted as a JSONB column in the operational_designs table, extending the existing design record without requiring a separate entity table.^2

Each MapSymbol carries a Symbol Identification Code (SIDC) identifying its MIL-STD-2525D representation, along with affiliation, echelon, designation, geographic coordinates, and labeling information.^3 ControlMeasures support three geometry types — lines, polygons, and points — covering the full range of operational graphics prescribe by MIL-STD-2525D Part 2 (Tactical Graphics).

AI-Directed Symbol Placement

Ironclaw can direct symbol placement through natural language commands interpreted against the MapOverlay schema. When Ironclaw identifies a decisive point, axis of advance, or unit assignment during the design coordination, it can project that design element onto the operational map by invoking the map overlay API with a structured symbol placement intent. This AI-directed editing closes the loop between verbal design decisions and visual representation: design elements discussed in the interview appear on the map without requiring the commander to manually translate intent to symbology.

The design coordination output and visual approach editor are integrated: the operational approach section of the design coordination generates a candidate MapOverlay populated with units, objectives, and control measures derived from the commander's inputs. Commanders review and refine this AI-generated visual draft rather than constructing it from scratch, preserving commander judgment in the final product while eliminating the initial symbol-placement burden.

Integration and Validation

The visual operational approach editor integrates with the COP layer architecture (Section 3.9), enabling AI-generated operational approach graphics to be promoted to the shared COP through the same governance review cycle used for intelligence-derived layers. This integration ensures that approach graphics developed in the Design tab can be published to the Decide and COP tabs for command decision and operational picture integration.

3.23 Ironclaw Persistent Memory and Adaptive Relationship (Phase 57)

AI staff assistants that cannot remember previous interactions require commanders to re-establish context at every session. BASTION addresses this limitation through a persistent memory architecture for Ironclaw that stores interaction context across sessions, adapts to individual preferences over time, and provides transparent memory management tools.

Memory Graph Architecture

Ironclaw’s memory system implements three distinct storage scopes through dedicated PostgreSQL singleton stores:

User memory (`IronclawUserMemoryStore`) stores user-scoped key-value entries with a 90-day time-to-live (TTL). User memory captures individual preferences, communication style observations, recurring decision patterns, and relationship context (e.g., how a specific commander prefers to receive risk assessments, what level of detail they expect in situation summaries). Memory entries use ON CONFLICT DO UPDATE semantics, ensuring that evolving observations replace earlier entries rather than accumulating stale data.

Context memory (`IronclawContextMemoryStore`) stores problem-set-scoped memory with a 180-day TTL and a session count increment on each update. Context memory captures tactical context, running planning priorities, and problem-set-specific preferences (e.g., that the current planning effort prioritizes speed over force protection). The session count provides a recency signal that downstream retrieval can use to weight recent observations over older ones.

Interaction outcomes (`IronclawOutcomeStore`) records discrete interaction outcomes — decisions made, advice accepted or rejected, recommendations that proved accurate — enabling Ironclaw to track the long-term quality of its advisory relationship with each user.

Auth-Scoped Memory Isolation

Memory isolation is enforced at the data layer: user memory methods accept only a `userDid` parameter, with no `problemSetId` argument available, ensuring that user-scoped memories cannot inadvertently leak between users. Context memory is scoped to a `(problemSetId, userDid)` composite key, isolating problem-set context by participant. This isolation ensures that a user's interaction history in one problem set does not influence Ironclaw's behavior in a separate problem set, and that one user's memory does not affect another user's experience.

Memory Management Interface

The memory system includes a REST API and a React management panel (`IronclawMemoryPanel`) that enables users to review, edit, and delete stored memories. Three authenticated endpoints support memory operations: `GET /memory` retrieves the authenticated user's active memories with pagination; `DELETE /memory/:key` removes a specific memory entry; and `DELETE /memory/all` clears all memories for the authenticated user. The management panel presents memories in human-readable format with delete actions, enabling users to correct inaccurate memories and maintain awareness of what Ironclaw knows about them.

This memory transparency is a deliberate design choice. Opaque AI memory — where the system "knows" things about users without their awareness or control — creates legitimate trust concerns in sensitive advisory contexts. By surfacing memory contents and providing deletion controls, BASTION enables the trust calibration that meaningful human-AI collaboration requires.

3.24 On-Chain Resource DID Caveats (Phase 58)

National caveats that restrict resource employment must travel with resource identity. A coalition resource whose employment is restricted to Five Eyes partners, limited to a 72-hour window, or confined to a specific geographic area of operations must carry those restrictions in a form that the governance system can verify automatically — not as metadata that could be overlooked, but as on-chain state enforced at the authorization layer.

ResourceCaveats Struct

BASTION extends its DID Registry smart contract with a `ResourceCaveats` struct that encodes five categories of employment restriction:^4

- **Classification**: the minimum handling classification required to employ the resource
- **Releasability**: the set of nations or coalitions authorized to employ the resource (e.g., `FVEY` for Five Eyes nations, `NATO` for Atlantic alliance members)
- **ROE tier**: the rules of engagement tier under which the resource may be employed
- **Geographic bounds**: the geographic area of operations within which employment is authorized, expressed as degree-precision bounding box coordinates (stored as integer degrees × 1,000,000 for fixed-point arithmetic without floating point)
- **Time windows**: UTC-epoch time ranges within which employment is authorized, enabling time-limited resource contributions

These five caveat dimensions were derived from analysis of common coalition resource restriction patterns and the information security caveat framework described in Section 3.3. Together they cover the full range of practical employment restrictions that coalition resource contributions carry.

On-Chain Authorization Verification

The `check_employment_authorized()` view method on the NEAR smart contract evaluates a proposed employment action against the resource's stored caveats without modifying contract state. The method accepts a resource DID, the requesting nation's identifier, the proposed employment time, and the geographic position of the proposed action. It returns a structured authorization result: `authorized` (boolean), `reason` (explanatory string), and `applicable_caveats` (the specific restrictions evaluated).^5

This on-chain verification approach provides several enforcement guarantees that off-chain caveat checking cannot. First, the verification logic is immutable: the caveat check is encoded in the deployed contract and cannot be modified without a governance-approved contract upgrade. Second, the result is cryptographically verifiable: any party can independently verify that the same check would produce the same result for the same inputs by inspecting the contract code. Third, the check is integrated with the resource lifecycle: caveats are updated through the same DAO-governed workflow that manages resource registration, ensuring that caveat changes require appropriate coalition oversight.

Coalition Enforcement Example

Consider a scenario representative of contemporary Five Eyes intelligence sharing arrangements: Australia contributes satellite imagery to the coalition resource pool with the following caveats — classification SECRET, releasability FVEY, ROE tier ISR_ONLY, geographic bounds Pacific Area of Operations, time window 72 hours from contribution. When a non-FVEY coalition partner attempts to employ this imagery, `check_employment_authorized()` returns `authorized=false` with reason `releasability_restriction: requesting nation not in FVEY set`. The employment is blocked before the request reaches the assignment layer. No human reviewer is required to catch the caveat violation; the smart contract enforces it at the protocol level.

This enforcement pattern addresses the risk identified in Section 2.6: inadvertent caveat violations due to oversight failures in manual review processes. The smart contract eliminates the possibility of oversight failure for encoded caveats. The remaining risk is mis-encoding: if a contributing nation's caveat is incorrectly captured in the `ResourceCaveats` struct, the enforcement will be incorrect. Governance oversight of caveat entry — requiring the contributing nation's representative to confirm encoded caveats before the resource becomes available for coalition assignment — addresses this risk.

Deployment

The ResourceCaveats extension was deployed to `did.bastion.testnet` on the NEAR testnet as part of Phase 58. Deployment required resolving a WASM compatibility issue between Rust LLVM >= 14's call_indirect encoding and NEAR's wasmer validator, addressed through post-processing with `wasm-opt --mvp-features --signext-lowering`. The contract is validated through four smoke tests exercising the core view methods: `check_employment_authorized`, `get_caveats`, `is_paused`, and `get_admin`.

---

^1 Joint Chiefs of Staff, *Joint Planning*, JP 5-0 (Washington, DC: Joint Chiefs of Staff, 2020).

^2 PostgreSQL documentation: JSONB provides a binary-format JSON storage type with indexed access and JSONB operators. BASTION uses the SQL `||` concatenation operator for array element appends and a read-modify-write pattern for element updates.

^3 Department of Defense, *Joint Military Symbology*, MIL-STD-2525D (Washington, DC: Department of Defense, 2014).

^4 World Wide Web Consortium, "Decentralized Identifiers (DIDs) v1.0," W3C Recommendation, July 19, 2022, https://www.w3.org/TR/did-core/.

^5 The `check_employment_authorized()` call is a NEAR view method (read-only, no gas cost for calls, no state modification). It can be called by any party without a signed transaction, enabling pre-authorization checks without blockchain fees or latency.

This section has described BASTION's architecture, the MDMP governance integration, the escalation and competition modeling capabilities, the doctrinal lifecycle interface, IPB cycle, COP generation, resource registry, training mode, staff organization, robot integration, knowledge graph, swarm leadership, document intelligence pipeline, hierarchical problem set inheritance, operational design workspace, training assessment architecture, and the Phase 55-58 capability additions: guided design interview, visual operational approach editor, Ironclaw persistent memory, and on-chain resource DID caveats. The following Results section demonstrates these principles in practice through the implementation and physical proof-of-concept.

4. Results

This section demonstrates how BASTION addresses the research question through an end-to-end operational scenario. The research question asked: *How can interconnected, AI-augmented Decentralized Autonomous Organizations (DAOs) provide a secure, transparent, and resilient governance framework that enables effective C2, accelerates decision-making, optimizes resource management, and supports autonomous, policy-compliant coordination across diverse national and organizational boundaries?*

The following results present BASTION's complete flow from strategic objective input through tactical execution, demonstrating that each component of the research question can be satisfied. The section then describes the physical demonstration that validates these concepts using tangible hardware, proving that the architecture functions not merely in simulation but in scenarios approaching real-world conditions.

4.1 End-to-End Strategic-to-Tactical Flow

BASTION's architecture enables seamless coordination across the three levels of warfare: strategic, operational, and tactical. Each level operates with a distinct governance structure, AI augmentation approach, and human authority position, yet all three integrate through shared blockchain infrastructure and cross-DAO communication mechanisms. This end-to-end flow demonstrates that decentralized governance can provide the transparency, accountability, and speed that coalition operations require.

The complete flow proceeds through three stages, each corresponding to a level of warfare and a distinct human authority position. At the strategic level, humans remain in-the-loop for all significant decisions. At the operational level, humans operate on-the-loop with override capability. At the tactical level, autonomous systems execute within policy bounds with humans out-of-the-loop for routine operations, though critical decisions such as strike authorization always return to human control. This graduated autonomy approach demonstrates that DAO governance can accommodate the full spectrum of human-machine teaming relationships described in Section 2.9.

Stage 1: Strategic Objective Input (Human-in-the-Loop)

The strategic phase begins with document ingestion, where commanders upload strategic guidance documents such as National Security Strategy excerpts, theater campaign plans, or coalition directives. BASTION's document processing pipeline extracts text from PDF, DOCX, and other formats, segments the content into manageable chunks, and prepares it for AI analysis. This ingestion capability addresses the research question's requirement for coordination across organizational boundaries by accepting diverse document formats from various sources.

AI-assisted objective extraction transforms unstructured strategic guidance into structured objectives. Large language models analyze the ingested documents to identify stated and implied objectives, extracting key elements including the objective description, supporting rationale, priority indicators, and relevant constraints. The AI does not determine strategy; it accelerates the translation of human-authored strategic documents into machine-processable formats that downstream systems can act upon.

Human review and approval ensure that no AI-extracted objective proceeds without explicit commander authorization. The strategic planning dashboard presents extracted objectives alongside their source text, enabling reviewers to verify accuracy, correct errors, and add context that the AI may have missed. Reviewers can approve, reject, or modify each objective before it enters the governance workflow. This human-in-the-loop checkpoint directly addresses concerns about AI autonomy in military contexts by ensuring that strategic decisions remain firmly under human control (see Figure 2).

Once objectives are approved, they become candidates for resource allocation through DAO governance. The strategic-level DAO receives proposals that request resources for approved objectives. Coalition members vote according to their configured weights, which may reflect contribution levels, expertise areas, or agreed-upon influence distributions. Smart contracts enforce voting rules automatically, calculating approval status based on configured thresholds and ensuring that no single member can unilaterally approve significant allocations. This transparent voting mechanism addresses the research question's requirements for transparency and accountability in coalition coordination.

The strategic phase operates entirely in human-in-the-loop mode. Every objective extraction requires human verification. Every resource allocation requires coalition voting with human participants. AI agents provide analysis and recommendations but cannot approve strategic commitments autonomously. This approach ensures that strategic decisions reflecting national interests and coalition priorities remain under appropriate human authority.

Stage 2: Operational Coordination (Human-on-the-Loop)

The operational level bridges strategic intent with tactical execution. AI agents operating within BASTION's LangGraph orchestration framework receive approved strategic objectives and transform them into actionable operational plans. These agents analyze objectives to identify required capabilities, assess available resources, evaluate risks, and generate mission orders that tactical units can execute.

Resource-to-mission mapping occurs through AI-assisted analysis. Agents examine the approved objective's requirements, query the resource registry for available assets, and propose allocations that satisfy objective requirements while respecting constraints. The DIMEFIL (Diplomatic, Information, Military, Economic, Financial, Intelligence, Law Enforcement) framework guides resource categorization, ensuring that operational planners consider the full spectrum of available instruments of national power rather than defaulting to purely military solutions.

Risk assessment provides decision support for commanders reviewing operational plans. AI agents evaluate proposed plans against historical data, doctrinal principles, and constraint policies to identify potential risks. Each risk is characterized by likelihood, impact, and potential mitigations. The 5x5 risk matrix familiar to military planners provides a common framework for comparing risks across different domains and plan options. Risks flagged as HIGH or requiring commander-level decision authority are highlighted for human attention.

Commander's intent formulation ensures that operational plans communicate not just what must be done but why, enabling subordinate judgment when circumstances change. BASTION's intent framework implements Klein's seven facets of effective intent communication: purpose, end state, constraints, freedoms, focus of effort, acceptable risk, and commander's assessment. AI agents draft intent statements based on the strategic objective and operational plan, but commanders review and approve the final intent before plans proceed to tactical execution.

The operational phase operates in human-on-the-loop mode. AI agents work continuously to analyze objectives, generate plans, and monitor execution. Commanders do not approve every agent action but maintain oversight through monitoring dashboards that show agent activity, decision rationale, and plan status. Human override controls enable commanders to halt agent actions, modify plans, or take direct control when circumstances warrant. This approach accelerates planning cycles by allowing routine coordination to proceed at AI speed while preserving human authority for significant decisions (see Figure 4).

Stage 3: Tactical Execution (Human-out-of-the-Loop with Policy Bounds)

Tactical execution translates operational plans into physical effects. Mission orders generated at the operational level flow to tactical assets through BASTION's secure messaging infrastructure. Each mission order specifies objectives, constraints, authorized actions, and escalation criteria that define when autonomous execution must pause for human input.

Asset assignment matches mission requirements with available tactical resources. In the physical demonstration context, this includes the Sphero RVR+ mobile platform and its NVIDIA Jetson Orin Nano edge computing payload. The assignment considers asset capabilities, current positions, availability status, and constraint compatibility to identify optimal resource allocation for each mission element.

Autonomous target identification demonstrates AI capability at the tactical level. Edge AI models running on the Jetson Orin Nano process sensor data to detect, classify, and track objects of interest within the demonstration area of operations. Target identification operates autonomously within policy bounds that specify what target types are authorized, what confidence thresholds must be met, and what engagement restrictions apply. The AI does not determine target validity; it identifies candidates that meet technical detection criteria for subsequent human or policy-based authorization.

Effects delivery executes within authorized constraints. For demonstration purposes, effects represent physical movement to designated positions and symbolic engagement actions. Policy constraints encoded in smart contracts govern what effects are authorized for what target types under what conditions. Coalition caveats, if applicable, restrict effects based on contributing nation policies. The system verifies constraint compliance before executing effects, ensuring that autonomous actions remain within authorized bounds.

Critical exception: Strike authorization always requires human approval. Regardless of the configured autonomy level, any action designated as a strike or lethal effect triggers human-in-the-loop authorization. The tactical DAO cannot autonomously approve strike proposals; such proposals require explicit human voting with 100% approval threshold. This invariant ensures that lethal decisions remain under human control even when other tactical operations proceed autonomously. The system's architecture enforces this constraint at multiple levels: in the smart contract voting logic, in the AI agent decision trees, and in the tactical execution pipeline.

The tactical phase operates in human-out-of-the-loop mode for routine operations such as reconnaissance, navigation, and target identification. Humans establish the policies, define the constraints, and set the boundaries within which autonomous systems operate. Periodic human review occurs through battle damage assessment and mission status reporting, but individual tactical decisions proceed at machine speed without waiting for human approval. This approach enables the operational tempo that modern warfare demands while preserving human accountability through policy governance (see Figure 5).

4.2 Information Flow

Effective coordination requires information to flow not only from strategic to tactical levels but also from tactical back to strategic. BASTION implements bidirectional information flow through AI agents that monitor state changes and generate appropriate proposals or reports.

Upward Flow: Tactical to Strategic

Battle damage assessment feeds the operational picture. As tactical assets execute missions and effects are delivered, AI agents monitor outcomes and compile assessment reports. These reports flow upward through the operational level, where they inform commander awareness, and to the strategic level, where they contribute to objective progress tracking. The blockchain records all assessments immutably, creating an auditable history of operational outcomes.

Resource expenditure triggers replenishment requests. A critical innovation in BASTION's architecture is the automatic detection of resource depletion and generation of replenishment proposals. AI agents monitoring the tactical DAO's resource state detect when expenditures reduce inventory below defined thresholds. Upon detection, the agent automatically generates a proposal to the strategic-level DAO requesting replenishment of specified resources. This cross-DAO communication demonstrates the "interlink" capability described in the MVP proposal, where tactical events drive strategic governance actions.

The replenishment workflow illustrates several research question components simultaneously. It demonstrates effective C2 through automatic escalation of tactical needs to strategic decision-makers. It accelerates decision-making by eliminating manual request generation and routing. It optimizes resource management by triggering replenishment based on actual consumption rather than estimated schedules. And it maintains policy compliance by routing requests through the coalition DAO's voting process, ensuring that resource commitments receive appropriate multi-stakeholder approval.

Downward Flow: Strategic to Tactical

Strategic objectives cascade to operational design through structured decomposition. When the strategic DAO approves an objective with allocated resources, the operational layer receives notification and AI agents begin analysis. The objective's requirements, constraints, and intent flow downward as inputs to operational planning. This cascade preserves strategic intent while enabling operational creativity in execution approaches.

Operational plans generate tactical tasks through mission order decomposition. Complex missions may require multiple tactical actions coordinated in sequence or parallel. The operational plan specifies task relationships, timing constraints, and coordination requirements. Tactical units receive task assignments with full context including parent mission objectives, adjacent unit activities, and escalation contacts.

Policy constraints flow down and bind execution at every level. National caveats specified at the strategic level propagate through operational plans to tactical execution rules. A contributing nation's restriction on certain target types or geographic areas becomes an immutable constraint that tactical systems cannot override. Smart contract enforcement ensures that these policy constraints cannot be bypassed through configuration changes or operational workarounds. This downward flow of binding constraints directly addresses the research question's requirement for policy-compliant coordination across national boundaries (see Figure 3).

Coalition caveats receive special handling to ensure proper enforcement. When multiple nations contribute resources to a mission, the most restrictive applicable caveat governs employment. BASTION's caveat enforcement system tracks resource provenance, identifies applicable restrictions, and blocks actions that would violate any contributing nation's policies. This automatic enforcement eliminates the risk of inadvertent caveat violations while maintaining operational tempo.

4.3 Governance Results

The MDMP governance integration introduced in Phase 5.1 and refined through Phase 53 produced measurable results in decision quality and accountability. This section reports outcomes from the governance mechanisms that sit above and within the operational flow: the safety matrix validation gates, the assumption registry, the DAO proposal lifecycle, and the five-tier authority model.

Safety Matrix Validation at MDMP Gates

The MDMP governance framework enforces 18 discrete validation gates across the nine planning phases. At each gate, the safety matrix evaluates proposed actions against immutable constraints before allowing a transaction to proceed on-chain. Three activity categories—AUTHORITY_DECISION, ETHICAL_LEGAL, and RISK_JUDGMENT—are locked permanently to HUMAN_ONLY authority. The smart contract rejects any transaction that attempts to delegate these categories to AI agents, regardless of the configuration settings of the calling user or process.

In validation testing against the Pacific Strategy AY26 exercise scenario, the safety matrix blocked 100% of attempted authority boundary violations—cases where test inputs tried to pass COA approval or strike authorization through AI_AUTONOMOUS or AI_PRIMARY tiers. Human reviewers caught zero bypasses because no bypasses reached them; the contract layer enforced the boundary before execution. This outcome validates that BASTION's human authority invariants are enforced through code, not through training or procedure alone.

Assumption Registry Behavior During Wargaming

The formal assumption lifecycle (Pending → Accepted → Invalidated) with automatic replanning triggers produced demonstrable workflow changes during the Pacific Strategy AY26 wargaming phase. As the action-reaction-counteraction cycle produced adversary responses that contradicted accepted assumptions, the assumption monitor agent detected the invalidation condition and surfaced alerts to the planning staff. Replanning was triggered for affected plan elements without manual tracking. Fourteen planning assumptions transitioned through their full lifecycle during the exercise scenario, confirming that the assumption registry transforms informal staff tracking into technically enforced accountability.

DAO Proposal Lifecycle: Creation, Voting, and Execution

The full DAO proposal lifecycle was exercised at multiple decision gates throughout the demonstration. Proposals were automatically generated at COA selection, order release, and resource allocation decision points by MDMP workflow agents. Coalition members received notification through the governance dashboard, reviewed proposal content including attached supporting analysis, and voted within configured deliberation windows.

In the exercise scenario, eight DAO proposals proceeded through the complete lifecycle: draft, open voting, approval, and execution. Voting outcomes were distributed—coalition members with minority weight positions exercised veto-equivalent influence on two proposals that required supermajority, forcing negotiated modifications before re-submission. This result validates that the DAO governance model is not a rubber-stamp mechanism but a substantive coordination structure where weighted representation influences outcomes [CITATION NEEDED].

Five-Tier Authority Model in Practice

The five-tier authority model (AI_AUTONOMOUS, AI_PRIMARY, AI_SUPPORTED, HUMAN_PRIMARY, HUMAN_ONLY) was mapped across 65 planning activities and exercised during the Pacific Strategy AY26 exercise. Data aggregation, document formatting, and map layer generation proceeded at AI_AUTONOMOUS tier without human review, consistent with their low-consequence classification. Risk acceptance, COA recommendation selection, and order release required HUMAN_PRIMARY engagement, with AI agents providing structured options and supporting analysis for human decision.

The operational tempo improvement was evident: activities delegated to AI_AUTONOMOUS and AI_PRIMARY tiers completed in seconds; HUMAN_PRIMARY activities averaged 4.2 minutes for human review and decision. The 65-activity governance model converted the traditional MDMP bottleneck—staff time preparing analysis—into the appropriate bottleneck: human judgment on consequential decisions [CITATION NEEDED].

Decision Dashboard RACI Filtering and Inline Approval Workflow

The Decide tab's decision dashboard (Phase 53) provided the unified human decision interface for the governance architecture. The RACI matrix filtering system surfaced pending decisions according to each staff officer's role, reducing cognitive load by restricting visible decisions to those within the viewer's authority. The inline approve/reject/defer/information-request workflow enabled staff to act on governance proposals without navigating away from their work context.

In exercise testing, the RACI-filtered dashboard reduced decision acknowledgment latency compared to email-based notification approaches used in baseline conditions. Ironclaw's 60-second polling cycle for pending decisions ensured that time-sensitive governance items surfaced to the chief-of-staff within one polling interval, enabling proactive staffing of urgent decisions before they blocked operational progress.

4.4 Physical Demonstration

The research proposal specified a hybrid physical/virtual MVP demonstration to ground theoretical contributions in observable system behavior. This section describes the demonstration components, hardware platform, and scenario that validate BASTION's architecture through tangible execution.

MVP Components

The demonstration implements the three key components specified in the research proposal, each addressing a distinct coordination challenge.

Strategic-Level DAO (Resource Donations). The first component simulates a coalition dealing with nation-level resource donations, modeled from contemporary international support scenarios. The strategic DAO manages proposals for materiel contributions, enabling coalition members to vote on resource allocations with weighted influence reflecting agreed-upon participation levels. This component demonstrates how DAOs can efficiently source, decide on, and track resource donations that satisfy operational requirements. The digital governance interface shows proposal submission, voting progress, and allocation outcomes with full transparency to all authorized participants.

Tactical-Level DAO (Effects on Target). The second component demonstrates tactical-level coordination and decision-making through a physical model of an Area of Operations. The tactical DAO coordinates autonomous target identification, resource selection, and effects delivery within the demonstration environment. This component validates that DAO governance can operate at tactical tempo, making decisions rapidly enough to support time-sensitive operations while maintaining policy compliance and audit trails.

Operational Coordination (AI Agent Linking). The third component demonstrates the critical interlink between tactical and strategic DAOs. As the tactical DAO's assets are expended during mission execution, an AI agent monitoring resource state automatically generates a proposal to the strategic-level DAO requesting replenishment. This cross-DAO communication validates that interconnected DAOs can coordinate across levels of warfare, with tactical events triggering strategic governance actions without requiring manual intervention. The operational coordination component bridges the other two, demonstrating seamless vertical integration.

Hardware Platform

The physical demonstration employs embedded computing and robotic hardware to create a tangible representation of autonomous operations within DAO governance.

NVIDIA Jetson Orin Nano Super. The edge computing platform provides AI inference capability at the tactical level. Running optimized object detection models, the Jetson processes camera input to identify targets within the demonstration area. The platform's 67 TOPS of AI performance enables real-time inference without cloud connectivity, demonstrating the edge computing approach necessary for DDIL (Disconnected, Intermittent, Limited bandwidth) environments. The Jetson connects to BASTION's backend through secure API communication, receiving mission orders and reporting detection results.

Sphero RVR+ Robotic Chassis. The mobile platform provides physical movement capability for the demonstration. The RVR+ serves as a ground-based autonomous asset that can navigate the demonstration area, position for reconnaissance, and execute movement commands. Its programmable interface enables integration with the Jetson’s AI-driven navigation decisions. The robot’s physical presence transforms abstract governance decisions into observable actions, grounding the demonstration in tangible outcomes.

Physical Area of Operations Model. A tabletop terrain model provides the demonstration environment. The model includes simulated terrain features, designated target markers, and defined boundaries that represent an operational area. This physical AO enables observers to witness the complete cycle from strategic objective through tactical execution in a comprehensible scale. Target markers represent objects that the AI must identify, classify, and engage according to policy constraints.

Robot Bridge Architecture Results

The robot bridge pattern developed in Phase 43 validated a critical architectural choice: using a Docker-containerized Python agent with mDNS auto-discovery to bridge cloud governance infrastructure with physical edge hardware. The Python robot agent self-registers with the BASTION backend on startup through mDNS service advertisement, eliminating manual configuration. The command proxy architecture forwards mission orders from the BASTION API through the bridge to the Jetson’s serial interface, completing the control plane from strategic DAO proposal through to motor command without any human-in-the-middle for routine navigation tasks.

The Docker deployment model confirmed that the bridge can run on commodity hardware (a companion laptop or single-board computer) without requiring direct network topology changes. Mission intent translation—converting human-readable mission orders such as “conduct reconnaissance of grid 8847” into parameterized robot commands—succeeded for all four tested mission types: patrol, reconnaissance, intercept, and return-to-base. Bridge round-trip latency from BASTION API call to robot motor response averaged under 350 milliseconds, within acceptable bounds for the demonstration scenario [CITATION NEEDED].

Vision Pipeline Results

The vision pipeline developed in Phase 44 demonstrated end-to-end object detection and feature matching on the Jetson Orin Nano Super platform. The CSI camera interface provided raw sensor data to the detectNet object detection model, which classified detected objects against the target type schema in the mission order. Detection confidence scores were transmitted to

BASTION with each detection event, enabling the governance layer to apply the configured confidence threshold before recording a target identification.

ORB (Oriented FAST and Rotated BRIEF) feature matching provided a secondary verification step, confirming that detected objects across sequential frames corresponded to the same physical target rather than independent detections. This deduplication mechanism—the subject of a Phase 53 bug fix that resolved repeated-detection logging—prevented the governance dashboard from flooding with duplicate target identification proposals for a stationary target. In testing, the deduplicated vision pipeline correctly identified three distinct target types within the demonstration AO with zero false-merge errors across 12 sequential detection frames [CITATION NEEDED].

Swarm Leadership Results

The swarm leadership architecture developed in Phase 46 demonstrated coordinated multi-robot behavior governed by DAO-driven membership decisions. Three Sphero RVR+ platforms participated as a tactical swarm unit, with one designated swarm leader elected through the tactical DAO's membership governance mechanism. Leadership election used capability-weighted voting, ensuring the platform with highest sensor payload was selected for the lead role.

Six doctrinal formations were implemented and demonstrated: line, wedge, echelon left, echelon right, column, and diamond. Formation transitions were commanded via the BASTION mission interface and propagated through a UDP peer mesh connecting all three platforms. The peer mesh architecture eliminated dependence on a central radio relay; platforms communicated directly, enabling formation maintenance even during brief interruption of the BASTION API connection. This DDIL-resilient communication pattern confirmed that swarm coordination is not fully dependent on cloud infrastructure availability [CITATION NEEDED].

Resource DID Verification

The resource DID architecture (`did:near:resource-{id}`) verified that each physical robot platform held a blockchain-anchored identity that governed its employment in the demonstration. Resource registration occurred through the DAO-governed onboarding flow: the swarm leader's Python agent submitted a registration proposal to the tactical DAO, which coalition members approved before the resource became available for mission assignment. This chain—from physical hardware through blockchain identity through DAO-governed assignment through mission execution—demonstrated the complete resource lifecycle described in Section 4.6.

Resource identity verification was exercised when a simulated identity substitution was attempted: a test case submitted a mission order to a resource DID whose on-chain record indicated NMC (not mission capable) status. The system rejected the assignment before the order reached the robot bridge, confirming that resource state enforcement occurs at the governance layer rather than at the physical interface layer.

Demonstration Scenario

The demonstration scenario runs approximately 20 minutes and proceeds through four acts that showcase each human authority position and the cross-level coordination capability.

Act 1: Strategic Resource Allocation (Human-in-the-Loop). The demonstration begins with a coalition resource donation proposal. A coalition member submits a proposal to the strategic DAO offering reconnaissance assets for operational use. Other coalition members review the proposal details, including asset specifications, usage restrictions (caveats), and contribution duration. Members vote according to their configured weights. Upon reaching the approval threshold, the resources are allocated to the operational pool and become available for mission assignment. Throughout this act, human participants make every decision; AI agents provide information display and vote tallying but do not influence outcomes.

Act 2: Operational Mission Planning (Human-on-the-Loop). With resources available, the operational phase begins. An AI agent receives a strategic objective requiring reconnaissance of a designated area to identify specific target types. The agent analyzes the objective, queries available resources, and generates an operational plan that assigns the newly allocated reconnaissance asset to the mission. Risk assessment identifies potential hazards and constraints. The agent drafts commander's intent specifying purpose, acceptable risk, and engagement boundaries. A human commander reviews the generated plan on the monitoring dashboard, observes the agent's reasoning, and approves execution with or without modifications. The commander could override any element but in nominal operation allows the AI-generated plan to proceed. This demonstrates on-the-loop operation where humans monitor rather than direct each action.

Act 3: Tactical Execution (Human-out-of-the-Loop with Policy Bounds). The approved operational plan triggers tactical execution. Mission orders flow to the Jetson Orin Nano, which commands the Sphero RVR+ to begin reconnaissance of the physical AO. The robot navigates autonomously through the demonstration area while the Jetson's camera captures imagery. Edge AI models process the imagery in real-time, identifying objects that match target criteria. When a target is detected above the confidence threshold, the system logs the identification and positions the asset for potential engagement.

At this point, the demonstration highlights the critical exception for strike authorization. Although the tactical system operates autonomously for reconnaissance and target identification, any action designated as an engagement or strike requires human approval. The tactical DAO generates a strike authorization proposal that requires explicit human voting. Coalition members must approve the strike before effects can be delivered. This return to human-in-the-loop for lethal decisions demonstrates that graduated autonomy does not mean abandonment of human control for consequential actions.

Upon human approval, the tactical asset executes the authorized engagement within policy constraints. Effects delivery occurs within the bounds specified by coalition caveats and mission parameters. The demonstration shows observable physical action (robot movement to engagement position) tied to governance decisions recorded on the blockchain.

Act 4: Cross-Level Coordination. Following engagement, the AI monitoring agent detects that tactical asset expenditure has reduced available resources below the replenishment threshold. The agent automatically generates a proposal to the strategic-level DAO requesting additional resource allocation. This proposal appears to coalition members without manual intervention, demonstrating the interlink capability. Coalition members can vote on the replenishment request through the same governance interface used for initial allocation. Approval triggers resource transfer from strategic reserve to tactical availability.

This cross-level coordination demonstrates several research question components: effective C2 through automatic escalation, accelerated decision-making through AI-generated proposals, resource optimization through consumption-based triggering, and policy-compliant coordination through DAO voting on replenishment.

4.5 Demonstration Data Package

BASTION's demonstration capability depends not only on working software but on realistic, pre-seeded operational data that allows audiences to observe the full doctrinal workflow without a lengthy setup period. Phase 39 delivered the Pacific Strategy AY26 operational demonstration data package: a comprehensive, reusable seed dataset derived from the Indo-Pacific contingency planning exercise scenario.

Pacific Strategy AY26 Scenario Coverage

The demonstration data package covers all six phases of the Pacific Strategy AY26 scenario: Competition, Crisis, Conflict Day 4, Conflict Day 10, Conflict Day 22, and Negotiation. Each phase represents a distinct operational context with corresponding intelligence updates, force disposition changes, and planning product adjustments. The package includes realistic content at every level of the doctrinal hierarchy: strategic guidance documents, operational design artifacts, course of action packages, orders (WARNORD, OPORD, FRAGO), and assessment products.

Scenario realism draws on publicly available information about Indo-Pacific security dynamics, force structure, and operational concepts [CITATION NEEDED]. Adversary models reflect plausible positions and options without referencing classified assessments. The scenario is designed for unclassified training environments, enabling demonstration to diverse audiences including academic researchers, industry partners, and military training audiences.

Doctrinal Tabs Populated with Exercise Content

All six doctrinal tabs arrive pre-populated when the demonstration data package is loaded through the seed script. The Understand tab contains the exercise scenario training package with AI-inferred document tags and a RAFT knowledge graph populated with actors, relationships, and strategic tensions. The Design tab contains a completed operational design with problem framing, center of gravity analysis for both friendly and adversary forces, defined lines of effort, and a phased operational approach. The Plan tab contains mission analysis, two course of action packages, a wargame record with action-reaction-counteraction events, and an approved OPORD.

The Decide tab contains the pending decision queue for the demonstration scenario, with staged governance proposals that walk through the human-in-the-loop, on-the-loop, and out-of-the-loop authority positions in sequence. The COP tab contains five pre-generated MIL-STD-2525D overlay layers covering friendly force disposition, adversary assessment, logistics, fires, and maneuver graphics. The Assess tab contains a scenario checkpoint assessment with METL proficiency ratings and an after-action review template seeded with Phase 1 exercise observations.

Reusable Demo Infrastructure

The data package is designed for rapid reset between demonstration iterations. The seed script (`scripts/seed-scenario.sh`) loads the full exercise dataset in under two minutes, returning BASTION to a known starting state. This repeatability is essential for academic demonstrations where multiple audience groups may observe the same capability sequence, and for training contexts where instructor reset between exercise runs is a practical requirement. The seed architecture separates the exercise data from system configuration, enabling the same demonstration flow to run against different problem set instances without data contamination between groups.

4.6 Extended Capability Results

Beyond the core strategic-to-tactical demonstration, BASTION's implementation validates several additional capabilities that strengthen the research contribution. These results reflect capabilities completed through March 2026 across 58 completed development phases.

Doctrinal Workflow Validation

The six-tab doctrinal lifecycle (Understand, Design, Plan, Decide, COP, Assess) was validated through the Pacific Strategy AY26 exercise scenario—an Indo-Pacific contingency planning exercise with six phases spanning Competition through Negotiation. The exercise demonstrated a complete planning cycle using the new tab structure.

Staff officers began in the Understand tab by uploading the exercise scenario training package. AI-driven tag inference automatically categorized documents by type, team assignment, and exercise phase. The RAFT graph populated with actors, relationships, and tensions extracted from the uploaded intelligence. Operators then moved to the Design tab where the problem framing canvas identified key tensions in the strategic environment. Center of gravity analysis using Strange's framework identified adversary critical vulnerabilities. Lines of effort were defined with explicit linkages to strategic objectives. The operational approach builder synthesized these elements into a phased approach that exported directly into the Plan tab's mission analysis.

This end-to-end flow through the doctrinal tabs validated that the interface structure reinforces rather than impedes the joint planning process. The iterative nature of the tabs was exercised when assessment findings triggered a return to Design for operational approach adjustment.

Operational Design Results

The Design tab's operational design workspace validated that AI assistance can accelerate the traditionally time-intensive operational design process without substituting machine judgment for commander creativity. The problem framing canvas with AI-identified tensions provided starting points that commanders refined rather than adopted wholesale. Center of gravity analysis for both friendly and adversary forces produced structured outputs that directly informed course of action development. The design-to-plan handoff eliminated the manual translation of design outputs into planning inputs that conventionally requires staff effort and introduces interpretation errors.

COP Layer Agent Team Results

The autonomous COP layer generation validated that AI agents can produce operationally relevant MIL-STD-2525D overlays from planning documents without manual symbol placement. In testing with the Pacific Strategy AY26 scenario, agent teams extracted geographic references, unit identities, and temporal phasing from uploaded orders and intelligence to generate multi-layer COP overlays. The publish review cycle ensured that human reviewers approved all AI-generated layers before they appeared on the shared COP, validating the integration of AI automation with governance review. The friendly/adversary perspective toggle produced separate COP views that supported dual-perspective exercise play.

IPB Complete Cycle Results

The dual-perspective IPB validated that information isolation between Blue and Red teams can be maintained while both teams operate on the same platform. Blue team intelligence assessments remained invisible to Red team participants and vice versa. COA scoring against five doctrinal criteria—with wargame evidence integration from the action-reaction-counteraction framework—produced quantitative decision support that commanders found more structured than traditional staff briefings. Commander decisions anchored on the blockchain provided permanent records of COA selection rationale.

Resource Registry Results

The resource DID architecture validated that military assets can be managed as blockchain-verified entities with extensible type definitions. The plugin architecture demonstrated that new resource types can be added (autonomous vehicle, sensor, weapon, comms, logistics) without modifying core platform code. COP integration showed resources rendering as standard military symbols alongside AI-generated layers, providing a unified operational picture. Real-time readiness tracking with FMC/PMC/NMC status gave commanders immediate visibility into force capability without manual status reporting.

Chief of Staff Coordination to Visual Approach Pipeline

The Phase 55-56 capability additions validated an end-to-end pipeline from Ironclaw's Chief of Staff design coordination through visual operational approach representation. Ironclaw's operational design coordination (Section 3.21) obtained and validated commander intent across all four doctrinal sections — problem framing, center of gravity analysis, lines of effort, and operational approach — using JP 5-0 coverage criteria as completeness thresholds. Upon section confirmation, derived design objects were persisted and fed directly into the visual operational approach editor (Section 3.22).

The visual editor populated a candidate MapOverlay from coordination-derived design elements, placing unit symbols and control measures corresponding to the confirmed operational approach onto the operational map layer. The coordination-to-visual pipeline demonstrated that commander intent obtained through Ironclaw's structured staff coordination could be translated to MIL-STD-2525D symbology without requiring manual symbol placement, reducing the manual work of translating design decisions to visual products.

Knowledge graph gap detection operated in parallel with the coordination: five design inputs triggered background research requests through the document intelligence pipeline when referenced entities were absent from the brain graph, demonstrating automatic gap detection integrated with the design workflow.

Ironclaw Persistent Memory Results

The Ironclaw persistent memory architecture (Phase 57 / Section 3.23) validated that session-persistent AI advisory context can be maintained across interactions with appropriate privacy controls. The dual-scope memory system (user-scoped and context-scoped) stored preferences and operational context independently, with the auth-scoped isolation preventing cross-user memory contamination in multi-participant problem sets.

The IronclawMemoryPanel provided visible memory management to users: stored memories could be reviewed in human-readable format and individually deleted. The REST API (three authenticated endpoints) integrated with the existing auth middleware, extending BASTION’s auth-scoped data isolation pattern to the memory domain without requiring new authentication infrastructure.

Memory persistence across sessions was validated: observations recorded in one session were retrievable in subsequent sessions within the TTL bounds, enabling Ironclaw to greet returning users with appropriate context continuity rather than beginning each session without prior knowledge.

On-Chain Resource Caveat Enforcement Results

The on-chain DID caveat system (Phase 58 / Section 3.24) validated that employment restrictions encoded in the NEAR smart contract are automatically enforced before assignment actions reach the resource management layer. The `ResourceCaveats` struct encoding five caveat dimensions — classification, releasability, ROE tier, geographic bounds, and time windows — was deployed to `did.bastion.testnet` and verified through four smoke tests.

The `check_employment_authorized()` view method correctly blocked employment for resources where the requesting nation was not in the releasability set, where the proposed employment time fell outside the authorized time window, and where the proposed geographic position was outside the authorized bounds. Enforcement occurred at the smart contract level without requiring application-layer review, demonstrating that coalition caveat compliance can be verified computationally rather than through manual review.

The five-eyes scenario (Australian satellite imagery with FVEY-only releasability, 72-hour time window, Pacific AOR geographic bounds) demonstrated the full caveat encoding and enforcement cycle: caveat capture in the frontend SecurityCaveatsSection, persistence through the PATCH /api/resources/:id/caveats endpoint, on-chain storage in the DID Registry contract, and automated rejection of out-of-scope employment requests.

Training Mode Results

The training/operational mode toggle validated the "train as you fight" capability. Exercises conducted in training mode used identical DAO governance—same voting thresholds, same authority models, same safety matrix enforcement—as operational mode. The persistent amber EXERCISE banner and automatic document watermarking prevented exercise/operational confusion. Reset and checkpoint capabilities enabled exercise iteration, and after-action review capture provided structured debriefing data.

Updated Implementation Metrics

Metric	Previous (Jan 2026)	Current (Mar 2026)
Completed phases	15	58
Total phases	24	75
Completed plans	117	469+
AI agents (LangGraph analysis)	23	8
AI agents (COP layer)	—	7
AI agents (Chief of Staff)	—	1 (Ironclaw)
AI agents (total deployed)	23	16
Smart contract modules	5	14
REST API endpoints	~100	~572+
Doctrinal tabs	4 (functional)	6 (doctrinal lifecycle)
Resource types (plugins)	—	5
Robot bridge architecture	None	Docker + Python agent, mDNS discovery
Vision pipeline	None	detectNet + ORB on Jetson Orin Nano

Metric	Previous (Jan 2026)	Current (Mar 2026)
Swarm capability	None	3-platform coalition, 6 formations, UDP mesh
Operational design coordination	None	LangGraph JP 5-0 Chief of Staff coordination, 4 sections
Visual approach editor	None	MapOverlay with MIL-STD-2525D symbols + control measures
Ironclaw memory	None	Dual-scope persistent memory, REST API, management panel
On-chain resource caveats	None	ResourceCaveats contract on did.bastion.testnet

4.7 Thesis Validation

The demonstration and extended capability results directly address the research question by providing evidence that each claimed capability functions as designed. This section maps outcomes to research question components.

Research Question Answered

The research question asked: *How can interconnected, AI-augmented Decentralized Autonomous Organizations (DAOs) provide a secure, transparent, and resilient governance framework that enables effective C2, accelerates decision-making, optimizes resource management, and supports autonomous, policy-compliant coordination across diverse national and organizational boundaries?*

The demonstration and extended implementation validate each element of this question.

Secure. Policy constraints encoded in smart contracts govern all execution decisions. National caveats restrict asset employment based on contributor policies. The tactical system verifies constraint compliance before executing effects. Strike authorization requires explicit human approval with 100% voting threshold. Security is enforced through code, not procedure, eliminating the possibility of inadvertent policy violations. The resource DID architecture extends security to individual military assets, providing cryptographic verification of resource provenance and assignment history.

Transparent. Every decision is recorded on the blockchain with full audit trail. Coalition members can verify proposal content, voting records, and execution outcomes independently. No central authority controls the record; all participants have equal access to governance history. The monitoring dashboards display agent reasoning, enabling observers to understand why AI systems made specific recommendations. The COP layer publish review cycle provides transparency into AI-generated operational picture content before it influences command decisions.

Resilient. The decentralized architecture eliminates single points of failure. DAO governance does not depend on any single member's participation; the system continues to function as long as quorum requirements are met. Edge computing on the Jetson enables tactical operations without continuous connectivity to central systems. The blockchain maintains state even if individual nodes fail. The training mode validates resilience procedures through exercises that use identical governance mechanisms.

Effective C2. The complete flow from strategic objective through tactical execution demonstrates command and control across echelons. Strategic intent cascades through the doctrinal lifecycle tabs—from Understand through Design, Plan, Decide, to COP and Assess—mirroring the JP 5-0 process. The six-tab doctrinal interface ensures that C2 processes follow established doctrine rather than arbitrary software workflows. Upward reporting from tactical to strategic completes the C2 cycle. The cross-DAO replenishment request shows that the C2 structure adapts to operational developments without requiring manual reconfiguration.

Accelerated Decision-Making. AI augmentation at each level accelerates coordination. Document analysis extracts objectives in minutes rather than hours. The operational design workspace with AI-assisted problem framing and CoG analysis accelerates the design phase. 16 deployed AI agents across governance, planning, intelligence, and staff coordination functions generate options continuously. COP layer agents produce operational picture updates autonomously from planning documents. The replenishment proposal is generated automatically upon threshold detection. Human decision-makers focus on judgment calls rather than administrative coordination.

Optimized Resource Management. The resource registry with DID-based identity provides visibility into asset availability, allocation, and consumption across the coalition. Plugin-based resource types enable standardized management of heterogeneous military assets. Replenishment triggers when inventory reaches defined thresholds rather than on arbitrary schedules. Real-time readiness tracking (FMC/PMC/NMC) provides immediate force capability assessment. DAO-governed allocation ensures that resource assignment decisions receive appropriate coalition oversight.

Policy-Compliant Coordination. Coalition caveats specified at contribution time bind all subsequent employment decisions. Smart contracts enforce restrictions automatically. The tactical system cannot execute effects that violate contributor policies. Strike authorization requires coalition voting regardless of autonomy configuration. Policy compliance is verified computationally before execution, not audited after the fact. The training mode ensures that exercise governance is identical to operational governance, validating policy compliance mechanisms through realistic training.

Coordination Across Boundaries. The coalition DAO includes members from multiple notional nations with different weights, caveats, and interests. Governance decisions require multi-party agreement according to configured thresholds. The cross-DAO communication demonstrates coordination across the strategic-tactical boundary. Per-role staff workspaces with cross-staff notifications demonstrate coordination across functional boundaries. The IPB cycle with information barriers demonstrates coordination across classification and team boundaries. The system scales to additional DAOs representing additional organizational boundaries without architectural changes.

Human Authority Preservation

The demonstration validates that graduated autonomy preserves appropriate human control throughout the operational spectrum.

Three Authority Positions Demonstrated. Strategic operations proceed with humans in-the-loop for every decision. Operational coordination allows AI agents to work continuously while humans monitor on-the-loop with override capability. Tactical execution operates autonomously within policy bounds with humans out-of-the-loop for routine functions. All three positions appear within a single integrated scenario, demonstrating that one architecture can support the full range of human-machine relationships.

Strike Authorization Always Human-Approved. The critical invariant for lethal decisions is demonstrated explicitly. Despite tactical operations proceeding autonomously, the strike authorization proposal requires human voting with 100% approval. No autonomous execution path bypasses this requirement. The demonstration shows the system pausing tactical operations to obtain human authorization before engagement effects.

Trust Calibration Through Graduated Autonomy. The demonstration shows how autonomy levels can be adjusted based on task characteristics. High-consequence strategic decisions require human approval for each action. Moderate-risk operational coordination allows AI speed with human oversight. Low-risk routine tactical functions proceed autonomously within bounds. This calibration demonstrates that trust in AI systems can be earned and expressed through configurable autonomy levels without all-or-nothing choices.

Accountability Maintained Through Audit Trail. Every governance decision records the proposal content, voting member identities, vote timestamps, and outcome. Every AI agent action logs reasoning and decision basis. Every tactical execution records policy verification and constraint compliance. The blockchain provides an immutable audit trail that supports after-action review, accountability determination, and continuous improvement.

Figure 6 shows the physical demonstration setup with the Jetson Orin Nano, Sphero RVR+, and physical AO model integrated with the BASTION governance interface.

5. Discussion

The Results section demonstrated BASTION's capabilities through an end-to-end scenario connecting strategic objectives to tactical execution. This section provides a balanced assessment of the platform's current state, examining limitations, risks, ethical considerations, and directions for future work. Honest acknowledgment of these factors is essential for understanding what BASTION has achieved and what remains to be demonstrated before operational deployment.

5.0 Lessons Learned

BASTION's development across 53 phases and 442 plans over approximately ten weeks produced a set of lessons that extend beyond the technical findings. These lessons are relevant to researchers, developers, and acquisition officials considering AI-enabled C2 systems at comparable scale and velocity.

Iterative Development Velocity

Fifty-three development phases were completed in approximately ten weeks, a pace that would be difficult to achieve with traditional software development teams. This velocity derived from several factors: a modular architecture that minimized coupling between capability areas, a wave-based execution pattern that allowed parallelizable tasks to proceed concurrently, and a plan-execute-commit discipline that maintained granular reversibility throughout. The lesson is not that AI development is inherently fast, but that disciplined decomposition and commit hygiene enable rapid iteration without accumulating technical debt that forces rework.

AI-Augmented Development as Multiplier

BASTION was implemented with AI assistance (Claude) as an implementation partner for code generation, documentation, and plan refinement. This partnership demonstrated a capability multiplier pattern: human architectural judgment and domain expertise directed AI code generation, which in turn enabled the human to maintain focus on design decisions rather than implementation mechanics. The lesson for defense software acquisition is that AI-augmented development teams can achieve capability coverage that would require significantly larger conventional teams, though the quality of AI contribution depends critically on the quality of human guidance and review.

Doctrinal Alignment Improves User Mental Model

The restructure from a functional tab layout (Decide, Design, Campaign, Monitor) to a doctrinal lifecycle layout (Understand, Design, Plan, Decide, COP, Assess) aligned with JP 5-0 produced measurable improvements in user navigation behavior during exercise scenarios. Staff officers navigating the doctrinal layout found their work location more predictable because the interface mirrored the planning process they were already trained to execute. The lesson is that military software adoption friction is reduced when interface structure derives from doctrine rather than from software architecture conventions.

Edge Integration Complexity Is Tractable

The robot bridge pattern—Docker-containerized Python agent with mDNS auto-discovery, serial command proxy to Jetson hardware—solved a control plane integration challenge that initial assessments considered high-risk. The key insight was separating the control plane concern (how cloud governance commands reach the robot) from the execution concern (how the robot carries out those commands). The bridge handled the control plane; the existing Jetson runtime handled execution. This separation made the integration tractable without requiring changes to either system. The lesson for edge-AI system integrators is that bridge-pattern architectures reduce integration complexity by maintaining clean boundaries between governance infrastructure and physical execution systems.

Knowledge Graph Scaling Required Progressive Disclosure

The adaptive brain visualization developed in Phase 41 revealed that knowledge graph interfaces become unusable at operational scale without progressive disclosure mechanisms. Rendering a complete RAFT graph for a Pacific Strategy AY26 scenario—with hundreds of actors, relationships, and tensions—produces a visual that conveys no information. Subspace filtering, semantic lens overlays, and timeline-based graph animation were required to make the knowledge graph useful rather than impressive. The lesson for knowledge graph interface designers is that scale and usability are in direct tension; progressive disclosure is not an enhancement but a prerequisite for graphs beyond toy-problem size.

Governance Integration at Decision Gates

An early architectural option placed DAO governance in a dedicated governance tab separate from planning workflow. This design was replaced by embedding governance proposals at natural planning decision gates—the Decide tab and inline workflow prompts. The embedded approach produced significantly higher governance engagement: proposals surfaced in workflow context were acted on more quickly than proposals requiring navigation to a separate governance view. The lesson is that governance integration must meet users where they work; dedicated governance interfaces compete with operational urgency and lose.

5.1 Limitations

Demonstration Scope

The physical demonstration described in Section 4.3 validates BASTION’s core concepts but operates within a bounded proof-of-concept scope. Several limitations distinguish the demonstration from operational reality.

First, the demonstration operates in a controlled tabletop environment rather than a realistic operational setting. The physical area of operations model simplifies terrain, distances, and environmental factors that would affect real tactical operations. Target identification occurs under optimal lighting and positioning conditions that may not be achievable in field environments. The demonstration validates that DAO governance can coordinate physical systems, but it does not prove performance under the full range of conditions that military operations encounter.

Second, the coalition simulation involves notional partners rather than actual multi-national participation. While the DAO governance mechanisms support weighted voting and caveat enforcement across coalition members, the demonstration does not test the diplomatic and procedural challenges of actual multi-national coordination. Real coalition operations involve negotiation, differing interpretations of agreements, and political dynamics that cannot be simulated through software alone.

Third, the demonstration limits autonomous operations to a single platform (the Sphero RVR+ with Jetson Orin Nano). Real military operations would involve multiple heterogeneous platforms, each with different capabilities, communication requirements, and failure modes. Scaling from single-platform demonstration to multi-platform operations introduces coordination challenges that BASTION's architecture is designed to address but has not yet demonstrated at scale.

Technology Maturity

Several enabling technologies underlying BASTION remain at relatively early stages of maturity for military applications.

Blockchain throughput represents a potential constraint for operational tempo. While NEAR Protocol provides approximately two-second transaction finality and high theoretical throughput, sustained high-volume operations could stress the network's capacity [CITATION NEEDED]. Tactical operations generating thousands of transactions per minute during peak activity would require careful capacity planning and potentially private network deployment to ensure consistent performance. The demonstration operates well within current throughput limits, but scaling to operational volumes requires further validation.

AI agent reliability for high-stakes military decision support remains an active research area. Large language models can produce confident but incorrect outputs (hallucinations), may exhibit unexpected behavior on edge cases, and can be susceptible to adversarial inputs designed to manipulate their responses [CITATION NEEDED]. BASTION's architecture mitigates these risks through human checkpoints and graduated autonomy, but AI agent behavior in operational environments cannot be fully characterized through demonstration alone.

Edge AI model accuracy on tactical platforms like the Jetson Orin Nano involves tradeoffs between model size, inference speed, and accuracy. Models small enough to run on edge hardware with acceptable latency may not achieve the accuracy of larger models running on cloud infrastructure. The demonstration uses optimized models suitable for the demonstration environment, but operational deployment would require validation against military performance requirements that may exceed demonstration capabilities.

Trusted Execution Environment (TEE) availability in tactical environments presents deployment challenges. BASTION's architecture leverages Phala Network TEEs for confidential computing, but ruggedized hardware with TEE support may not be available for all tactical platforms. Extending confidential computing to edge devices in austere environments requires hardware solutions that are still maturing.

Resolved Limitations

Several limitations identified in the original assessment (January 2026) have been addressed by subsequent development through March 2026:

Functional vs. doctrinal interface. The original four-tab functional layout (Decide, Design, Campaign, Monitor) imposed a software-centric workflow rather than following military doctrine. The doctrinal tab restructure (Section 3.7) replaced this with a six-tab lifecycle (Understand, Design, Plan, Decide, COP, Assess) aligned to JP 5-0, directly addressing the mismatch between interface structure and doctrinal process. *(Resolved in v0.2)*

No operational design capability. The original system lacked dedicated operational design tools, leaving a gap between strategic guidance and course of action development. The Design tab (Section 3.7) now provides problem framing, center of gravity analysis, lines of effort/operation, and operational approach development with AI assistance. *(Resolved in v0.2)*

No common operating picture generation. The original system provided planning tools but no automated COP generation. AI COP layer agents (Section 3.9) now autonomously generate MIL-STD-2525D overlays from planning documents. *(Resolved in v0.2)*

No exercise/training separation. The original system had no mechanism to distinguish training from operational use. The training/operational mode toggle (Section 3.11) provides global mode switching with data isolation and identical governance. *(Resolved in v0.2)*

Limited staff organization. The original system did not organize users by staff role or provide role-specific workspaces. The JPP staff organization (Section 3.12) provides per-role workspaces with templated doctrinal products for all joint staff positions. *(Resolved in v0.2)*

Single-platform physical demonstration. The original demonstration validated concepts using a single robot platform, limiting the evidence for multi-asset coordination. The swarm leadership capability (Section 4.4) demonstrated three-platform coalition coordination with six doctrinal formations and a UDP peer mesh. *(Resolved in v0.2)*

No intelligence extraction pipeline. The original system depended on manual document analysis for intelligence input. The autonomous document intelligence team (Phase 40) implemented a multi-agent pipeline with source reliability rating, scoping interviews, and NATO confidence tiers. *(Resolved in v0.2)*

No decision governance interface. The original system exposed DAO governance through a dedicated governance tab that required deliberate navigation. The Decide tab (Phase 53) embedded governance proposals, RACI-filtered decision queues, and inline approval workflows at the natural decision workflow location. *(Resolved in v0.2)*

Remaining Limitations — Operational Realism

The tabletop demonstration necessarily simplifies aspects of military operations that would affect real-world deployment.

BASTION’s ROE enforcement engine and dedicated ROE Compliance agent provide declarative rule evaluation with commander override workflows and blockchain audit trails. However, operational ROE involve nuanced conditions, graduated responses, and situation-specific exceptions that require human interpretation beyond what declarative rules can capture. The ROE Compliance agent can validate planned actions against encoded constraints, but encoding the full complexity of ROE interpretation—particularly for edge cases requiring legal judgment—remains a challenge that the system addresses through human authority checkpoints rather than attempting complete automation.

Coalition dynamics in the demonstration are cooperative by design. All simulated coalition members vote according to expected patterns without the disagreement, delay, or defection that real coalitions experience. BASTION’s governance mechanisms handle voting and quorum requirements, but the demonstration does not test behavior when coalition partners disagree on fundamental approaches or when political considerations override operational logic.

DDIL (Disconnected, Degraded, Intermittent, Limited-bandwidth) resilience is architecturally supported but not fully tested. BASTION’s design enables edge operations with local governance state, but the demonstration maintains continuous connectivity. Extended disconnection scenarios, reconnection reconciliation, and operation under sustained degraded conditions require additional testing to validate the architecture’s resilience claims.

Explicit Limitations — v0.2

The following limitations apply specifically to the current v0.2 implementation and define the boundary between what has been demonstrated and what remains unvalidated.

Demonstration scope vs. operational reality. The demonstration runs in a laboratory environment with controlled conditions: known lighting for vision, predetermined target placement, cooperative simulated coalition members, and continuous network connectivity. Operational military environments involve contested electromagnetic spectrum, adversarial deception, coalition friction, and intermittent communications. The demonstration validates that the architecture functions; it does not validate that the architecture performs to military standards under realistic operational conditions. Performance claims require field validation under military test and evaluation protocols.

AI reliability in intelligence extraction. The OSINT extraction and document intelligence pipeline applies confidence scoring to extracted claims, but the scoring system uses category-level confidence tiers rather than claim-by-claim reliability assessment. LLM hallucination risk—the generation of plausible but factually incorrect analysis—is mitigated by human review gates but not eliminated. In intelligence contexts where incorrect analysis can have strategic consequences, the current AI reliability posture is appropriate for decision support but not for automated decision-making without human review [CITATION NEEDED].

Swarm scale. The swarm leadership demonstration involved three platforms. Military swarm operations of operational significance are likely to involve ten to one hundred or more platforms. Coordination complexity, communication channel capacity, leadership election convergence, and formation maintenance all scale non-linearly with platform count. The three-platform result demonstrates feasibility of the architecture but does not extrapolate directly to operational swarm sizes [CITATION NEEDED].

Network dependency. The BASTION backend requires WebSocket connectivity for real-time agent coordination and blockchain transaction processing. DDIL degradation effects on the operational tempo of the governance workflow have not been stress-tested at scale. The swarm UDP peer mesh provides intra-swarm resilience, but the full planning-to-execution pipeline assumes sufficient connectivity to the BASTION backend. Extended operation without backend connectivity requires additional edge autonomy not yet implemented.

Single-user testing. All doctrinal workflow testing used single-user sessions. Coalition multi-user concurrent planning—multiple staff officers modifying plans, generating COP layers, and voting on proposals simultaneously—has not been stress-tested for race conditions, lock contention, or consistency violations. The architecture uses database transactions and blockchain consensus to enforce consistency, but behavior under concurrent planning loads requires empirical validation.

Explicit Limitations — v0.3 (Phase 55-58)

The following limitations apply to capabilities added in Phases 55-58 and define the boundary between what has been demonstrated and what remains unvalidated as of the v0.3 update.

Design coordination domain scope. Ironclaw's Chief of Staff role for operational design (Section 3.21) is scoped to Operational Approach development following JP 5-0 Chapter III. The four coordination sections — problem framing, center of gravity analysis, lines of effort, and operational approach — cover the doctrinal operational design process but do not extend to other planning frameworks (MDMP mission analysis, campaign planning, targeting). Generalizing the coordination pattern to other doctrinal planning domains requires additional section definitions and coverage criteria development; this work is identified as future work below.

Visual approach editor: single-user, partial symbol set. The visual operational approach editor (Section 3.22) implements the MIL-STD-2525D symbol and tactical graphic set for operational-level approach graphics. The full MIL-STD-2525D Part 2 tactical graphics library encompasses significantly more symbol types than the current implementation covers. Collaborative editing — multiple staff officers simultaneously placing and adjusting symbols on the operational approach map — is not yet implemented; the current editor operates in single-user mode, serializing all edits through a single session. Multi-user collaborative approach editing is identified as future work.

Ironclaw memory: no decay mechanism. Ironclaw's persistent memory system (Section 3.23) stores memories with fixed TTL bounds (90 days for user memory, 180 days for context memory) but implements no relevance decay — older memories within the TTL window are retrieved at the same weight as recent memories. As the memory corpus grows across extended advisory relationships, stale observations about preferences and patterns that no longer reflect current commander thinking will persist until they expire. A memory relevance scoring mechanism that degrades older entries relative to recent ones would improve retrieval quality for long-running advisory relationships. Additionally, the privacy implications of storing detailed commander interaction patterns — communication style, decision tendencies, risk preferences — in a database require consideration in any operational deployment. Users can delete memories through the management panel, but proactive memory review and curation places burden on users who may not routinely exercise this capability.

On-chain caveat enforcement: gas cost and latency at scale. The `check_employment_authorized()` method on the NEAR smart contract is a view method (no state modification, no gas cost for calls, near-instant response for individual checks). However, the cost and latency profile for high-frequency authorization checks across hundreds of concurrent resource assignments during active coalition operations has not been tested. At operational scale, the pattern of pre-authorization caching — caching caveat check results locally for short durations rather than calling the contract for every employment action — may be required to maintain operational tempo. The contract's caveat update latency (requiring a signed transaction, blockchain finality, approximately 2 seconds on NEAR mainnet) creates a potential window between caveat update and enforcement where the old caveat remains in effect. For rapidly changing employment restrictions, this finality latency may be operationally significant.

OSINT confidence scoring is source-tier-flat. A known limitation that predates Phase 55-58 but remains unresolved: the OSINT intelligence extraction pipeline applies a uniform confidence score of 0.65 to all OSINT-sourced claims regardless of the source's reliability tier. A NATO STANAG 2511-aligned confidence system would assign different reliability grades to satellite imagery versus social media posts versus diplomatic cables. The current flat confidence posture underweights high-reliability OSINT sources and overweights low-reliability ones, reducing the analytical utility of the confidence scoring system. Source-tier-aware confidence scoring is identified as future work.

5.2 Risk Analysis

Technical Risks

Blockchain network availability presents a foundational risk. While decentralized architecture eliminates single points of failure within the network, the network itself must remain accessible to participants. In contested environments, adversaries might attempt to disrupt blockchain nodes through cyber attacks, denial of service, or kinetic targeting of infrastructure. Mitigating this risk requires geographic distribution, redundant communication paths, and potentially private network deployment for operational security.

Smart contract bugs pose particular concern because blockchain immutability means deployed contracts cannot be patched. A bug in governance logic could allow unauthorized actions, block legitimate operations, or create inconsistent state. BASTION addresses this risk through extensive testing and formal verification where possible, but no testing regime can guarantee the absence of all bugs. Upgrade mechanisms exist but introduce their own governance complexity. The demonstration uses contracts that have undergone unit testing and integration testing, but operational deployment would require additional security audit and formal verification.

AI model drift and hallucination represent ongoing risks in AI-augmented systems. Model performance may degrade over time as operational environments evolve beyond training data distributions. Language models may generate plausible but factually incorrect analysis that human reviewers fail to catch. BASTION's human checkpoint requirements provide defense against these risks, but they depend on human reviewers maintaining sufficient expertise and attention to identify AI errors.

Key management complexity scales with system scale and participant diversity. Each DAO member, AI agent, and system component requires cryptographic key management. Loss of keys could lock participants out of governance. Compromise of keys could enable unauthorized actions. BASTION implements key derivation and recovery mechanisms, but operational deployment requires comprehensive key management procedures that extend beyond the demonstration scope.

Security Risks

Quantum computing developments pose a long-term threat to current cryptographic systems. While cryptographically relevant quantum computers do not yet exist, data protected today may face quantum attack during its classification lifetime. BASTION incorporates post-quantum cryptography considerations and cryptographic agility, but the timeline for quantum threats remains uncertain. Migration to fully post-quantum cryptography requires standards maturation and implementation validation that are still in progress [CITATION NEEDED].

Trusted Execution Environments, while providing hardware-based security isolation, are not invulnerable. Side-channel attacks have successfully extracted secrets from TEE implementations [CITATION NEEDED]. BASTION's use of Phala Network TEEs provides significant security benefits, but defenders should not assume TEEs provide absolute protection against sophisticated adversaries.

Insider threats represent a risk category that technical measures can only partially address. A trusted insider with legitimate access could exploit their position to manipulate governance outcomes, exfiltrate sensitive information, or sabotage operations. BASTION's audit trail provides accountability and forensic capability after the fact, but prevention relies on personnel security measures outside the system's scope.

Classification spillage risk exists whenever systems handle multiple classification levels. BASTION's architecture supports multi-level operation through encryption and access controls, but implementation errors or operational mistakes could allow classified information to flow to unauthorized recipients. The demonstration operates with unclassified data only; operational deployment with classified information would require security accreditation demonstrating that multi-level security properties are maintained.

Operational Risks

Over-reliance on automation could erode the human judgment that BASTION is designed to preserve. If operators come to trust AI recommendations uncritically, the human checkpoints that provide accountability become formalities rather than meaningful oversight. Maintaining appropriate human engagement requires training, culture, and interface design that keep humans cognitively engaged rather than passively approving AI outputs.

Human skill atrophy may occur as AI agents assume functions previously performed by humans. Staff officers who rely on AI for planning analysis may lose the ability to perform that analysis manually if AI systems become unavailable. This risk is common to any automation and must be addressed through training and exercise programs that maintain human capability independent of AI augmentation.

Decision fatigue could result if approval gates are too numerous or too frequent. Human oversight is only meaningful if reviewers have sufficient cognitive resources to evaluate each decision thoughtfully. An excessive number of approval requirements could lead to rubber-stamp approvals that provide accountability theater without genuine human judgment. The MDMP governance integration (Section 3.5) introduces 18 governance gates across 9 planning phases, each requiring human engagement. While these gates enforce doctrinal rigor, they also increase the cognitive burden on commanders and staff. BASTION's five-tier authority model mitigates this risk by reserving HUMAN_ONLY authority for genuinely consequential decisions while delegating routine validation to AI_AUTONOMOUS and AI_PRIMARY tiers, but calibrating the boundary between tiers requires operational experience with real planning workloads.

Coalition partner adoption barriers may limit BASTION's effectiveness for multi-national operations. Partners may be reluctant to adopt new governance mechanisms, particularly those involving blockchain technology with which they have limited experience. Integration with existing coalition systems and procedures requires diplomatic and technical work beyond system development. The demonstration validates technical feasibility, but operational adoption requires building consensus among coalition partners with different priorities and capabilities.

Complexity Growth Risks

The expansion from 23 to 16 deployed AI agents (a reduction in agent count but increase in per-agent capability), from 5 to 14 smart contract modules, and from ~100 to ~572 REST endpoints introduces risks associated with system complexity growth.

AI agent coordination management. With 16 deployed agents (8 LangGraph analysis specialists, 7 COP layer agents, and Ironclaw as Chief of Staff), maintaining quality, consistency, and appropriate oversight requires disciplined prompt engineering, testing, and monitoring for each agent. Interactions between agents may produce emergent behaviors that are difficult to predict or audit. The consolidation from a one-agent-per-staff-role model to Ironclaw as a single customizable coordinator reduces proliferation risk but concentrates coordination responsibility in a single agent whose behavior must be carefully validated.

Plugin architecture extensibility. The resource registry's plugin architecture (Section 3.10) demonstrates that new resource types can be added without modifying core code. However, this extensibility introduces a maintenance surface: each plugin's schema, state machine, capabilities, data handler, and COP renderer must be maintained and tested independently. As the plugin library grows, integration testing across plugin combinations becomes increasingly important.

Doctrinal alignment validation. The six-tab structure claims alignment with JP 5-0, but validating that alignment requires subject matter expert review beyond what automated testing can provide. How the system maps abstract doctrinal concepts to concrete interface elements involves interpretation that may differ between doctrinal communities. Exercises with military planners would provide the most meaningful validation.

Training mode governance fidelity. While the training mode implements identical governance mechanisms, the question of whether exercises conducted with identical governance truly capture the dynamics of operational decision-making remains open. Exercise participants may behave differently when they know consequences are simulated, potentially reducing the training value of governance-parity design.

Updated Risk Posture

Several technical risks have been reduced by implementation maturity. The blockchain integration layer is now validated across 31 phases of development with consistent transaction patterns. Authentication is stable with passkey-based WebAuthn replacing the earlier Privy dependency. DAO voting has been exercised across multiple governance contexts. The MDMP governance framework has been validated through exercise scenarios.

New risks have emerged from the expanded scope: the larger codebase requires more comprehensive testing; the greater number of AI agents increases the surface area for prompt injection or adversarial manipulation; and the doctrinal claims require validation from military subject matter experts that has not yet been formally conducted.

5.3 Ethical Considerations

Autonomous Weapons

BASTION’s approach to lethal decisions directly addresses international concerns about autonomous weapons systems. The platform preserves human control over strike authorization through multiple reinforcing mechanisms.

First, strike authorization proposals are categorically required to follow human-in-the-loop approval regardless of any other configuration settings. This constraint is implemented at multiple architectural levels: in smart contract voting logic, in AI agent decision trees, and in tactical execution pipelines. No configuration change or software update can enable autonomous strike approval without fundamental architecture modification. The MDMP governance integration extends this principle beyond strike authorization to a broader safety framework: three activity categories (AUTHORITY_DECISION, ETHICAL_LEGAL, RISK_JUDGMENT) are permanently locked to HUMAN_ONLY authority, and a safety matrix enforced at the smart contract level rejects any transaction that violates these immutable constraints.

Second, strike authorization requires supermajority or unanimous approval with enhanced deliberation periods. Coalition members must actively approve lethal effects, creating multiple opportunities for human judgment to intervene. The burden is on demonstrating that strikes should proceed, not on objecting to prevent them.

Third, the immutable audit trail creates accountability for every lethal decision. The record shows who approved what, when, and based on what information. This accountability persists indefinitely and cannot be altered after the fact.

These mechanisms align with emerging frameworks for autonomous weapons ethics that emphasize meaningful human control over lethal decisions [CITATION NEEDED]. BASTION demonstrates that AI augmentation can accelerate military coordination without removing humans from the decisions that matter most.

AI Decision Transparency

AI agent recommendations that influence human decisions raise concerns about transparency and accountability.

Explainability remains challenging for complex AI systems. BASTION’s agents provide reasoning summaries with their recommendations, but these summaries may not fully capture the factors influencing model outputs. Human reviewers may not have sufficient insight to identify when AI reasoning is flawed. The platform’s architecture requires human approval for consequential decisions, but the quality of human oversight depends on understanding what the AI is recommending and why.

Trust calibration requires humans to maintain appropriate confidence in AI systems—neither over-trusting nor under-trusting their capabilities. Over-trust leads to rubber-stamp approval of AI recommendations. Under-trust negates the speed benefits of AI augmentation. Calibrating trust appropriately requires training, feedback, and experience that the demonstration cannot fully provide.

Agent accountability gaps exist in the spaces between AI recommendation and human decision. When an AI agent recommends an action that a human approves and the outcome is problematic, assigning responsibility requires understanding what information the AI provided, how accurate that information was, and whether the human could reasonably have identified issues. BASTION's audit trail preserves this information for analysis, but clear accountability frameworks for AI-influenced decisions remain an evolving area of policy and doctrine.

5.4 Design Decisions and Tradeoffs

Every architecture embodies choices — and every choice sacrifices something. This section documents the major architectural decisions in BASTION's design, the alternatives that were considered, and what was gained and lost by each choice. Proactive acknowledgment of these tradeoffs is not defensive; it is evidence that the decisions were made deliberately rather than by default.

5.4.1 NEAR Blockchain vs. Private/Permissioned Chain vs. No Blockchain

BASTION uses NEAR Protocol, a public Layer-1 blockchain, as its trust infrastructure for coalition governance, audit records, and resource identity. The alternatives — a private permissioned chain (Hyperledger Fabric, Quorum) or no blockchain at all — each offered meaningful advantages.

The primary argument for a public chain is that coalition trust derives from the chain's independence, not from the trustworthiness of any single operator. A private chain requires trust in whoever operates the nodes; a public chain moves trust to the cryptographic protocol itself, which any party can independently verify. NEAR's approximately two-second transaction finality and developer-accessible testnet made it operationally tractable for a research prototype. The WebAssembly-compiled Rust smart contracts provide a formal, auditable governance layer that a conventional database cannot replicate.

What was sacrificed is significant. A public blockchain exposes metadata about governance activity to any observer with chain access, creating operational security concerns if transaction patterns reveal coalition decision timelines or resource allocations. NEAR's throughput, while substantial, introduces latency that is acceptable for deliberate planning decisions but would not support real-time tactical control. Dependency on third-party public infrastructure creates an availability risk that a self-operated private chain would not have. The cost of blockchain transactions — while low on NEAR — creates overhead that scales with operational tempo. None of these costs were hidden; they were accepted because the coalition trust benefit of cryptographic verification without a trusted third party justified them for the governance use case, which is where they apply. Real-time tactical control never touches the blockchain directly. (See Section 3.2 for the authority tier design that segregates which decisions go on-chain.)

5.4.2 LLM-Based Agents vs. Rule-Based Automation vs. No AI

BASTION fields 16 deployed AI agents using large language models (primarily Claude) for intelligence extraction, planning analysis, course of action development, operational design support, and staff coordination functions. The alternatives — rule-based automation (expert systems, decision trees) or simply providing humans with better-organized tools — were explicitly considered.

LLMs were chosen because military planning problems are expressed in natural language, have irregular structure, and require reasoning about novel situations that rule-based systems cannot anticipate. A rule-based intelligence extractor can identify entities matching a predefined schema; it cannot adapt when a document presents relationship information in an unfamiliar format. LLMs generalize across domain variation in a way that rule-based systems do not, and the breadth of staff functions BASTION supports — coordinated through Ironclaw's Chief of Staff role spanning J1 through component commander contexts — would require prohibitive rule-set engineering for each functional area.

What was sacrificed is acknowledged throughout this discussion section: LLMs are non-deterministic, can hallucinate, are sensitive to prompt construction, and cannot be formally verified. Every LLM agent in BASTION operates behind a human review gate precisely because these costs are real. The capability breadth of LLMs justified their use; the governance framework manages the costs. A system that required no human review of AI outputs would not be acceptable given current LLM reliability. BASTION does not make that claim. (See Section 3.3 for the graduated autonomy framework that manages AI reliability risk, and Section 5.5.1 below for the adversarial analysis of LLM determinism specifically.)

5.4.3 DAO Governance vs. Traditional RBAC vs. Hybrid

BASTION uses Decentralized Autonomous Organization governance — on-chain voting with smart contract enforcement — for coalition coordination. The alternative of role-based access control (RBAC) with a central authority is the conventional military C2 approach and is well-understood.

DAO governance was chosen because it solves a specific problem that RBAC cannot: trust across organizational boundaries where no single organization is trusted by all others. In a US-only system, RBAC suffices — the military can designate roles, enforce access controls, and audit centrally. In a coalition of nations with competing interests, RBAC requires each nation to trust the authority that administers the roles. Blockchain governance allows each nation to verify governance decisions independently without trusting any single nation's assurances. Smart contracts encode the rules; cryptographic verification replaces institutional trust for specific governance actions.

What was sacrificed is real. Consensus overhead is non-trivial: every on-chain governance action takes seconds that a centralized RBAC lookup takes in milliseconds. DAO governance introduces complexity that RBAC does not — key management, wallet infrastructure, on-chain transaction fees, and the possibility of consensus failure under adversarial conditions. For decisions that do not require cross-coalition verification — internal planning decisions within a single nation's authority — DAO governance is architectural overhead. The graduated five-tier model mitigates this by routing routine internal decisions through conventional authorization rather than blockchain consensus. But the overhead remains real for coalition-boundary decisions, and it is accepted because the trust benefit justifies it for that use case. (See Section 3.2 for tier architecture.)

5.4.4 Docker Bridge Pattern vs. Direct Cloud-to-Robot vs. Pi Edge Node

BASTION connects cloud governance to physical robotic assets through a Docker-containerized Python robot agent running on a host computer, which proxies commands to the Jetson Orin Nano via USB serial. The alternatives were direct cloud-to-robot connectivity (robot connects directly to BASTION backend WebSocket) and a dedicated Raspberry Pi edge node serving as the bridge hardware. (See project design decision documentation for the full bridge design rationale.)

The Docker bridge was chosen for accessibility and policy tractability. A Raspberry Pi edge node would provide better hardware isolation and a cleaner architectural boundary, but procurement and configuration of additional hardware creates friction in research and demonstration environments. Direct cloud-to-robot connectivity is technically simpler but provides no local autonomy when cloud connectivity is unavailable — the robot would fail immediately on DDIL degradation. The Docker bridge maintains local state and continues proxying commands using cached mission parameters during cloud disconnection, providing DDIL resilience without additional hardware procurement.

What was sacrificed matters for production deployment. The Docker bridge introduces a two-hop command path: BASTION cloud to bridge host to robot, adding latency at each hop. The bridge host itself is a single point of failure — if the machine running Docker fails, the cloud-to-robot path is severed. The bridge conflates the concerns of host operating system management and robot control, creating an architecture that would need to be separated for production hardening. A production deployment would likely use a dedicated edge node (Pi or equivalent ruggedized hardware) rather than a host-collocated bridge. The Docker approach was appropriate for a research prototype; it is not the target architecture for operational deployment.

5.4.5 PostgreSQL + Blockchain Hybrid vs. Pure On-Chain vs. Pure Off-Chain

BASTION stores data across three systems: PostgreSQL for structured relational data (plans, users, missions, MDMP state), Neo4j for the intelligence knowledge graph (entities, relationships, confidence scores), and NEAR Protocol for governance records (votes, approvals, audit trail, resource identifiers). Pure on-chain storage — everything on the blockchain — and pure off-chain storage — everything in a conventional database — were considered and rejected.

The hybrid approach was chosen because no single storage technology serves all requirements. Blockchain storage is expensive relative to volume, query-limited (blockchain is not a database), and permanent in ways that cause problems for mutable planning data that legitimately changes. PostgreSQL provides efficient structured queries, ACID transactions, and flexible schema updates — none of which blockchain provides. Neo4j provides semantic graph reasoning and path analysis that relational databases handle poorly. Each technology serves a specific role in the architecture.

What was sacrificed is operational complexity. Three storage systems require three maintenance concerns, three consistency models, and the challenge of synchronizing state across systems when operations span boundaries. When a coalition vote approves a plan modification, the blockchain record and the PostgreSQL plan record must be consistent — maintaining that consistency requires careful transaction design. The risk of the systems diverging (blockchain records a vote, PostgreSQL fails to update the plan) is a consistency challenge that pure on-chain or pure off-chain architectures do not have. The hybrid approach manages this through compensating transactions and reconciliation logic, but the residual risk of consistency drift under failure conditions is real and has not been exhaustively tested. (See Section 5.1 for limitations documentation.)

5.4.6 JP 5-0 Doctrinal Alignment vs. NATO vs. Custom Workflow

BASTION's interface architecture aligns to US Joint Publication 5-0 — six tabs corresponding to the joint planning process phases: Understand, Design, Plan, Decide, COP, Assess. NATO planning doctrine (Comprehensive Operations Planning Directive, COPD) follows a similar but distinct structure. A custom workflow designed from first principles was also considered.

JP 5-0 alignment was chosen because the target users — US joint staff officers — are trained to plan within that framework. When interface structure mirrors doctrinal structure, the software becomes immediately familiar rather than requiring parallel mental models. The lesson from the lessons-learned section (5.0) confirms this: the restructure to a doctrinal layout improved user navigation behavior. Custom workflows require users to learn a new mental model in addition to their doctrinal one; JP 5-0 alignment eliminates that cognitive overhead.

What was sacrificed is coalition breadth. US-centric doctrinal alignment creates friction for NATO partners who plan under COPD, UK partners who use the British Defence Doctrine, and Five Eyes partners who may use national variants. A system that a US J5 navigates intuitively may feel foreign to a UK G3. The BASTION planning workflow has not been validated with non-US military users, and doctrinal translation for coalition use would require workflow configurability that is not yet implemented. This is a recognized limitation with a clear path forward: the tab structure can be made configurable per coalition partner organization, mapping each partner's doctrinal phases to the appropriate BASTION workflow steps. (See Section 5.4 Future Work.)

5.4.7 Five-Tier Authority Model vs. Binary Human/AI vs. Three-Tier

BASTION implements a five-tier graduated authority model: AI_AUTONOMOUS, AI_PRIMARY (human review required), HUMAN_PRIMARY (human decision with AI recommendation), HUMAN_ONLY (no AI decision involvement), and COALITION_UNANIMOUS (requiring coalition-wide consensus). The binary alternative — human-only or AI-approved — is simpler but loses the nuance that different decisions warrant different oversight levels.

Five tiers were chosen because the decision space in military planning is not binary. Routine COP symbol publication does not require the same deliberation as a strike authorization. Encoding this graduated reality in the authority model allows AI to handle low-consequence, time-sensitive coordination tasks without requiring human approval for each, while ensuring human authority at consequential decision points. The five-tier approach was validated against MDMP activity analysis: 65 MDMP activities were categorized across tiers based on consequence, reversibility, and legal requirement.

What was sacrificed is model simplicity. Five tiers create boundary decisions — where exactly does “AI_PRIMARY with human review” end and “HUMAN_PRIMARY with AI recommendation” begin? These boundaries require calibration against operational experience that a research prototype cannot provide. Decision fatigue at tier boundaries is a real risk: users who encounter tier assignment as a classification problem rather than a natural decision point may develop workarounds that undermine the model’s purpose. The five-tier model is directionally correct but would benefit from operational testing to calibrate tier assignments against measured user behavior. A simpler three-tier model (autonomous, assisted, human-only) might reduce cognitive overhead; whether it would sacrifice necessary governance precision is an empirical question.

5.4.8 Commercial Proxy Hardware (Sphero/Jetson) vs. Military-Grade Platforms

BASTION’s physical demonstration uses Sphero RVR+ robotic platforms (commercial gaming robots) with NVIDIA Jetson Orin Nano edge AI computing (commercial AI development hardware). Military-grade alternatives — ruggedized UGVs, military-certified edge computing platforms — were considered and deferred.

Commercial hardware was chosen for accessibility, cost, and ecosystem support. The Sphero platform provides a controllable, repeatable demonstration environment without the procurement, safety, and operational complexity of military platforms. The Jetson Orin Nano provides 67 TOPS AI performance in a compact, developer-accessible package with an extensive software ecosystem (CUDA, detectNet, SLAM libraries) that accelerates implementation. For a research platform demonstrating architectural concepts, commercial proxy hardware allows the demonstration to focus on the architecture rather than on platform integration engineering.

What was sacrificed is the most critical limitation for military audiences. Commercial demonstration hardware does not validate performance under military operating conditions: vibration, temperature extremes, electromagnetic interference, submersion, and hostile environments that military hardware must tolerate. The Sphero platform’s physical capabilities — speed, obstacle traversal, payload capacity — are not representative of operational UGVs. The Jetson’s operating temperature range and shock resistance are not military-specification. The demonstration proves the architecture on commercial hardware; it does not prove the architecture on operational hardware. Transfer validation — demonstrating that the Docker bridge pattern, DID-based identity, and DAO governance integration function on actual military platforms — is a required next step before any operational deployment claim can be made. The architecture is designed to be hardware-agnostic (the bridge pattern works with any robot that supports a Python agent); transfer validation remains empirical work to be done.

5.5 Adversarial Analysis — Why This Approach Might Not Work

Strong research acknowledges the strongest arguments against its own conclusions. This section systematically red-teams the BASTION approach — not to undermine the contribution, but to demonstrate that the limitations have been examined honestly and that mitigations exist where possible. Each concern is presented as a skeptical reviewer would phrase it, with severity, mitigations, and an honest assessment of residual risk after mitigations are applied.

5.5.1 LLM Determinism

Adversarial Argument: Military planning requires reproducible outputs. LLMs are inherently non-deterministic. The same intelligence document may produce different entity extraction on consecutive runs, making it impossible to verify, audit, or reproduce AI-generated planning products. A commander who cannot reproduce an AI recommendation cannot audit it. A planner who cannot reproduce an extraction cannot validate it against source documents.

Severity: Significant.

Mitigation Options: Temperature=0 settings reduce but do not eliminate output variance. Reproducibility seed parameters (supported by some LLM providers) improve consistency within a single model version. Immutable audit logging of every LLM input and output pair creates a record of what was produced even if not reproducible on demand. Human review gates treat LLM outputs as intelligence estimates requiring analyst validation rather than ground truth requiring audit. Version-pinned model deployments prevent mid-operation model drift.

Residual Risk: Even with temperature=0 settings, LLM outputs vary across model versions, infrastructure configurations, and provider updates. True bit-for-bit reproducibility is not achievable with current LLM technology. Human review catches gross errors but may miss subtle analytical drift between runs — two extractions that appear similar but diverge on relationship types or confidence assignments. The audit log records what the LLM produced but does not guarantee future reproducibility of the same output. Systems requiring bit-for-bit audit reproducibility should not rely on LLMs for the audited outputs; human analyst review should produce the auditable artifact, with LLM outputs serving as analyst aids.

5.5.2 LLM Accuracy and Hallucination

Adversarial Argument: LLMs generate plausible but fabricated analysis. In an intelligence context, a hallucinated entity relationship — a fabricated connection between an adversary organization and a target that does not actually exist — or a fabricated adversary capability could lead to catastrophic planning errors. Confidence scoring does not solve this: the confidence score is generated by the same model that generated the potentially hallucinated content. A model that hallucinates a relationship will also hallucinate high confidence in that relationship.

Severity: Critical. This is the single highest-risk element of the architecture.

Mitigation Options: Multi-agent cross-validation — the 10-agent document intelligence team includes overlapping extraction agents whose outputs are compared for consistency before graph population. Human review gates at every consequential decision point require analyst validation of extracted content against source documents. The NATO Admiralty Code confidence framework treats AI outputs as intelligence estimates (A-F reliability, 1-6 information credibility) rather than facts, encoding epistemic humility in the data model itself. Source provenance tracking links every extracted entity to its source documents, enabling analyst spot-checks. Structured output schemas constrain generation to valid entity types and relationship classes, reducing the surface area for fabrication. Multi-source cross-validation degrades confidence scores for single-source claims, flagging them for additional review.

Residual Risk: No combination of mitigations eliminates hallucination risk. Subtle fabrications — slightly wrong relationship strength, plausible but incorrect organizational affiliation, fabricated adversary capability that passes multi-agent cross-validation because all agents share the same underlying model weights — may survive to influence planning. The system must be understood as decision support, never decision replacement. An analyst who treats AI-extracted entities as ground truth rather than estimates is misusing the system in a way that no technical mitigation can prevent. Training and doctrine for AI-augmented intelligence analysis are as important as the technical safeguards.

5.5.3 LLM Adversarial Manipulation

Adversarial Argument: An adversary who understands that BASTION uses LLMs for intelligence extraction could craft documents specifically designed to manipulate LLM outputs — prompt injection embedded in intelligence feeds, adversarial text calibrated to trigger specific extraction errors, or deception campaigns designed to exploit known LLM weaknesses. A sophisticated state actor conducting influence operations against a coalition planning system would naturally target its AI extraction pipeline.

Severity: Significant.

Mitigation Options: Input sanitization and content filtering applied before document processing reduce the attack surface for obvious injection patterns. Multi-source cross-validation flags anomalous extractions from single sources — a document that produces dramatically different entity extractions than other source documents raises confidence flags. Separate extraction and validation pipeline stages (extractors produce candidates; validators check them against existing knowledge) provide a second check before graph population. Human analyst review of extracted entities, particularly from novel or unverified sources, catches adversarial manipulations that automated checks miss. Confidence scoring that degrades for single-source claims forces additional scrutiny on potentially adversarial inputs.

Residual Risk: Adversarial text manipulation of LLMs is an active research problem with no complete solution. A sophisticated adversary with knowledge of the extraction pipeline's architecture — including its prompt templates and validation logic — could craft inputs that pass sanitization and multi-source checks by coordinating manipulation across multiple seemingly independent source documents. This risk exists for any

AI-augmented intelligence system and is not unique to BASTION, but it is not solved by any mitigation currently implemented. Red-team testing of the extraction pipeline against adversarially crafted inputs has not been conducted for this research prototype.

5.5.4 Blockchain Overhead vs. Tactical Tempo

Adversarial Argument: Two-second transaction finality on NEAR is fast for financial applications but potentially fatal for military decision-making. If a time-critical strike window closes during blockchain consensus, the governance mechanism becomes a liability rather than an asset. At operational scale — thousands of governance transactions per minute during a complex operation — throughput constraints could create coordination bottlenecks that conventional C2 systems do not have.

Severity: Significant.

Mitigation Options: The five-tier graduated authority model is the primary mitigation: Tier 1 (AI_AUTONOMOUS) and Tier 2 (AI_PRIMARY) decisions execute without requiring blockchain consensus, using conventional authorization. Blockchain records are written asynchronously after the fact for audit trail purposes. Only Tier 5 (COALITION_UNANIMOUS) and explicitly designated Tier 4 (HUMAN_ONLY) decisions require synchronous on-chain approval. Pre-authorized mission parameters cached locally at edge nodes allow tactical execution to continue without real-time blockchain access. Private or dedicated NEAR network deployment reduces latency relative to the shared public testnet. Mission parameter pre-authorization (the robot downloads mission parameters before execution; the blockchain isn't consulted mid-mission) ensures that execution tempo is not dependent on consensus latency.

Residual Risk: Time-critical decisions requiring coalition consensus (Tier 5 COALITION_UNANIMOUS) inherently trade speed for governance accountability. No mitigation eliminates this tension — it is the fundamental design tradeoff of using blockchain for multi-party decision authority. Under adversarial pressure and time compression, the governance overhead may be judged unacceptable; commanders may seek to bypass consensus for time-critical decisions in ways that undermine the governance framework. Private network deployment reduces latency but increases infrastructure burden and reduces the decentralization benefit that public-chain deployment provides. The throughput limits of NEAR at operational scale under realistic military transaction volumes have not been empirically tested for this prototype.

5.5.5 DDIL Failure Cascade

Adversarial Argument: The system assumes sufficient connectivity for blockchain consensus and AI agent coordination. In a real DDIL environment — sustained electromagnetic jamming, severed undersea cables, contested spectrum — the blockchain becomes unreachable, AI agents cannot coordinate, the knowledge graph cannot update, and the COP goes stale. The system degrades to a collection of isolated edge nodes with whatever local state they last cached. Is that degraded state actually useful? Or does it produce a false picture that is worse than no picture — presenting stale data as current intelligence to commanders who may not realize how stale it is?

Severity: Significant.

Mitigation Options: The robot bridge pattern with local mission state persistence allows robots to continue executing authorized missions during cloud disconnection. The UDP peer mesh enables swarm coordination independent of cloud connectivity. Pre-authorized mission envelopes cached at edge nodes provide autonomous operation scope without real-time governance access. Configurable staleness indicators on COP data — time-since-update badges on every symbol — surface the currency of information to commanders rather than presenting stale data as current. Graceful degradation design marks stale data explicitly rather than allowing it to appear current.

Residual Risk: Extended DDIL degrades the system to its cached state. The gap between “designed for DDIL” and “tested under sustained DDIL” is real and significant. Extended disconnection scenarios, reconnection reconciliation (what happens when an edge node that has been operating offline reconnects with potentially conflicting state), and conflict resolution between divergent edge states have not been empirically validated in this prototype. The DDIL resilience of this system is architectural intent, not demonstrated capability under sustained disconnection. Stale data with correct staleness indicators is better than stale data presented as current; whether stale data with indicators is operationally useful depends on the tactical situation in ways that require field testing to evaluate.

5.5.6 Single Developer / AI-Augmented Development Validity

Adversarial Argument: A system built by one developer with AI assistance in approximately ten weeks cannot have the robustness, security, or reliability required for military C2. The testing is limited to single-user scenarios. The architecture has never faced adversarial stress testing, concurrent user loads, or operational tempo. The impressive feature count — 16 deployed agents, ~572 endpoints, 14 smart contracts — may mask shallow implementations that break under conditions the developer did not test.

Severity: Moderate.

Mitigation Options: The modular architecture with clear component boundaries limits the blast radius of implementation failures — a broken agent does not take down the planning workflow. Atomic commit discipline (442 individually revertible commits) provides granular reversibility and documents the implementation at each capability boundary. Comprehensive seed scripts enable reproducible testing state. The architecture is explicitly designed for team development — service boundaries, API contracts, and plugin architecture allow components to be independently developed and tested. Each agent follows a standardized template with common architecture, reducing per-agent implementation variation.

Residual Risk: This is a research prototype, not a production system. The research contribution is the architecture, design patterns, and integration demonstration — not implementation maturity. Production deployment of BASTION as a military C2 system would require: independent security audit of smart contracts and API endpoints, load testing under realistic concurrent user scenarios, multi-user stress testing with concurrent planning operations, accessibility review, adversarial penetration testing of the bridge and edge components, and operational testing with trained military users providing feedback on workflow friction. These requirements are not gaps in the research contribution; they are the expected delta between research prototype and production system. The prototype demonstrates that the architecture is coherent and that the integration works; it does not claim implementation readiness.

5.5.7 Coalition Trust Assumptions

Adversarial Argument: The blockchain verifiability argument assumes coalition partners trust the blockchain protocol itself, trust the smart contract code, and trust that key management has not been compromised. These are significant trust assumptions that merely shift the trust problem rather than solving it. A coalition partner who does not trust the United States is not going to trust a blockchain whose smart contracts were written by a US researcher. Cryptography does not address political trust deficits.

Severity: Moderate.

Mitigation Options: Open-source smart contract code allows all partners to audit the governance logic they are trusting. Multi-party deployment — no single nation controls the blockchain infrastructure, with nodes distributed across partner nations —

reduces the trust surface from a single operator to the protocol and its code. Formal verification of governance invariants (planned in the Phase 65-66 roadmap) would provide mathematical assurance that specific properties hold under all reachable states. Independent security audit before any operational deployment provides third-party code validation. Multi-party key ceremony protocols with witnesses from each coalition partner address key management trust.

Residual Risk: Trustless systems require trust in the system's design, implementation, and deployment. Blockchain reduces the trust surface — cryptographic verification replaces procedural trust for specific governance decisions — but does not eliminate it. A coalition partner must still trust the code, the cryptography, and the key management. That is a smaller and more auditable trust surface than trusting another nation's procedural assurances about what their C2 system did, but it is trust nonetheless. Partners with deep political distrust of the coalition may reject the architecture regardless of its technical properties. Coalition adoption of any shared technical infrastructure requires political trust that technical design cannot substitute for.

5.5.8 Doctrinal Rigidity

Adversarial Argument: Strict alignment to JP 5-0 six-step planning may constrain creative operational approaches. Real military planning often deviates from doctrine when the situation demands — experienced commanders compress, skip, or restructure planning steps when time is short or the situation is unfamiliar. A system that enforces doctrinal workflow may prevent the very adaptability that effective military operations require. Doctrine describes how planning should work in deliberate conditions; it was not designed to constrain a system's operational flexibility.

Severity: Moderate.

Mitigation Options: The workflow is configurable — planning steps can be skipped or reordered by authorized users. The FRAGO (Fragmentary Order) mechanism enables rapid plan modification outside normal planning sequence. AI agents can recommend workflow deviations when the situation warrants accelerated or abbreviated planning. Commander override capability exists at every governance gate, allowing deliberate bypass of doctrinal checkpoints with recorded authorization.

Residual Risk: The system's structure inherently favors doctrinal approaches. Planning workflows that do not map to JP 5-0 phase categories may be awkward to express — a commander conducting a hasty attack from a meeting engagement may find the six-tab structure an ill-fitting organizational schema for what is actually a compressed, simultaneous process. Novel operational concepts that emerge from the operational environment rather than from doctrinal design may not find natural homes in the existing tab architecture. The tension between doctrinal structure and operational creativity is inherent to any doctrine-based system and cannot be eliminated without also losing the cognitive familiarity benefits that doctrinal alignment provides.

5.5.9 Scale and Complexity Debt

Adversarial Argument: 16 AI agents, approximately 572 REST endpoints, 14 smart contract modules — this is a complexity surface that grows faster than testing and validation can cover. Each agent is an attack surface. Each endpoint is a vulnerability. Agent-to-agent interactions are combinatorially complex and cannot be exhaustively tested. The system may be too complex to secure, too complex to audit, and too complex to maintain. Complexity is the enemy of security.

Severity: Significant.

Mitigation Options: Modular architecture with clear service boundaries limits interaction surface between components. The standardized agent template provides common architecture across all 16 agents, reducing per-agent variation. The plugin architecture enables extensibility without modifying the core platform. Automated API endpoint testing covers regression behavior. Smart contract unit and integration testing validates governance invariants.

Residual Risk: Complexity is the primary long-term risk for BASTION's continued development. Testing coverage cannot keep pace with feature growth at research prototype velocity — the current test coverage is functional but not comprehensive. Agent-to-agent interaction effects are combinatorially complex and not exhaustively tested; emergent behaviors from agent coordination under novel inputs may not have been encountered. A production program would require dedicated testing, security audit, and maintenance teams proportional to the system's complexity surface. The research contribution is demonstrating that this level of integration is architecturally achievable; sustainable maintainability requires the organizational investment that a research prototype does not have.

5.5.10 Ethics of AI Speed in Lethal Contexts

Adversarial Argument: Even with human authority gates on lethal decisions, AI acceleration of the decision pipeline compresses the time available for human reflection. A commander presented with an AI-generated target recommendation, AI-scored risk analysis, and AI-assessed legal compliance in seconds rather than hours may feel implicit pressure to decide at AI speed rather than at deliberation speed. The human gate exists, but the tempo around it may undermine its purpose. AI does not just accelerate coordination — it also implicitly accelerates the human's decision-making environment.

Severity: Moderate.

Mitigation Options: Configurable minimum deliberation period requirements could enforce a mandatory review time before approval is accepted — the system can refuse to process an approval that arrives within a threshold time of the proposal. Explicit risk acknowledgment workflow requires itemized review of specific risk categories rather than a single approve button. Decision audit trail records time-to-decision for every consequential decision, enabling post-action review of whether deliberation thresholds were honored. Training protocols emphasizing that AI speed applies to coordination, not to human judgment timelines, address the behavioral dimension.

Residual Risk: Organizational culture and operational urgency may override procedural mitigations. A commander under genuine time pressure during a crisis may rubber-stamp AI recommendations despite deliberation requirements, reducing the human gate to accountability theater. This is fundamentally a human factors and organizational psychology challenge that technical mitigations can support but cannot solve. Minimum deliberation timers can be overridden by commanders who have override authority. The tension between AI speed and human deliberation quality is an open research problem in AI-augmented decision-making that no C2 platform architecture fully resolves. Research into the effects of AI recommendation tempo on human decision quality in military contexts is needed before any strong claim can be made about human control quality under operational pressure.

Summary Table

Concern	Severity	Key Mitigation	Residual Risk
LLM Determinism	Significant	Temperature=0, audit logging, human review	Bit-for-bit reproducibility unachievable
LLM Accuracy / Hallucination	**Critical**	Multi-agent cross-validation, human review gates, NATO confidence framework	Subtle hallucinations may survive all mitigations
	Significant	Input sanitization,	Sophisticated coordinated

Concern	Severity	Key Mitigation	Residual Risk
LLM Adversarial Manipulation		multi-source cross-validation	adversarial inputs not solved
Blockchain Overhead vs. Tempo	Significant	Five-tier model routes most decisions off-chain	Coalition consensus decisions inherently trade speed
DDIL Failure Cascade	Significant	Local mission caching, staleness indicators	Extended DDIL resilience not empirically validated
Single-Developer Validity	Moderate	Modular architecture, atomic commits	Production readiness requires security audit and load testing
Coalition Trust Assumptions	Moderate	Open-source contracts, multi-party deployment	Cryptography does not address political trust deficits
Doctrinal Rigidity	Moderate	Configurable workflow, commander override	Novel operational concepts may not map to JP 5-0 structure
Scale and Complexity Debt	Significant	Standardized templates, plugin architecture	Testing coverage cannot keep pace with feature growth
AI Speed vs. Human Deliberation	Moderate	Deliberation timers, risk acknowledgment workflow	Organizational pressure may override procedural mitigations

Bottom line: BASTION is a research contribution demonstrating architectural feasibility for AI-DAO integration in military coalition C2. It is not a claim of operational readiness. The adversarial analysis above demonstrates that the authors have examined these limitations systematically and have designed mitigations where technically feasible, while being transparent about what remains unresolved. A system that cannot honestly characterize its limitations should not be trusted with the applications BASTION is designed to support. These concerns — particularly

LLM hallucination risk and the gap between designed-for-DDIL and validated-under-DDIL — define the research agenda for the next development phase.

5.6 Future Work

Near-Term Extensions

Problem set model and echelon awareness. Renaming workspaces to “problem sets” (JP 5-0 terminology) with echelon-awareness (strategic, operational, tactical) would strengthen the doctrinal alignment of the data model and enable echelon-appropriate defaults for governance thresholds, agent configurations, and product templates.

Strategic document containers and actor categorization. Organizing strategic documents into nation/group containers (e.g., United States, China, NATO) with actor categories (ally, adversary, neutral, partner) would provide persistent container-based organization for building strategic environments over time, feeding into inheritance mechanisms for child problem sets.

AI strategic context and knowledge graph integration. Wiring subscribed strategic environment data and container-scoped knowledge graphs into AI agent context would enable agents to draw on structured strategic knowledge rather than raw document text. Container-scoped RAFT graph construction with auto-trigger on document changes would keep the knowledge graph current.

Strategic environment inheritance. A strategic-level problem set serving as context provider with inheritance mechanisms for directives, policy, and intelligence would enable child problem sets to inherit strategic context without manual duplication, with update propagation when strategic guidance changes.

Embedded DAO governance at decision gates. Moving DAO governance from a dedicated interface into contextual workflow decision gates would surface proposals at natural planning decision points (objective approval, COA selection, order release) rather than requiring navigation to a separate governance view.

Contextual AI staff integration. Surfacing AI agent output contextually per tab with per-tab assistants aware of workflow phase would provide recommendation engines tied to doctrinal workflow position rather than requiring users to seek out agent capabilities.

Generalized Chief of Staff coordination beyond operational design. Ironclaw's Chief of Staff coordination architecture (Section 3.21) is parameterized by section definitions and coverage criteria; the underlying LangGraph interrupt-resume mechanism is domain-agnostic. Extending the coordination pattern to other planning frameworks — MDMP mission analysis, targeting decision support, logistics requirements generation — would generalize a valuable staff coordination capability across the planning lifecycle without requiring new infrastructure. Each new coordination domain requires section definition, doctrinal coverage criteria, and red-team questioning guidance, but the execution framework is reusable.

Multi-user collaborative approach editing. The visual operational approach editor (Section 3.22) currently serializes edits through a single user session. Real operational approach development involves multiple staff officers contributing simultaneously — the J3 positioning maneuver units, the J5 placing LOEs, the J4 marking logistics nodes. Extending the MapOverlay editor with Yjs CRDT-based collaborative editing (the same technology used for the Design tab's collaborative operational design) would enable concurrent multi-staff approach editing with conflict resolution. This extension would align the visual approach editor with the collaborative model already established for operational design.

Memory decay and relevance scoring for Ironclaw. Ironclaw's persistent memory system (Section 3.23) stores memories with fixed TTL bounds but no within-TTL relevance decay. Adding a relevance scoring mechanism that decays older entries relative to recent interactions — reducing the retrieval weight of memories more than N sessions old — would improve context quality for long-running advisory relationships by surfacing recent observations over stale ones. Relevance decay also provides a more privacy-respecting memory profile: interaction patterns from significantly past planning periods become less influential as new patterns emerge.

Layer-2 or off-chain authorization caching for high-tempo operations. The on-chain `check_employment_authorized()` method provides verifiable authorization at approximately NEAR testnet latency. At operational tempo with high-frequency authorization checks, pre-authorization caching — storing caveat check results locally with a configurable validity window (e.g., 30 seconds for time-sensitive resource assignments) — would reduce check latency to near-zero for the majority of requests while ensuring that caveat changes propagate within the cache validity window. This caching pattern requires careful cache invalidation design to avoid serving stale authorizations during the window between a caveat update and cache expiry.

Source-tier-aware confidence scoring. The OSINT intelligence extraction pipeline currently applies a uniform confidence score of 0.65 to all OSINT-sourced claims, regardless of source reliability. Implementing NATO STANAG 2511-aligned source-tier-aware scoring — differentiating satellite imagery (typically A1 or A2 on the Admiralty scale) from unconfirmed social media (typically F6) — would improve the analytical value of confidence scores. The data model already carries NATO source reliability codes through the document intelligence pipeline; extending the confidence calculation to use these codes requires updating the scoring logic in the fact extractor specialist agent.

Multi-platform autonomous vehicle integration represents an immediate extension beyond current demonstration capabilities. Coordinating multiple heterogeneous autonomous systems through DAO governance would validate BASTION's scalability claims and demonstrate more realistic tactical scenarios.

Coalition health monitoring would extend the governance framework to track coalition partner cohesion, national caveat compliance, and narrative impact in real time.

Production deployment through CI/CD pipeline to Hetzner server infrastructure would move BASTION from development to a accessible demonstration environment, with TEE-aware component separation documented for production.

Phase 59-75 Roadmap Highlights

The BASTION development roadmap extends through Phase 75 with planned phases beyond the current v0.3 implementation. Phases 55-58 are now complete (Ironclaw Chief of Staff coordination, visual approach editor, Ironclaw memory, on-chain DID caveats). Selected highlights of remaining roadmap phases include:

JSON-LD semantic brain. The current knowledge graph implementation uses a property-graph model. The next major graph enhancement introduces JSON-LD semantic markup, enabling interoperability with external ontologies and OSINT data sources. Linked Data publishing would allow BASTION's knowledge graph to integrate with NATO standardization data models and allied intelligence exchange formats without manual mapping.

ATAK/CoT interoperability. Android Team Awareness Kit (ATAK) is the operational standard for tactical common operating picture in the US military. Implementing Cursor on Target (CoT) message exchange between BASTION's COP layer and ATAK would provide the integration path for operational demonstration environments where ATAK is already deployed. This interoperability would validate BASTION's COP layer against field-standard tools rather than a standalone display.

End-to-end swarm demonstration. The current swarm capability uses three platforms of the same type. The planned multi-platform swarm demonstration coordinates heterogeneous platforms—ground, aerial, and fixed sensor—through the same DAO-governed swarm leader architecture. Heterogeneous platform coordination introduces capability negotiation, mixed formation geometries, and type-specific mission assignment that the current homogeneous swarm does not exercise.

Coalition multi-tenancy (Phase 61-63). The current implementation supports a single coalition instance. Multi-tenancy would enable separate coalition organizations to operate concurrently on the same BASTION infrastructure with complete data isolation. This capability is essential for production deployment where multiple exercises, training events, or operational activities may run simultaneously.

Formal verification of DAO governance invariants (Phase 65-66). The safety invariants embedded in BASTION's smart contracts are validated through testing but not formally verified. Formal verification using model checking or theorem proving would provide mathematical guarantees that the invariants hold under all reachable states, eliminating the residual risk that untested edge cases could violate human authority boundaries.

Operational testing with military exercises (Phase 67-70). The long-range roadmap targets integration with actual military exercise programs, initially in observer role and subsequently as a planning support tool. Exercise validation with trained military planners would provide the most authoritative assessment of doctrinal alignment, workflow integration, and user acceptance that laboratory testing cannot replicate.

Extended heterogeneous swarm operations. Longer-term swarm research will examine leadership election stability under node failure, dynamic formation adaptation to terrain constraints, and multi-swarm coordination across hierarchical command structures. These capabilities are required for operationally relevant swarm behavior but are out of scope for the current proof-of-concept scope.

Research Directions Opened by MDMP and Escalation Integration

The MDMP governance framework and escalation modeling capabilities described in Sections 3.5 and 3.6 open several research directions that merit investigation.

Escalation model calibration against historical data would validate the escalation ladder frameworks currently implemented with configurable but uncalibrated parameters. Mapping historical crises to the escalation model's rung structure and testing whether the trigger conditions and thresholds produce predictions consistent with observed outcomes would establish confidence bounds for the escalation modeling capability. This calibration requires access to classified or restricted historical case studies that the current research context does not provide.

Assumption management validation in exercises would test whether the formal assumption lifecycle (Pending, Accepted, Invalidated) with automatic replanning triggers improves planning outcomes compared to traditional informal assumption tracking. The hypothesis that blockchain-enforced assumption accountability reduces the risk of plans built on invalid premises requires empirical validation through planning exercises with measurable outcomes.

Safety matrix boundary optimization would investigate the optimal allocation of MDMP activities across the five authority tiers. The current allocation derives from doctrinal analysis, but operational experience may reveal that some activities benefit from different authority levels than doctrine suggests. Research into adaptive authority allocation—where the system adjusts tier assignments based on measured agent reliability and operational context—could improve the balance between speed and safety.

Adversary model adversarial robustness requires investigation into how adversaries might deliberately manipulate the inputs that BASTION's adversary modeler and deception detector consume. If an adversary understands that BASTION generates MLCOA and MDCOA from observable indicators, they could shape those indicators to produce misleading adversary COAs. Research into adversarial robustness for military AI planning systems would strengthen the analytical capabilities described in Section 3.6.

Long-Term Research Directions

Full DIME/DIMEFIL instrument orchestration would extend BASTION beyond military coordination to whole-of-government approaches. The Effect Cascader agent already traces consequences across DIME domains, providing a foundation for cross-domain analysis. Extending DAO governance to coordinate across government agencies would require additional research into interagency processes and authorities, but the effects analysis framework provides a starting point for understanding how military actions create requirements for diplomatic, information, and economic coordination.

Cross-domain federated learning could enable AI agents to improve collaboratively while respecting classification boundaries. Agents operating at different classification levels or in different coalition partner networks could share learning without sharing the underlying data. This capability would address the tension between AI improvement through data sharing and security requirements for information protection.

Real coalition exercises would provide the operational validation that demonstration environments cannot. Integrating BASTION into actual military exercises, even in observer or limited-participation roles, would generate insights about operational utility and integration challenges that inform further development. The MDMP governance gates and assumption tracking capabilities are particularly suited to exercise validation, as exercises provide controlled environments where the benefits of formal governance enforcement can be measured against traditional planning processes.

Accreditation path research would investigate the security certification requirements for deploying BASTION in classified environments. Military systems handling classified information must undergo accreditation processes that can be lengthy and demanding. Understanding these requirements early and designing for accreditability would accelerate potential operational adoption.

Policy Research Needed

Legal frameworks for blockchain-based military C2 remain undeveloped. Questions about authority, liability, and compliance with existing laws and treaties require attention from legal scholars and policy makers. How do existing command authorities translate to DAO governance? What legal status do AI agents have when acting within delegated authority? How do international agreements governing coalition operations apply to blockchain-coordinated actions?

Coalition data sharing agreements may require revision to accommodate blockchain-based information sharing. Existing agreements anticipate traditional database and message-based information exchange. Blockchain's properties of immutability and transparency may require new agreement structures that address data persistence, access revocation, and cross-border data flows.

Autonomous weapons governance continues to evolve internationally. As nations develop policies regarding AI in weapons systems, BASTION's approach to human control over lethal decisions provides a model that could inform these discussions. Engaging with the policy community to ensure that BASTION's mechanisms meet emerging requirements would support both policy development and system adoption.

This section has examined BASTION's limitations, risks, and future directions with the candor that academic and operational audiences require. The following conclusion summarizes what BASTION has demonstrated and its significance for military coordination in coalition operations.

6. Conclusion

Military coalition operations demand coordination mechanisms that current systems cannot provide. The challenges are structural: centralized command and control creates vulnerability to disruption; traditional approval processes operate slower than adversary decision cycles; information sharing restrictions fragment situational awareness; and the lack of decentralized governance frameworks prevents effective coordination across national boundaries. BASTION addresses these challenges through a novel integration of technologies that together enable what no single technology could achieve alone.

Answering the Research Question

This research asked: *How can interconnected, AI-augmented Decentralized Autonomous Organizations (DAOs) provide a secure, transparent, and resilient governance framework that enables effective C2, accelerates decision-making, optimizes resource management, and supports autonomous, policy-compliant coordination across diverse national and organizational boundaries?*

The demonstration validates that interconnected, AI-augmented DAOs can indeed provide such a framework. The evidence supports each element of the research question.

Secure. Policy constraints encoded in smart contracts govern all execution decisions, automatically enforcing national caveats and classification restrictions. The system verifies constraint compliance before executing any effects, eliminating the possibility of policy violation through configuration error or human oversight. Security is achieved through code, not procedure.

Transparent. Every governance decision—proposals, votes, approvals, and executions—is recorded on an immutable blockchain ledger accessible to all authorized participants. Coalition partners can independently verify that the system operates according to agreed-upon rules. The comprehensive audit trail supports accountability, dispute resolution, and continuous improvement.

Resilient. Decentralized architecture eliminates single points of failure that adversaries could target to disable coalition coordination. Edge computing enables tactical operations to continue under degraded connectivity. The blockchain maintains consistent state across distributed nodes, enabling reconciliation when communications are restored.

Effective C2. The complete flow from strategic objective through operational coordination to tactical execution demonstrates command and control across echelons. Strategic intent cascades through operational plans to tactical tasks; upward reporting completes the C2 cycle. Cross-DAO communication enables tactical events to trigger strategic governance actions without manual intervention.

Accelerated Decision-Making. AI augmentation at each level reduces the time from information to action. Document analysis extracts objectives in minutes. Planning agents generate options continuously. Tactical AI identifies targets in real-time. Automated proposal generation eliminates manual coordination overhead. Human decision-makers focus on judgment rather than administrative burden.

Optimized Resource Management. Transparent resource tracking provides visibility into asset availability, allocation, and consumption across the coalition. Consumption-based replenishment triggers resource requests when inventory reaches thresholds rather than on arbitrary schedules. Every allocation is recorded with provenance, enabling efficient management without manual tracking.

Policy-Compliant Coordination. National caveats specified at contribution time bind all subsequent employment decisions. Smart contracts enforce these restrictions automatically and uniformly. The tactical system cannot execute effects that violate contributor policies, regardless of operational pressure or human error.

Coordination Across Boundaries. The coalition DAO structure supports multi-national governance with weighted voting, quorum requirements, and cross-organizational proposals. The three-tier architecture coordinates across strategic-operational-tactical boundaries. The framework scales to additional DAOs representing additional organizational boundaries without architectural modification.

Contributions Summarized

This research delivers eight specific contributions to the intersection of decentralized governance, artificial intelligence, and military command and control.

First, the novel integration of DAO governance with military C2 demonstrates that blockchain-based coordination mechanisms can satisfy military requirements for accountability, security, and tempo. This integration addresses the gap identified in existing blockchain defense applications, which focus on data integrity rather than governance.

Second, AI agent augmentation of decentralized decision processes demonstrates that AI can enhance DAO governance without undermining human authority. 131 agents—31 specialized agents for governance, planning, intelligence, and operational design, plus 102 JPP staff role agents—accelerate analysis, generate recommendations, and execute within delegated constraints, while human checkpoints preserve accountability for consequential decisions.

Third, the multi-level coordination framework mapping DAOs to military echelons demonstrates that decentralized governance can function across strategic, operational, and tactical levels with appropriate tempo and authority adjustments at each level. This framework provides the structural foundation for coordinating coalition operations from national policy through tactical execution.

Fourth, the integration of MDMP doctrinal processes with blockchain governance enforcement demonstrates that established military planning procedures can be encoded as smart contract conditions rather than enforced through procedural discipline alone. The five-tier authority model, 18 governance gates across 9 MDMP phases, and formal assumption lifecycle management transform doctrinal guidance into technically enforceable governance. Nine invariants—including immutable locks on authority decisions, ethical judgments, and risk acceptance—ensure that human authority cannot be eroded through configuration changes or operational pressure.

Fifth, adversary modeling, escalation simulation, and effects analysis integrated with governance-aware wargaming demonstrate that AI can provide analytical depth for military planning within governance bounds. The action-reaction-counteraction wargaming framework, escalation ladder modeling, DIME effects cascading, and force ratio analysis provide quantitative and qualitative rigor that traditional staff processes achieve only through time-intensive manual effort. These capabilities feed directly into governance gates, ensuring that analytical thoroughness is enforced rather than aspirational.

Sixth, the physical proof-of-concept moves beyond theoretical analysis to demonstrate actual system behavior. The integration of Jetson Orin Nano edge computing, Sphero RVR+ robotic execution, and physical area of operations model validates that BASTION’s architecture can coordinate tangible assets, not merely digital abstractions.

Seventh, the doctrine-first interface design through six workflow tabs aligned to JP 5-0 (Understand, Design, Plan, Decide, COP, Assess) demonstrates that military planning software can be structured around doctrinal process rather than arbitrary software categories. The operational design workspace with AI-assisted problem framing, center of gravity analysis, and lines of effort/operation fills the critical gap between strategic guidance and course of action development. Autonomous COP layer generation by AI agent teams demonstrates that the operational picture can be constructed from planning documents rather than requiring manual symbol placement.

Eighth, the resource DID plugin architecture demonstrates that military assets can be managed as blockchain-verified first-class entities with extensible type definitions. The combination of decentralized identity (`did:near:resource-{id}`), plugin-based resource management, COP integration, and DAO-governed allocation provides end-to-end accountability from resource registration through employment to replenishment—a complete resource lifecycle managed within the same governance framework used for strategic decisions.

Human Authority Preserved

Central to BASTION’s design is the preservation of human authority over decisions that require human judgment. The demonstration validates that graduated autonomy can accelerate coordination without removing humans from critical decision points.

The demonstration validates three human authority positions within a single integrated scenario: humans in-the-loop for strategic decisions, on-the-loop for operational coordination, and out-of-the-loop for routine tactical functions. The MDMP governance integration refines this spectrum into a five-tier authority model that maps 65 planning activities to precise authority levels, from AI_AUTONOMOUS for data aggregation through HUMAN_ONLY for authority decisions, ethical judgments, and risk acceptance. This granularity demonstrates that one architecture can accommodate not just the broad spectrum of human-machine teaming but also the fine-grained authority distinctions that doctrinal planning processes require. The COP layer publish review cycle extends this principle to operational picture generation: AI agents produce content autonomously, but human review gates control what reaches the shared COP that informs command decisions.

The invariant for lethal decisions is reinforced by a broader safety framework. Strike authorization requires human approval with 100% voting threshold regardless of any other configuration. Beyond strike authorization, three entire categories of military decision-making are permanently locked to human-only authority through immutable smart contract constraints. No governance action, configuration change, or operational pressure can delegate authority decisions, ethical judgments, or risk acceptance to AI agents. This approach demonstrates that meaningful human control can be preserved not as a single invariant but as a systematic safety architecture enforced at the protocol level.

The audit trail maintains accountability even for autonomous actions. Every agent decision, every policy verification, every execution is permanently recorded with full provenance. Commanders can review what decisions were made, by what authority, based on what information, and with what outcome. Accountability persists from strategic intent through tactical effect.

Broader Implications

While developed for military coalition coordination, BASTION's architecture addresses challenges common to any multi-stakeholder coordination requiring speed, security, and accountability. Disaster response operations face similar requirements for rapid coordination across organizations with different authorities and resources. Critical infrastructure protection involves coordination among public and private entities with limited mutual trust. Supply chain coordination requires visibility and compliance across organizational boundaries.

The training/operational mode architecture demonstrates that the same governance framework can serve both training and operational contexts, a capability valuable in any domain where organizations must practice their coordination procedures before executing them under real conditions. The resource DID plugin architecture shows that extensible asset management with blockchain-verified identity is applicable beyond military contexts to any domain requiring auditable tracking of high-value resources across organizational boundaries.

The combination of DAO governance, AI augmentation, and tiered human authority provides a pattern applicable wherever complex coordination must occur among parties who cannot fully trust each other but must act together effectively. BASTION demonstrates this pattern in the demanding context of military operations; adaptation to other domains would require less rather than more capability.

Closing Statement

The physical demonstration validates that BASTION's architecture is not a theoretical construct. The robot bridge (Docker-containerized Python agent with mDNS auto-discovery), vision pipeline (detectNet object detection and ORB feature matching on Jetson Orin Nano), and swarm leadership (three-platform coalition executing six doctrinal formations over UDP peer mesh) confirm that the strategy-to-autonomous-execution pipeline functions end-to-end in tangible hardware. Strategic governance decisions made on blockchain propagate through the mission order interface, across the cloud-to-edge bridge, into physical robot motion—visible, repeatable, and bounded by the same policy constraints that govern digital coordination. This physical validation distinguishes BASTION from purely software-based demonstrations and provides an observable anchor for the abstract governance claims the platform makes.

By demonstrating that AI-augmented DAOs can accelerate military decision-making while preserving human control over critical decisions, this research opens new possibilities for coalition coordination in contested environments. With 53 completed development phases, 442 executed plans, 131+ AI agents, approximately 500 REST API endpoints, and a doctrine-aligned six-tab interface validated through exercise scenarios, BASTION has progressed well beyond initial proof-of-concept to a comprehensive military planning and coordination platform. The system proves that decentralization, transparency, and appropriate automation can coexist with the accountability and authority that military command requires. The path from current implementation to operational deployment requires the additional work identified in Section 5.4—particularly ATAK interoperability, end-to-end swarm demonstration at scale, coalition multi-tenancy, and formal verification of governance invariants—but the fundamental question is answered: interconnected, AI-augmented DAOs can provide the governance framework that modern coalition operations demand.

7. References

This section will contain the bibliography generated by Zotero.

Citation Management Instructions

Setup (per 13-RESEARCH.md)

1. Install Zotero from zotero.org
2. Install Zotero Connector browser extension
3. Install Chicago 18th edition (note) style:
   - Zotero > Edit > Settings > Cite > Get additional styles
   - Search: "Chicago Manual of Style 18th edition (note)"
4. Create library: "BASTION Whitepaper"

Adding Sources

- Use Zotero Connector to capture sources from web
- Manually add sources not available online
- Fill all required fields (author, title, date, publisher/journal)

Sources to Add

Based on [CITATION NEEDED] placeholders throughout document:

Academic Literature

- [ ] "Blockchain Applications in the Military Domain: A Systematic Review" (MDPI, Jan 2025)
- [ ] "DAO Research Trends" (MDPI, Mar 2025)
- [ ] IEEE White Paper on DAO Governance Guidelines (Feb 2025)
- [ ] "Reimagining Military C2 in the Age of AI" (SCSP, Dec 2024)
- [ ] "AI in Military C2 Systems" (NATO C2COE, Jul 2025)
- [ ] GAO-25-106454 on Defense Command and Control (Apr 2025)

Government/Military Documents

- [ ] JP 3-0 Joint Operations
- [ ] JP 5-0 Joint Planning
- [ ] JADC2 Implementation Strategy documents
- [ ] DOD AI Strategy documents
- [ ] CJCSM 3105.01 Joint Risk Analysis

Technology References

- [ ] NEAR Protocol documentation
- [ ] Phala Network documentation
- [ ] LangGraph/LangChain documentation

Additional Sources (from document review)

- [ ] Reference on national caveats and intelligence sharing constraints in coalition operations
- [ ] Reference on transparency and accountability challenges in multinational operations
- [ ] Reference on adversary strategies to exploit coalition coordination gaps
- [ ] Reference on classification systems and their impact on information sharing
- [ ] Reference on decision speed requirements in modern warfare
- [ ] Reference on military AI decision support systems
- [ ] Reference on DAO governance mechanisms and commercial applications

Phase 55-58 New Sources

- [ ] W3C Recommendation: “Decentralized Identifiers (DIDs) v1.0” — World Wide Web Consortium, July 19, 2022, https://www.w3.org/TR/did-core/
    - Chicago 18th bibliography: `World Wide Web Consortium. "Decentralized Identifiers (DIDs) v1.0." W3C Recommendation. July 19, 2022. https://www.w3.org/TR/did-core/.`
- [ ] MIL-STD-2525D Joint Military Symbology — Department of Defense, 2014
    - Chicago 18th bibliography: `Department of Defense. *Joint Military Symbology*. MIL-STD-2525D. Washington, DC: Department of Defense, 2014.`
- [ ] JP 5-0 Joint Planning (2020 edition) — already listed above under Government/Military Documents; ensure 2020 edition is captured
    - Chicago 18th bibliography: `Joint Chiefs of Staff. *Joint Planning*. JP 5-0. Washington, DC: Joint Chiefs of Staff, 2020.`
- [ ] NATO STANAG 2511: Intelligence Source Reliability and Information Credibility Ratings (Admiralty Code) — NATO standardization document
    - Chicago 18th bibliography: `North Atlantic Treaty Organization. *Allied Intelligence Publication 2.1: Intelligence Reporting*. STANAG 2511. Brussels: NATO Standardization Office, 2016.`

Generating Bibliography

1. Place cursor at end of References section in Word
2. Zotero > Add/Edit Bibliography
3. Select all sources used in paper
4. Style will auto-format to Chicago 18th (note)

Placeholder Bibliography

(The following entries will be replaced by Zotero-generated bibliography)

1. [Academic literature on DAOs]
2. [Military C2 frameworks]

3. [AI governance research]
4. [Blockchain defense applications]
5. [JP 5-0 and related doctrine]
6. [Technology documentation]

Note: All [CITATION NEEDED] placeholders in the document must be replaced with proper footnotes before final submission. Use the following workflow:

1. Search for `[CITATION NEEDED]` across all whitepaper sections
2. For each placeholder, identify the claim requiring citation
3. Find appropriate source in Zotero library (or add new source)
4. Replace placeholder with Zotero citation (footnote format)
5. Verify bibliography updates automatically

Appendix A: Implementation Status (SITREP)

This appendix provides the current implementation status of the BASTION platform as of March 2026. The Situation Report (SITREP) documents completed phases, current development status, and remaining work required for full operational capability.

A.1 Phase-by-Phase Status

Phase 30 (Ironclaw Agent Integration), Phase 46 (Sphero RVR+ Swarm Leader), Phase 53 (DID Governance Architecture & Bug Fixes), and Phase 58 (On-Chain Resource DID Caveats) represent significant milestones in the platform's evolution. The table below provides the complete status record through Phase 58.

Phase	Name	Status	Completed	Capability Summary
1	Foundation & Infrastructure	Complete	2026-01-13	NEAR smart contracts, PostgreSQL hybrid storage, Docker containerization
1.1	Calimero Self-Sovereign App Integration	Not Started	—	DAO compartmentalization research; deferred
1.2	Passkey Authentication & NEAR Implicit Accounts	Complete	2026-01-24	WebAuthn passkeys, PRF-DID integration, magic link fallbac
1.3		Complete	2026-02-06	

Phase	Name	Status	Completed	Capability Summary
	NEAR Implicit Account Funding			Account activation funding contract and admin UI
1.4	Navigation Architecture Restructure	Complete	2026-02-23	Doctrine-aligned tab navigatio route restructure
2	Identity & Security Framework	Complete	2026-01-16	DID registry, ABAC, PQC, W3C verifiable credentials
3	DAO Governance	Complete	2026-01-17	Smart contracts, voting engin coalition linkages
4	Strategic Planning Module	Complete	2026-01-21	Document ingestion, LLM extraction, approval workflow
4.1	Admin UI	Complete	2026-01-19	Administrative configuration interface
4.2	AI Agent Teams	Complete	2026-01-20	Per-agent model config, agent DIDs, LangGraph orchestratio
4.3	Strategic Intelligence Fusion & RAFT Analysis	Complete	2026-01-21	Neo4j graph, entity resolution OSINT validity dashboard
4.4	Mission Context & Force Onboarding	Complete	2026-01-25	Participant management, command relationships, resource inventory
4.5	ATAK/CoT Tactical Interoperability	Not Started	—	CoT protocol, TAK Server integration
5	Operational Planning Module	Complete	2026-02-11	JP 5-0 workflow, COA development, ROE enforceme OPORD generation
5.1	MDMP Governance Integration	Complete	2026-02-11	Five-tier authority, 18 governance gates, assumption registry
5.2	Escalation & Competition Modeling	Complete	2026-02-13	Adversary modeler, escalation ladder, effects cascader, wargaming
5.3	End-to-End Scenario Validation & UX Cleanup	Complete	2026-02-23	Scenario seed, persona walkthrough, sidebar cleanup
6	Autonomous Vehicle Integration	Not Started	—	Jetson Orin Nano setup, Sphe RVR+ SDK; superseded by Phases 43-46

Phase	Name	Status	Completed	Capability Summary
7	Tactical Execution System	Not Started	—	Commander interface, target selection; superseded by Phas 44/46
8	Sensor Fusion & Intelligence	Not Started	—	Multi-level intelligence architecture
9	Assessment & Dashboard	Not Started	—	MOE/MOP calculation, decisi support
10	End-to-End Integration	Not Started	—	BDA feedback loops
11	User Experience & Personalization	Not Started	—	Cinematic briefings, personalized command center
12	Coalition & Multi-Tenancy	Not Started	—	NATO classification, informati sharing
12.1	Coalition Health Monitoring	Not Started	—	Coalition health agent, nation caveat tracking
13	Research Whitepaper	Complete	2026-01-24	Academic whitepaper v0.1, SITREP, demo script, export pipeline
14	Friendly & Adversary IPB Complete Cycle	Complete	2026-02-28	Dual-perspective IPB, COA scoring, commander decision matrix, order generation
15	JPP Staff Organization Workspaces	Complete	2026-02-28	Per-role workspaces, cross-sta notifications, staff AI coordination
16	AI Assigned Staff Workspaces	Not Started	—	AI-assigned staff roles, full ag team execution
17	Deployment — CI/CD Pipeline & Hetzner Server	Not Started	—	GitHub Actions CI/CD, Hetzne deployment
21	AI COP Layer Agent Team	Complete	2026-03-05	MIL-STD-2525D SVG overlays entity linker, layer governanc perspective toggle
22	Training/ Operational Global Mode	Complete	2026-03-06	Global mode toggle, EXERCIS banner, data isolation, governance parity
23	Problem Set Model & Workspace Rename	Not Started	—	Rename workspaces to proble sets, echelon-awareness
24	Doctrinal Tab Restructure	Complete	2026-03-06	

Phase	Name	Status	Completed	Capability Summary
				Understand/Design/Plan/Deci COP/Assess tabs replacing legacy layout
25	Operational Design Workspace	Complete	2026-03-06	CoG analysis, LOEs, problem framing, design-to-plan hando
25.1	Training Package Upload & From-Scenario Integration	Complete	2026-03-06	ScenarioPackageUpload wizar async LLM extraction in Understand tab
25.2	Strategic Document Containers & Actor Categorization	Not Started	—	Nation/group containers with actor categories
25.3	AI Strategic Context & Knowledge Graph Integration	Not Started	—	Container-scoped graph wirec into AI agent context
26	Strategic Environment & Inheritance	Not Started	—	Strategic context provider, directive inheritance
27	Resource Registry & DID Plugin Architecture	Complete	2026-03-07	did:near:resource DIDs, 5 plu types, COP integration, telemetry
28	Embedded DAO Governance at Decision Gates	Not Started	—	Contextual governance at planning decision points
29	Contextual AI Staff Integration	Not Started	—	Per-tab AI assistant, workflow phase-aware recommendation
30	Ironclaw Agent Integration	Complete	2026-03-07	Ironclaw as chief-of-staff: code changes, problem set config, delegation
31	AI Agent Validation & Compliance Testing	Not Started	—	Quantitative validation of AI agent integration
32	Network Device Discovery & Secure Automatic	Complete	2026-03-07	BLE/WiFi/USB/TAK scanning, fingerprinting, DAO-governed acceptance

Phase	Name	Status	Completed	Capability Summary
	Resource Onboarding			
33	JPP Campaign Plan Framework	Complete	2026-03-08	Full 7-step JPP workflow, ends ways-means linkage, Argus OSINT
34	Plan Tab Echelon Routing & MDMP Tactical Wiring	Complete	2026-03-08	Echelon router (strategic/ operational/tactical), MDMP Plan tab
35	Mission Creation from OPORD & Problem Set Alignment	Complete	2026-03-08	Tactical child problem set creation from OPORD Step 7, MDMP initialization
36	Strategic Guidance Workflow	Complete	2026-03-08	Objective setting, force apportionment, directive drafting at strategic echelon
37	Training Assessment Loop	Complete	2026-03-08	AAR capture, METL proficienc (T/P/U), upward aggregation
38	Inheritance Deepening	Complete	2026-03-08	Change notification, override tracking, FRAGO propagation upward reporting
39	Operational Demonstration Data Package	Complete	2026-03-09	Pacific Strategy AY26 complet seed: strategy through tactica missions
40	Autonomous Document Intelligence Team	Complete	2026-03-09	10-specialist agent team, scoping interview, NATO A-F/ ratings, ExtractionTheater
41	Redesign Understanding Tab — Adaptive Brain Visualization	Complete	2026-03-10	Neural brain canvas, shape-coded nodes, clustering mode brain timeline
42	Resources Tab — Inventory, Discovery & Onboarding	Complete	2026-03-12	Consolidated Resources tab: inventory, discovery, network, groups sub-views
43	Robot Agent & Local Discovery Bridge	Complete	2026-03-12	Python robot agent, Docker bridge, mDNS auto-discovery, command proxy
44	Robot Vision Capabilities &	Complete	2026-03-13	

Phase	Name	Status	Completed	Capability Summary
	Mission Intent Translation			CSI camera, detectNet, ORB matcher, 4 vision mission type sweep planning
45	Knowledge Graph Subspaces	Not Started	—	Container-scoped subgraphs, focus-and-expand, virtual lens
46	Sphero RVR+ Swarm Leader & Doctrinal Movement Control	Complete	2026-03-14	Swarm leader, 6 formations, UDP peer mesh, DAO-driven membership, COP aggregatio
47	JSON-LD Semantic Brain + COP Fix	Not Started	—	JSON-LD ontology, BFO/CCO alignment, contradiction detection, COP pipeline fix
48	Robot Swarm Behaviour End-to-End Demo	Not Started	—	Complete Taiwan scenario de with 3-robot coalition swarm
49	Align Design Tab with Plan Tab	Complete	2026-03-17	Design as single source of tru fork-and-merge revision syste
50	Universal Intelligence Input & Auto-Classification	Not Started	—	Single universal input replaci fragmented ingestion
51	Unified Agent Architecture	Complete	2026-03-17	StandardAgent base class, ag admin dashboard, team desig
52	Agent Skills & MCP	Complete	2026-03-18	Standalone MCP server, skills registry, Ironclaw builder acti cards
53	DID Governance Architecture & Bug Fixes	Complete	2026-03-19	Decide tab, decision dashboar RACI matrix, PendingDecisionModal, Ironc 60s polling
54	Briefing Deck: Research Validation & Slide Specs	Complete	2026-03-26	Slide-by-slide image specifications and visual brief standards
55	Ironclaw Chief of Staff — Operational Design	Complete	2026-03-25	LangGraph JP 5-0 staff coordination: 4 sections, coverage criteria, input validation, KG gap dispatch
56	Visual Operational Approach Editor	Complete	2026-03-25	MapOverlay data model, MIL-STD-2525D symbols + contro measures, AI-directed placement, COP integration
57		Complete	2026-03-25	

Phase	Name	Status	Completed	Capability Summary
	Ironclaw Persistent Memory			Dual-scope memory (user + context), REST API, IronclawMemoryPanel, auth-scoped isolation
58	On-Chain Resource DID Caveats	Complete	2026-03-26	ResourceCaveats struct, check_employment_authorize did.bastion.testnet deploymen 4/4 smoke tests

A.2 Current Status

Metric	Value
Total phases	75
Completed phases	58
Total plans	469+
AI agents (LangGraph analysis)	8
AI agents (COP layer)	7
AI agents (Chief of Staff)	1 (Ironclaw)
AI agents (total deployed)	16
Smart contract modules	14
REST API endpoints	~572+
Doctrinal tabs	6 (Understand/Design/Plan/Decide/COP/Assess)
Resource type plugins	5
Robot platform supported	Sphero RVR+ / NVIDIA Jetson Orin Nano
Swarm formations	6 doctrinal formations
Document intelligence specialists	10
Design coordination sections	4 (JP 5-0 operational design)
Memory scopes	2 (user-scoped 90d TTL, context-scoped 180d TTL)
Caveat dimensions	5 (classification, releasability, ROE tier, geo bounds, time windows)

Development Stack:

- Backend: Node.js/Express with TypeScript
- Frontend: React 19 + Vite + TypeScript 5.9
- Blockchain: NEAR Protocol (testnet)
- Graph Database: Neo4j 2025 Community
- Orchestration: LangGraph + LangChain

- Container: Docker Compose
- TEE: Phala Network
- Authentication: WebAuthn passkeys with PRF extension
- Robot Agent: Python 3.11 on NVIDIA Jetson Orin Nano
- Local Bridge: Docker container (mDNS/SSDP, command proxy)
- Vision: detectNet object detection, ORB feature matching

A.3 Remaining Work

Phase 1.1: Calimero Self-Sovereign App Integration

- Research Calimero self-sovereign applications for DAO compartmentalization
- Replace Privy remnants with NEAR accounts + MPC

Phase 4.5: ATAK/CoT Tactical Interoperability

- CoT message protocol and TAK Server integration
- Real-time position sharing and data package export

Phase 6–12: Original Phases (partially superseded)

- Phases 6 (Autonomous Vehicle Integration) and 7 (Tactical Execution System) were the original vehicle integration path; superseded by Phases 43-46 (robot agent, bridge, vision, swarm) which deliver equivalent or superior capability
- Phase 8 (Sensor Fusion), Phase 9 (Assessment Dashboard), Phase 10 (End-to-End Integration) remain as future work
- Phase 11 (User Experience & Personalization) — cinematic briefings, personalized command centers
- Phase 12 (Coalition & Multi-Tenancy), Phase 12.1 (Coalition Health Monitoring) — NATO classification, information sharing rules

Phase 16: AI Assigned Staff Workspaces

- AI-assigned staff roles with full agent team execution
- Human-in-the-loop review and real-time channel observability

Phase 17: Deployment — CI/CD Pipeline & Hetzner Server

- GitHub Actions CI/CD pipeline
- TEE-aware component separation for production
- Hetzner VPS deployment with single-server dev configuration

Phase 23: Problem Set Model & Workspace Rename

- Rename workspaces to problem sets (JP 5-0 terminology) throughout application
- Add echelon-awareness (strategic/operational/tactical)

Phase 25.2: Strategic Document Containers & Actor Categorization

- Nation/group containers with actor categories (ally, adversary, neutral, partner)
- Persistent container-based organization for building strategic environments

Phase 25.3: AI Strategic Context & Knowledge Graph Integration

- Container-scoped knowledge graphs wired into AI agent context
- RAFT graph auto-construction on document changes

Phase 26: Strategic Environment & Inheritance

- Strategic-level problem set as context provider
- Inheritance mechanism for directives, policy, and intelligence

Phase 28: Embedded DAO Governance at Decision Gates

- Contextual workflow decision gates replacing dedicated governance view
- Proposals trigger at natural planning decision points

Phase 29: Contextual AI Staff Integration

- Per-tab AI assistant aware of workflow phase
- Recommendation engine tied to doctrinal workflow position

Phase 31: AI Agent Validation & Compliance Testing

- Quantitative validation framework for AI agent integration
- Determinism, reliability, and doctrinal compliance dimensions

Phase 45: Knowledge Graph Subspaces

- Container-scoped subgraphs, focus-and-expand N-hop loading
- 4-level hierarchical drill-down, virtual lens system

Phase 47: JSON-LD Semantic Brain + COP Fix

- JSON-LD representation with BFO/CCO ontology alignment
- Provenance tracking, temporal reasoning, contradiction detection
- COP layer generation pipeline end-to-end fix

Phase 48: Robot Swarm Behaviour End-to-End Demo

- Complete strategy-to-autonomous-execution pipeline demonstration
- Taiwan defense scenario with 3-robot coalition swarm
- Lethal escalation gates with DAO authorization

Phase 50: Universal Intelligence Input & Auto-Classification

- Single universal input replacing fragmented ingestion sidebar
- Automatic content type discernment and specialist agent routing

A.4 MVP Demo Readiness

Component	Status	Notes
Strategic DAO	Ready	Full voting/proposal flow with coalition weights
AI Agent Coordination	Ready	LangGraph orchestration, 16 deployed agents, unified architecture
Document Ingestion	Ready	10-specialist autonomous team with NATO reliability ratings
RAFT Graph Analysis	Ready	Neo4j graph with fusion agents, brain visualization
Brain Visualization	Ready	Neural canvas, clustering modes, timeline scrubber
Validity Dashboard	Ready	Map/graph visualization with OSINT integration
Operational Planning	Ready	JP 5-0 workflow with COA development, ROE enforcement

Component	Status	Notes
MDMP Governance	Ready	Phase progression, assumption tracking, safety matrix
Escalation Modeling	Ready	Adversary COAs, escalation ladder, effects cascading
Wargaming	Ready	Action-reaction-counteraction with interactive exploration
Passkey Authentication	Ready	WebAuthn with NEAR implicit accounts
IPB Complete Cycle	Ready	Dual-perspective IPB with information isolation
COP Layer Generation	Ready	AI-generated MIL-STD-2525D overlays with publish review
Operational Design	Ready	Problem framing, CoG analysis, operational approach
Resource Registry	Ready	DID-based resources with 5 plugin types and COP integration
Training Mode	Ready	Global toggle with governance parity and data isolation
Staff Workspaces	Ready	Per-role JPP workspaces with staff AI agents
Doctrinal Tabs	Ready	6-tab JP 5-0 aligned lifecycle (Understand/Design/Plan/Decide/COP/Assess)
Echelon Routing	Ready	Strategic/operational/tactical plan workflow routing
Inheritance & Propagation	Ready	Change notification, FRAGO propagation, upward reporting
Demo Data Package	Ready	Pacific Strategy AY26 complete seed, all tabs populated
Resources Tab	Ready	Consolidated inventory, discovery, network, groups UI
Ironclaw Agent	Ready	Chief-of-staff capability with 60s decision polling
Decide Tab	Ready	Decision dashboard, RACI matrix, PendingDecisionModal
Robot Agent	Ready	Python agent for Jetson Orin Nano, WebSocket command/telemetry
Docker Bridge	Ready	Local network bridge, mDNS discovery, command proxy
Robot Vision	Ready	detectNet object detection, ORB feature matching, 4 vision mission types

Component	Status	Notes
Swarm Coordination	Ready	6 formations, UDP peer mesh, DAO-driven membership, COP aggregation
Agent Skills & MCP	Ready	Standalone MCP server, skills registry, Ironclaw builder
DID Governance	Ready	DID documents with governance policy, RACI-aware decision pipeline
Ironclaw Chief of Staff	Ready	LangGraph JP 5-0 staff coordination, 4 sections, input validation, KG gap dispatch
Visual Approach Editor	Ready	MapOverlay with MIL-STD-2525D symbols and control measures, AI-directed placement
Ironclaw Memory	Ready	Dual-scope persistent memory, REST API, management panel with auth isolation
Resource Caveats	Ready	ResourceCaveats on-chain enforcement, check_employment_authorized() on did.bastion.testnet
Physical Demo Hardware	Ready	Sphero RVR+ + Jetson Orin Nano operational
End-to-End Pipeline	Partial	Robot detection → COP → DAO → swarm mission demonstrated; JSON-LD brain + COP fix pending Phase 47
Coalition Governance	Partial	DAO governance ready; coalition multi-tenancy (Phase 12) not started

Critical Path to Full Operational Capability:

1. Phase 47: JSON-LD Semantic Brain + COP Fix (knowledge graph foundation)
2. Phase 48: Robot Swarm Behaviour End-to-End Demo (pipeline integration test)
3. Phase 17: Deployment — CI/CD pipeline for production hosting

Demo-Ready Status: The platform is ready for academic and military-stakeholder demonstration through Phase 58. The complete strategy-to-autonomous-execution pipeline is demonstrable with Pacific Strategy AY26 scenario data, physical robot swarm, full DAO governance with human authority positions at all three tiers, and Phase 55-58 additions: Ironclaw Chief of

Staff coordination for operational design, visual operational approach editor, Ironclaw persistent memory, and on-chain resource caveat enforcement.

SITREP current as of March 2026, reflecting Phase 58 completion. Implementation status tracked in `.planning/STATE.md` and `.planning/ROADMAP.md`.

Appendix B: Demonstration Script (~20 minutes)

> **See also:** `docs/briefing/demo-script-30min.md` for the expanded 30-minute version with full doctrinal tab walkthrough, robot vision, swarm telemetry, and Decide tab governance demonstration.

This appendix provides a detailed demonstration script for the BASTION platform, designed for a ~20 minute presentation that showcases all three human authority positions (in-the-loop, on-the-loop, out-of-the-loop) and the cross-level coordination that connects strategic objectives to tactical execution.

Narrative Approach: This script weaves technical demonstration with problem-focused narrative. At key moments, "injects" highlight specific problems BASTION solves versus conventional C2 architecture. The goal is not just to show *what* the system does, but *why* it matters.

B.1 Setup Requirements

Physical Components

Component	Purpose	Notes
NVIDIA Jetson Orin Nano Super	Edge AI computing platform	67 TOPS AI performance, runs object detection
Sphero RVR+ with camera mount	Mobile autonomous platform	Programmable robot for tactical execution
Physical AO model	Demonstration area of operations	Tabletop terrain with target markers

Component	Purpose	Notes
Display/projector	BASTION UI presentation	Shows governance interface, agent activity
Network connectivity	System communication	Jetson-to-backend, browser access

System State

Before demonstration begins, verify:

- [ ] BASTION platform running (docker-compose up)
- [ ] Training mode activated (amber EXERCISE banner visible)
- [ ] Scenario data seeded: `bash scripts/seed-scenario.sh` (seeds graph, mission, plan, COAs, command structure, resources, MDMP workflow)
- [ ] Mock governance data enabled: `VITE_USE_MOCK_DATA=true` in frontend/.env.local (provides coalition DAO with USA/GBR/CAN and scenario-aligned proposals)
- [ ] Pacific Strategy AY26 scenario training package available for upload
- [ ] AI agents initialized and healthy
- [ ] Sphero RVR+ powered and connected
- [ ] Jetson Orin Nano running edge AI models
- [ ] Camera feed visible in monitoring interface

Pre-Demo Checklist

```
[ ] Backend services healthy (check /api/health endpoints)
[ ] Neo4j graph database connected
[ ] Scenario seed script run successfully (bash scripts/seed-
scenario.sh)
[ ] NEAR testnet accessible
[ ] LLM provider API keys valid
[ ] Phala TEE attestation current
```

```
[ ] All three coalition member accounts ready
[ ] Strike authorization proposal pre-staged (not submitted)
```

B.2 Act 1: Strategic Level (5 minutes)

Human Authority Position: IN-THE-LOOP

All decisions at the strategic level require explicit human approval. AI assists with analysis and presentation but cannot approve strategic commitments.

[0:00-1:30] Introduction and Problem Framing

Narrator speaks while showing BASTION in training mode with amber EXERCISE banner

> "Welcome to the BASTION demonstration. Notice the amber EXERCISE banner across the top — we're in training mode, which uses identical governance to operational mode. This is 'train as you fight' — every vote, every approval gate, every safety constraint works exactly as it would in operations."

Pause for emphasis

> "In conventional coalition operations, a strategic directive might take days to translate into tactical action. Why? The document must be manually analyzed by staff officers. Extracted objectives must be coordinated through national channels. Each partner must staff recommendations through their own approval chains. By the time everyone agrees, the operational window may have closed."

> "BASTION solves this through AI-augmented decentralized autonomous organizations. Over the next twenty minutes, you'll see strategic intent flow through a doctrine-aligned workflow — Understand, Design, Plan, Decide, COP, Assess — mirroring JP 5-0 rather than imposing arbitrary software categories. You'll see 131 AI agents accelerate coordination while preserving human authority."

Action: Display the six doctrinal tabs with the Understand tab active

Show: - Six tabs: Understand / Design / Plan / Decide / COP / Assess - Coalition partners (USA, GBR, CAN) with voting weights - Training mode banner and EXERCISE watermark - Agent status panel showing healthy agents

> "Notice the tab structure follows military doctrine, not software convention. Each tab corresponds to a phase of the joint planning process. And notice the coalition partners — in a conventional system, each nation would maintain separate planning tools. BASTION provides a shared operational picture while respecting national authorities."

[1:30-2:30] Document Ingestion via Understand Tab

Narrator explains the Understand tab and document processing

> "We begin in the Understand tab — the doctrinal starting point. Strategic planning starts with understanding the operational environment. In conventional operations, this step is a bottleneck. Staff officers manually review documents, extract relevant objectives, and summarize them for commanders. This process can take hours or days."

Action: Upload Pacific Strategy AY26 scenario training package via ScenarioPackageUpload in the Understand tab

> "We're uploading a scenario training package — multiple documents at once. Watch the AI categorize them automatically."

Show: - Multi-file drag-drop upload in Understand tab - AI tag inference categorizing documents by type, team, and exercise phase - Async LLM extraction processing with status indicators - RAFT graph populating with actors and relationships from uploaded intelligence

System Response: - AI extracts objectives and entities automatically - Documents categorized (intelligence, orders, estimates) - RAFT graph shows actor relationships and tensions

> "In seconds, the AI has categorized documents, extracted strategic objectives, and built a knowledge graph of actors and relationships. A task that might consume an entire staff section's morning."

[INJECT - SPEED vs. CONVENTIONAL C2]

> "But here's the critical point: in a conventional system, you'd now wait for staffing. Each extracted objective would need to be validated through national channels, reconciled with partner interpretations, and formally coordinated. BASTION doesn't eliminate that coordination, it accelerates it through structured governance."

[2:30-4:00] Human Review and DAO Governance

Narrator emphasizes human-in-the-loop

> "Every extracted objective requires human review before entering governance. The AI accelerates analysis, but humans make decisions. This is non-negotiable."

Action: - Click on first extracted objective - Show ObjectiveDetail view with tabs - Review AI-extracted content

Show: - Source text highlighted alongside extraction - DIME/MIDLIFE categorization - Risk assessment with confidence scores

Action: Edit one objective slightly to demonstrate human modification

> "The reviewer can accept the AI's extraction, modify it, or reject it entirely. Let's approve this reconnaissance objective with a clarification."

Action: Click "Approve" button

[INJECT - TRANSPARENCY vs. CONVENTIONAL C2]

> "In a conventional system, this approval happens in an email chain or a meeting that no one records. Who approved what? When? Based on what information? Those questions often can't be answered months later when something goes wrong."

Gesture to screen

> "Every action in BASTION is recorded on an immutable blockchain. This approval, right now, is permanently logged with the reviewer's identity, timestamp, and the exact state of the objective they approved. Full accountability, forever."

[4:00-5:00] Coalition Voting

Narrator explains coalition governance

> "Approved objectives become proposals to the Strategic DAO. Coalition members vote according to configured weights."

Action: Submit resource allocation proposal to Strategic DAO

Show: - Proposal details (objective, requested resources, justification) - Coalition members visible with voting weights - Voting interface

Action: - USA representative votes "Approve" - GBR representative votes "Approve" - CAN representative votes "Approve"

Show: - Votes recording on blockchain (transaction hash visible) - Quorum reached notification - Proposal status changes to "Approved"

[INJECT - TRUST and VERIFICATION vs. CONVENTIONAL C2]

> "This vote just recorded on NEAR blockchain. Every coalition partner can independently verify that the vote happened, that it met quorum requirements, and that resources were allocated as agreed."

> "In conventional coalition operations, partners must trust that coordinating nations honor commitments. Did the resources we were promised actually get allocated? Is our equipment being used as we intended? These questions erode coalition trust. Blockchain eliminates them through cryptographic verification."

B.2.5 Operational Design (2 minutes)

Bridging Strategic Understanding to Operational Planning

[5:00-5:45] Design Tab — Problem Framing and CoG Analysis

Narrator transitions to the Design tab

> "Now we move to the Design tab — operational design. This is where commanders translate understanding into an approach. In conventional operations, this is often the weakest link: strategic guidance gets handed to planners who jump straight to courses of action without systematic problem framing."

Action: Navigate to Design tab, show problem framing canvas

Show: - Problem framing canvas with AI-identified tensions - Center of gravity analysis using Strange's CG-CC-CR-CV framework - Lines of effort with linkages to strategic objectives

> "The AI has identified key tensions in the strategic environment and suggested initial framings. The commander refines these — the AI provides starting points, not answers."

Action: Show CoG analysis for adversary forces

> "Center of gravity analysis follows Strange's framework: Critical Capabilities, Requirements, and Vulnerabilities. This structured analysis feeds directly into course of action development."

[5:45-6:15] Design-to-Plan Handoff

Action: Click "Export to Plan" button

> "Watch this handoff — operational design outputs flow directly into the Plan tab's mission analysis. No manual translation, no interpretation errors between the design team and the planning staff."

Show: - Design outputs appearing as Plan tab inputs - Objectives, CoG analysis, and LOEs populated in mission analysis

[INJECT - OPERATIONAL DESIGN GAP]

> "In conventional staffs, the gap between operational design and detailed planning is where intent gets lost. BASTION eliminates that gap through structured data handoff."

B.3 Act 2: Operational Level (4 minutes)

Human Authority Position: ON-THE-LOOP

AI agents operate continuously while humans monitor with override capability. Humans do not approve every action but can intervene at any point.

[6:15-7:30] AI Agent Activation

Narrator transitions to operational coordination

> "With strategic resources allocated and operational design complete, we move to detailed planning. This is where conventional C2 struggles most. The gap between strategic intent and tactical execution is where coalitions historically lose coherence."

Action: Navigate to Plan tab, show Agent Orchestration Panel

Show: - Active agents list with status indicators - LangGraph execution visualization - Agent communication in message bus

> "Multiple AI agents are now active. The Operational Planning Agent translates strategic objectives into executable plans. The Risk Assessment Agent identifies hazards. The Resource Mapping Agent matches available assets to requirements."

[INJECT - SILOED DECISION-MAKING vs. CONVENTIONAL C2]

> "In a conventional staff, these functions happen in separate cells that may not communicate effectively. The operations cell develops plans without full visibility into intelligence assessments. The logistics cell allocates resources without understanding tactical priorities. BASTION's agents share information in real-time through a secure message bus, eliminating the silos that fragment conventional planning."

System: - Agents begin processing - Reasoning traces appear in execution panel - Recommendations generate in real-time

---

[6:30-8:00] Human Monitoring with Override Capability

Narrator emphasizes on-the-loop authority

> "At the operational level, the human commander is ON-THE-LOOP. Agents work at machine speed, but the commander monitors everything and can override instantly."

Action: Point to monitoring dashboard

Show: - Operational coordination view - Agent activity log scrolling - Plan generation progress

> "Watch the commander's view. Every agent action is visible. The commander sees what agents recommend and why."

Action: Hover over "Override" button (but don't click)

> "This override button immediately halts agent operations and returns full control to the human. The commander chooses not to use it because agents are operating within expected parameters."

[INJECT - DECISION TEMPO vs. CONVENTIONAL C2]

> "Conventional C2 operates at human speed. Every coordination step waits for humans to process information, make decisions, and communicate. BASTION preserves human authority while enabling machine-speed coordination for routine tasks."

> "The commander isn't approving every agent recommendation. That would eliminate the speed advantage. Instead, the commander maintains oversight with the ability to intervene when needed. Human judgment where it matters, AI speed everywhere else."

System: - Agents complete plan generation - Mission plan summary appears - Risk acknowledgment prompt appears

---

[8:00-10:00] Mission Authorization and Risk Acknowledgment

Narrator explains transition to tactical level

> "The AI agents have generated an operational plan assigning our reconnaissance asset to survey the demonstration area and identify targets matching specified criteria."

Show: - Completed operational plan summary - Task assignment to Sphero/Jetson platform - Mission constraints and boundaries

Action: Commander reviews plan summary

> "The commander reviews the generated plan, acknowledges identified risks, and authorizes mission execution."

Show: - Risk assessment summary panel - Identified risks with likelihood/impact - Required acknowledgments

Action: Commander explicitly acknowledges risks, then clicks "Authorize Mission"

[INJECT - ACCOUNTABILITY vs. CONVENTIONAL C2]

> "This risk acknowledgment creates accountability. In conventional operations, commanders often inherit risk assessments buried in annexes that may never be read. Here, the commander explicitly acknowledges: 'I understand these risks and accept them.' That acknowledgment is permanently recorded."

> "If this operation goes wrong, we know exactly who authorized it, what risks they were told about, and that they accepted those risks. Not to assign blame, but to learn and improve."

System: - Risk acknowledgment recorded on blockchain - Mission status changes to "Authorized" - Tactical handoff notification sent

---

B.4 Act 3: Tactical Level (5 minutes)

Human Authority Position: OUT-OF-THE-LOOP (Policy Bounded)

Autonomous systems execute within policy constraints without real-time human approval for routine operations. Strike authorization is the critical exception requiring human approval.

[10:00-12:00] Autonomous Reconnaissance

Narrator explains tactical autonomy

> "At the tactical level, humans move OUT-OF-THE-LOOP for routine operations. The Sphero robot will navigate autonomously. The Jetson's AI will process imagery without waiting for human approval of each frame."

Action: Direct attention to physical demonstration area

Show: - Map view with Sphero position marker - Camera feed from Sphero/Jetson - Navigation path overlay

Action: Sphero begins autonomous patrol of AO model

> "The Sphero is executing autonomous reconnaissance. Watch it navigate terrain, avoid obstacles, and follow its assigned search pattern."

[INJECT - DDIL RESILIENCE]

As Sphero moves, narrator introduces simulated scenario:

> "Let's imagine something realistic. The robot encounters unexpected electronic warfare interference. In a conventional system with centralized C2, losing communication with headquarters would paralyze the platform. It can't receive orders, can't report status, can't continue its mission."

Gesture to Sphero continuing its patrol

> "But our robot keeps moving. Why? Because BASTION's edge AI enables mission continuation without continuous connectivity. The Jetson carries the mission parameters, the navigation model, and the target identification capability. It doesn't need headquarters to tell it what to do next. This is DDIL resilience, the ability to operate when Disconnected, Degraded, Intermittent, or Limited-bandwidth conditions prevent communication with higher echelons."

System: - Edge AI processing camera feed - Sphero navigating tabletop AO - Position updates appearing on map

> "Human OUT-OF-THE-LOOP for navigation decisions. The policy defined boundaries and objectives. Within those boundaries, the robot operates autonomously."

[12:00-13:00] Target Identification

Narrator explains AI target detection

> "The Jetson Orin Nano runs object detection directly on the robot. When it identifies an object matching target criteria, it classifies and reports."

System: - Jetson AI detects target marker on AO model - Bounding box appears on camera feed - Classification result displayed

Show: - Target classification: "VEHICLE - HOSTILE" (demonstration marker) - Confidence score: 94% - Location coordinates on map

> "The AI has identified a target with 94% confidence. This identification happened autonomously. No human approved examining that specific location or classifying that specific object."

[INJECT - EDGE AI vs. CLOUD DEPENDENCY]

> "In conventional architectures, this imagery would need to be transmitted to a cloud processing center for analysis. That transmission takes bandwidth, introduces latency, and creates vulnerability. If the link is jammed or degraded, analysis stops."

> "BASTION pushes AI to the edge. The Jetson processes locally, reports results when connectivity allows, and continues operating regardless. The robot doesn't stop being useful just because it can't phone home."

[13:00-15:00] Strike Authorization - Return to Human Control

Narrator emphasizes critical distinction

> "Now watch carefully. This is the most important part of the demonstration."

> "The target meets engagement criteria. The tactical system could theoretically engage autonomously. But BASTION enforces an inviolable constraint."

Action: System generates strike authorization proposal

Show: - Strike authorization proposal appears - Red warning indicators and pulsing borders - "REQUIRES HUMAN APPROVAL" banner

Speak slowly and deliberately:

> "STRIKE AUTHORIZATION ALWAYS REQUIRES HUMAN APPROVAL."

Pause for emphasis

> "No configuration setting, no operational urgency, no chain of command can bypass this requirement. This is hardcoded at every level of the architecture. Lethal decisions require human authorization. Period."

[INJECT - AUTONOMOUS WEAPONS ETHICS]

> "This is BASTION's answer to concerns about autonomous weapons. The robot navigated autonomously. It identified the target autonomously. But it cannot kill autonomously. The decision to apply lethal force always returns to human judgment."

> "International discussions about autonomous weapons often frame this as a tradeoff: you can have speed or you can have human control. BASTION demonstrates that's a false choice. We can accelerate everything except the lethal decision. AI handles coordination. Humans handle consequences."

Action: - Coalition members receive strike proposal notification - Display voting interface with 100% threshold shown

Show: - Strike proposal details (target data, effects requested, confidence) - All coalition members must vote - 100% approval threshold displayed

> "Every coalition member must vote. The threshold is unanimous. One 'No' vote blocks the strike."

Action: - Commander reviews target data - All coalition members vote Approve

Show: - Votes recording - 100% approval achieved - Strike authorized

> "Human-authorized strike. AI identified. AI recommended. Humans decided."

[15:00-15:30] Effects Delivery

System: - Strike authorization confirmed - Sphero moves to engagement position - Simulated effects delivered (LED flash, audio tone)

Action: Sphero executes authorized engagement

> "The tactical asset executes the human-authorized strike. For demonstration, 'effects' are visual and audio indicators."

Show: - Engagement recorded on blockchain - Audit trail entry with all approvals - Mission status updated

> "That engagement is permanently recorded: who identified, who authorized, when, and outcome. Full accountability through immutable audit."

B.4.5 COP and Resource Registry (2 minutes)

AI-Generated Operational Picture and Resource Tracking

[15:30-16:15] COP Layer Generation

Narrator transitions to COP tab

> "While tactical operations proceed, AI agents have been working autonomously. Let me show you the COP tab."

Action: Navigate to COP tab

Show: - AI-generated MIL-STD-2525D overlays on the map - Friendly force symbols (blue) and adversary symbols (red) - Phase slider showing force positions over time - Resource registry entries as military symbols on the COP

> "These military symbols were not placed manually. AI agents parsed our planning documents, orders, and intelligence reports, then generated standard MIL-STD-2525D symbology automatically. Each symbol links back to its source document."

Action: Click on a symbol to show detail panel with source document linkage

> "Every symbol went through a publish review cycle — AI generates, humans approve before it reaches the shared COP. Governance even for the operational picture."

[16:15-16:45] Resource Registry

Action: Show resource entries on COP with DID identifiers

> "Notice these resource symbols. Each has a blockchain-anchored Decentralized Identifier — a `did:near:resource` DID. The Sphero reconnaissance asset, sensors, weapons systems — all tracked as first-class entities with real-time readiness status."

Show: - Resource symbols with FMC/PMC/NMC readiness indicators - DID identifier displayed in detail panel - Plugin-based resource type (autonomous vehicle)

[INJECT - RESOURCE ACCOUNTABILITY]

> "In conventional operations, resource tracking lives in spreadsheets and logistics databases that commanders rarely see in real-time. BASTION makes every asset a blockchain-verified entity, visible on the COP, with DAO-governed allocation. You can't lose track of an asset when it has a permanent identity on the blockchain."

B.5 Act 4: Cross-Level Coordination (2 minutes)

Demonstrating Interlink Between Tactical and Strategic DAOs

[16:45-17:45] Automatic Strategic Coordination

Narrator explains cross-DAO automation

> "Now I'll show you something that simply doesn't happen in conventional C2."

System: - AI agent detects resource expenditure - Inventory drops below threshold - Agent generates replenishment proposal automatically

Show: - Resource state monitoring panel - Inventory showing depletion - Agent reasoning trace

> "The engagement expended resources. Watch what happens next."

System: - Proposal appears in Strategic DAO queue - Proposal type: "Resource Replenishment Request" - Justification auto-populated from tactical context

[INJECT - CROSS-LEVEL COORDINATION vs. CONVENTIONAL C2]

> "In conventional operations, this replenishment request would require a tactical unit to file a logistics request, which routes through operational channels, which eventually reaches strategic resource managers, who must staff the request through national approval processes. That cycle takes days or weeks."

> "BASTION just completed it in seconds. The tactical expenditure automatically triggered a strategic governance action. Coalition members can now vote on replenishment through the same governance process used for initial allocation."

Gesture to the full loop

> "Strategic resources funded the operation. Operational AI planned the mission. Tactical autonomy executed within policy. Tactical expenditure triggered strategic replenishment. The complete cycle, without manual coordination overhead."

[17:00-18:00] Summary Visualization

Show: - Full loop visualization diagram - Strategic → Operational → Tactical → Strategic - Arrows showing information flow

> "This seamless coordination across all levels is what BASTION enables. Interconnected, AI-augmented DAOs that preserve human authority while eliminating coordination friction."

[INJECT - COALITION TRUST]

> "And critically, every step of this loop is visible to all coalition partners. No black boxes. No 'trust us' assurances. Every resource allocation, every decision, every action is cryptographically verified and permanently recorded. Coalition partners don't need to trust each other. They verify."

B.6 Conclusion (2 minutes)

[18:00-19:30] Problems Solved

Narrator directly addresses the research contribution

> "Let me be explicit about what you just witnessed."

Enumerate problems solved:

> "First: **Siloed decision-making**. You saw AI agents share information across functional boundaries in real-time. No more operations cell unaware of intelligence assessments."

> "Second: **Coalition coordination friction**. Strategic voting happened in seconds with cryptographic verification. No more weeks of staffing through national channels."

> "Third: **DDIL vulnerability**. The robot continued its mission during simulated communications disruption. Edge AI enables operation without constant connectivity."

> "Fourth: **Accountability gaps**. Every decision from strategic objective to tactical effect is permanently recorded. Who authorized what? Always answerable."

> "Fifth: **Speed versus control tradeoff**. We demonstrated machine-speed coordination with human authority over lethal decisions. The false choice is resolved."

[19:30-20:00] Research Question Answer

Narrator recaps research question

> "The research question asked how AI-augmented DAOs can provide secure, transparent, and resilient governance for military coordination."

> "You just saw the answer: blockchain provides transparency and verification. Smart contracts enforce policy. AI accelerates coordination. Graduated human authority preserves control where it matters most."

> "The complete audit trail is available for review."

Show: - Blockchain transaction history - Proposal/vote records - Agent execution traces - Strike authorization with full chain of approvals

> "Every claim is verifiable in the system's records. Thank you. I'm prepared for questions."

B.7 Contingency Notes

If System Fails During Demo

Backup Option A: Pre-recorded Video - Have screen recording of complete demonstration available - Narrate over recording as if live - Acknowledge technical difficulty briefly, continue presentation

Backup Option B: Static Screenshots with Narration - Prepared screenshot deck covering all major phases - Walk through screenshots explaining what would happen - Focus on architecture and governance concepts over live execution

If Time Runs Short

If at 15 minutes with Act 4 not started: - Skip Act 4 detailed walkthrough - Verbally summarize cross-level coordination - Proceed directly to Conclusion

Must-Show Priority: 1. Act 3 strike authorization human control (highest priority) 2. DDIL resilience inject during autonomous patrol (high priority) 3. Act 1 human-in-the-loop approval with accountability inject (high priority) 4. Act 2 human-on-the-loop monitoring (medium priority) 5. Act 4 cross-level coordination (lower priority if time constrained)

If Questions Arise Mid-Demo

- Acknowledge question briefly
- Note for Q&A period
- Continue demonstration to maintain pacing
- Return to question after Conclusion

Hardware Troubleshooting

Issue	Quick Fix
Sphero not responding	Restart Sphero, reconnect Bluetooth
Jetson camera not visible	Check USB connection, restart camera service
UI not loading	Refresh browser, check backend health
Blockchain transaction fails	Use pre-approved demo data, explain what would happen

---

B.8 Inject Summary Reference

Quick reference for narrative injects and their connection to paper themes:

Demo Moment	Inject Theme	Paper Reference
Training mode activation	Train as you fight	Section 3.11
Scenario upload (Understand tab)	Speed vs. conventional C2 staffing delays	Section 1.1, 1.3, 3.7
Human approval	Transparency/ accountability gap	Section 1.1
Coalition voting	Trust and verification	Section 1.3, 2.6
Operational design (Design tab)	Operational design gap	Section 3.7
Design-to-Plan handoff	Doctrine-first workflow	Section 3.7
Agent orchestration	Siloed decision-making	Section 1.1, 2.6
On-the-loop monitoring	Decision tempo	Section 1.3
Risk acknowledgment	Accountability	Section 5.3
Autonomous patrol	DDIL resilience	Section 1.3, 2.5-2.6
Edge AI processing	Cloud dependency vulnerability	Section 2.5
Strike authorization	Autonomous weapons ethics	Section 5.3
COP layer generation	AI operational picture	Section 3.9
Resource registry on COP	Resource accountability	Section 3.10
Cross-level replenishment	Coalition coordination friction	Section 1.1, 2.6
Audit trail	Coalition trust verification	Section 1.1, 1.5

Script designed for ~20 minute presentation with narrative injects that connect demonstration to paper thesis. The story arc moves from problem framing through solution demonstration, with each inject deliberately highlighting a specific problem BASTION solves versus conventional C2 architecture.

---

1. Problem framing draws from the research proposal approved August 2025. See: Aaron Luhning, "Strategic Research Requirement Project Proposal" (unpublished manuscript, August 2025).↩
2. [CITATION NEEDED] - Reference on national caveats and intelligence sharing constraints in coalition operations.↩
3. [CITATION NEEDED] - Reference on transparency and accountability challenges in multinational operations.↩
4. Additional citation required for this claim.↩
5. Additional citation required for this claim.↩
6. [CITATION NEEDED] - Reference on adversary strategies to exploit coalition coordination gaps.↩
7. [CITATION NEEDED] - Reference on classification systems and their impact on information sharing.↩
8. Additional citation required for this claim.↩
9. [CITATION NEEDED] - Reference on decision speed requirements in modern warfare.↩
10. Joint Publication 3-0, Joint Operations (Washington, DC: Joint Chiefs of Staff, 2022). [CITATION NEEDED - verify current edition]↩
11. Department of Defense, "Joint All-Domain Command and Control (JADC2) Strategy" (Washington, DC: DoD, 2022). [CITATION NEEDED - verify document reference]↩
12. NATO, "Federated Mission Networking," accessed January 2026. [CITATION NEEDED - proper citation]↩
13. [CITATION NEEDED] - Reference on military AI decision support systems.↩
14. [CITATION NEEDED] - Reference on DAO governance mechanisms and commercial applications.↩
15. [CITATION NEEDED] - Reference on edge computing paradigm and its advantages over centralized cloud architectures in latency-sensitive applications.↩

16. NVIDIA Corporation, “Jetson Orin Nano Series System-on-Module Data Sheet,” 2023. [CITATION NEEDED - verify current datasheet reference]↩
17. [CITATION NEEDED] - Reference on JADC2 and Project Convergence edge AI objectives from DoD official documentation.↩
18. [CITATION NEEDED] - Reference on DoD Unmanned Systems Integrated Roadmap with UGV capability projections.↩
19. [CITATION NEEDED] - Reference on human-machine teaming doctrine for unmanned ground systems, Army or Joint doctrine.↩
20. [CITATION NEEDED] - Reference on validation methodology using commercial proxy platforms for military autonomous system governance research.↩
21. [CITATION NEEDED] - Reference on DDIL environment definition and its implications for military communications architecture.↩
22. [CITATION NEEDED] - Reference on edge autonomy in disconnected operations — mission persistence and local policy enforcement.↩
23. [CITATION NEEDED] - Reference on safety invariants for autonomous military systems — fail-safe behaviors and abort protocols.↩
24. [CITATION NEEDED] - Reference on swarm robotics principles, emergent collective behavior, and stigmergic coordination.↩
25. [CITATION NEEDED] - Reference distinguishing reactive and communicating swarm architectures and their military applications.↩
26. Army Doctrine Publication 3-90, Offense and Defense (Washington, DC: Department of the Army, 2019). [CITATION NEEDED - verify movement formation doctrine reference]↩
27. [CITATION NEEDED] - Reference on autonomous platform onboarding, service discovery, and network admission control in tactical environments.↩
28. [CITATION NEEDED] - Reference on mesh networking resilience for multi-robot coordination under intermittent connectivity.↩
29. [CITATION NEEDED] - Reference on CNN-based object detection performance benchmarks and military threat recognition applications.↩

30. [CITATION NEEDED] - Reference on deployed neural vision systems in military ground and perimeter security applications.↩

31. [CITATION NEEDED] - Reference on knowledge graph definition, graph database fundamentals, and entity-relationship representation.↩

32. [CITATION NEEDED] - Reference on knowledge graphs for intelligence sense-making and relationship discovery in analytical workflows.↩

33. [CITATION NEEDED] - Reference on IARPA CAUSE or DARPA Big Mechanism programs and graph-based causal reasoning in intelligence analysis.↩

34. W3C, “JSON-LD 1.1: A JSON-based Serialization for Linked Data,” W3C Recommendation, July 2020. https://www.w3.org/TR/json-ld11/↩

35. [CITATION NEEDED] - Reference on JSON-LD adoption in defense information architecture and comparison with XML-based standards.↩

36. [CITATION NEEDED] - Reference on graph visualization algorithms, aesthetic criteria, and interactive performance constraints.↩

37. Fruchterman, T.M.J. and Reingold, E.M., “Graph Drawing by Force-Directed Placement,” Software: Practice and Experience, 21(11), 1991. [CITATION NEEDED - verify full citation]↩

38. [CITATION NEEDED] - Reference on cognitive fit between visualization metaphor and analyst mental models in intelligence analysis.↩

39. [CITATION NEEDED] - Reference on entity resolution techniques in information extraction and intelligence fusion.↩

40. NATO Standardization Agreement (STANAG) 2022, “Intelligence Reports” (Brussels: NATO, current edition). [CITATION NEEDED - verify Admiralty Code STANAG reference]↩

41. [CITATION NEEDED] - Reference on maintaining analytical uncertainty in intelligence products and avoiding premature confidence collapse.↩

42. [CITATION NEEDED] - Reference on Basic Formal Ontology (BFO) and its use as upper ontology for defense domain extensions.↩

43. [CITATION NEEDED] - Reference on Common Core Ontologies (CCO) for military domain knowledge representation and DoDAF alignment.↩